Section 1 Overview

NOTE

This document describes the specification of several products. Because one or more products are under development, the specification of the following product(s) is tentative and subject to change.

– U2B24, U2B20, U2B10

The RH850/U2B-EVA, U2B-FCC, U2B24, U2B20, U2B10 and U2B6 are products of the single-chip microcontroller RH850 series from Renesas Electronics.

This section gives an overview of the RH850/U2B-EVA, U2B-FCC, U2B24, U2B20, U2B10 and U2B6.

1.1 Outline

This RH850/U2B is a 32-bit single-chip microcontroller with multiple CPUs, Code Flash, Data Flash, RAM modules, DMA controllers, A/D converters, timer units and many communication interfaces that are used in the automotive applications. This microcontroller conforms to the Automotive Safety Integrity Level (ASIL) that is highly demanded in the recent automotive field (ASIL D level).

RH850/U2B main features are as follows:

(1) **RH850 multi-core CPU**
This microcontroller contains multi RH850G4MH2.1 cores support RISC-type instruction sets and have significantly improved the instruction execution speed with basic instructions (one clock cycle per instruction) and the optimized 10-stage pipeline configurations. Furthermore, this product also supports multiplication instructions using a 32-bit hardware multiplier, saturated product-sum operation instructions, and bit manipulation instructions as instructions best suited for various fields.In addition, this product also support CPU virtualization function.
Two-byte basic instructions and high-level language instructions improve object code efficiency for the C compiler and reduce the program size. Furthermore, this product is suited for advanced real-time control applications by offering a high-speed response time including the processing time of the onchip interrupt controller.

(2) **On-Chip Code Flash and Data Flash**
This microcontroller has high-speed Code Flash from which CPU can fetch the instructions and the constant data. Code Flash with a capacity of up to 24MB can be reprogrammed when the chip is implemented in the application system.
This chip also has Data Flash capable of EEPROM emulation with a capacity of up to 512 KB and up to 160 KB exclusively for ICUMHB.

(3) **Rich peripheral functionality**
This microcontroller supports common communication interfaces such as SPI as well as automotive-oriented communication interfaces such as Ethernet, RHSIF, FlexRay, CAN-FD, LIN, SENT and PSI5. As internal peripheral modules, this microcontroller incorporates A/D Converter, System Timer, Generic Timer Module, and a dedicated Peripheral Interconnection module which connects the functionalities of these peripherals.

(4) **Functional Safety support**
This microcontroller includes several dedicated functionalities such as Dual-Core Lockstep

configuration for CPU, the memory protection with ECC/EDC on data and the address feedback mechanism, the bus protection with ECC/EDC on data and address, the peripheral module protection, and clock monitors to support the functional safety standard (ISO26262) required in the automotive applications.

(5) **Security support**
This microcontroller supports various security features. The Intelligent Cryptographic Unit - Master (ICUMHB) has a dedicated secure CPU (RH850 G3K) and some secure peripherals such as AES engines, a public key cryptography coprocessor, an engine that supports a HASH function based on SHA and Random Number Generator (RNG). This microcontroller also realizes the HW-level domain separation between non-secure and secure domains. The internal resources such as Code and Data Flash can be assigned to either a non-secure or secure domain, and the secure domain is protected against non-secure accesses by the HW mechanism. This microcontroller also has the protection scheme for debug and test functionality.

1.2 Features

Item	Features
Pin and Port Functions	• Most of the digital pins have selectable functionalities and a general purpose I/O port • Selectable drivability for output pins • Selectable pull-up/pull-down/none for input pins • Selectable Inversion/non-inversion for output pins • General-purpose I/O ports can be accessed per-pin basis by multiple CPUs without mutual exclusion • General-purpose I/O ports can operate in the following three different modes: – Port mode • The pin operates as a general-purpose I/O port. • The selection between input or output is selected by S/W via control bits. – S/W I/O control alternative mode • The pin is operated by an alternative function in S/W I/O control alternative mode • The selection between input or output is selected by S/W via control bits. – Direct I/O control alternative mode The pin is operated by an alternative function in direct I/O control alternative mode In contrast to S/W I/O control alternative mode, input/output is directly controlled by the alternative function in this mode.
CPU Subsystem	• G4MH2.1 CPU – High-performance 32-bit architecture for embedded control – 2x 32-bit internal data busses – Thirty-two 32-bit general purpose registers – RISC-type instruction set • Long/short type load/store instructions • Three-operand instructions • Instruction sets based on the C language – CPU operating modes: User mode / Supervisor mode – Address space: 4-GB linear address space for data and instructions • Floating Point Unit (FPU) – Each CPU has its own FPU. Supports single precision (32 bits) and double precision (64 bits). – Supports IEEE754-compliant data types and exceptions. – Rounding modes: Nearest, 0 direction, +∞ direction and -∞ direction – Handling of non-normalized numbers: Truncation to zero / IEEE754-compliant exception • An extended floating-point operation coprocessor (FXU) – Supports 4 single-precision (32-bit) parallel operations (FP-SIMD) – Supports IEEE754-compatible data types and exceptions – Rounding modes: Nearest, 0 direction, +∞ direction, and ∞ direction – Handling of non-normalized numbers: These are truncated to 0, or an exception is reported to comply with IEEE754. • Interrupt Handling – 64-level interrupt priority that can be specified for each channel – Two methods of interrupt handler address calculation: • Direct vector method • Table reference method – PUSHSP and POPSP instructions for high-speed context switch – Support for high-speed context save on interrupts by using the register bank feature – Support for restoration from the register bank using a dedicated instruction (RESBANK)

<table>
<tr><th>Item</th><th>Features</th></tr>
<tr><td></td><td>

Protection Mechanisms

Memory Protection Unit (MPU) protects memory regions against illegal accesses by software. (32 areas settable)
PE Guard (PEG) protects Local RAM and internal peripheral registers against illegal accesses by external bus masters.
Cluster RAM Guard (CRG) protects Cluster RAM regions against illegal accesses by all bus masters.

Instruction Cache

Each CPU has its own instruction cache for the Code Flash.
Parity-based way Selection (PBS) 4-way cache
Cache line size is 32 bytes and it is transferred from Code Flash in 128-bit x 2-burst manner.

Local RAM

Each CPU has its own Local RAM.
Up to two 32-bit accesses into Local RAM in a cycle

Cluster RAM

Each CPU can access Cluster RAM.
Up to four 32-bit accesses into Cluster RAM in a cycle

Inter-processor Communication and Mutual Exclusion Mechanisms

Inter-processor interrupt registers (IPIR)
Barrier-Synchronization (BARR)
Time Protection Timer (TPTM)
Three types of atomic instructions
• CAXI (Compare and Exchange for Interlock)
• SET1, CLR1 and NOT1 for atomic one-bit set and clear
• LDL (Load Linked) and STC (Store Conditional)

</td></tr>
<tr><td>Operating Modes</td><td>This MCU has multiple operating modes, which can be selected by the pins and option byte settings.
Normal Operation Mode :
After release from the reset state, an instruction fetch is carried out from the user area.
User Boot Mode :
After release from the reset state, an instruction fetch is carried out from the user boot area.
Serial Programming Mode:
After release from the reset state, the MCU boots up from the on-chip boot program and starts connection in the specified protocol for flash programming.</td></tr>
<tr><td>Interrupt Functions</td><td>1 non-maskable interrupt (NMI pin)
1 FE level interrupt
1024 maskable interrupts (high-speed: 32, low-speed: 992)
Simultaneous distribution of interrupt sources to multiple cores (each CPU)
Applicable sources: non-maskable interrupt (NMI pin), FE level interrupt, 32 high-speed maskable interrupts
External interrupt input function (IRQ pins)
Software interrupt function (SINT)
Inter-processor interrupt function (IPIR)
64-level priority specifiable for maskable interrupts
For RH850G4MH2 exceptions, see Section 3.2.2.3, Exceptions and Interrupts.</td></tr>
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Item	Features
Direct Memory Access	• Two types of DMA controller: DTS and sDMAC • DTS – Transfer information is stored in the dedicated SRAM (DTSRAM) – Up to 128 channels – Selectable transfer size: 1, 2, 4, 8, 16 bytes – Dual-address transfer mode – Address reloading function – Chain transfer function – Three transfer modes settable • Single transfer • Block transfer 1 (specified by the number of transfers) • Block transfer 2 (specified by address count) – Transfer target • On-chip memory • On-chip peripheral modules (excluding DTS) – Transfer requests can be set by interrupt sources and the software. • sDMAC – Transfer information is stored in the sDMAC control registers and can be loaded from the dedicated descriptor memory. – Up to 32 channels can be active at the same time. – Address space: 4 GB – Source and destination addresses are 32-bit wide – Selectable transfer size: 1, 2, 4, 8, 16, 32, 64 bytes – Supports scatter/gather and descriptor function. – Parallel reads and writes (fly-by) – Address mode: Dual address mode – Transfer request: Two request modes are available. Requests are triggered by on-chip modules and auto request. – Bus mode: Each channel can be set either normal speed mode or slow speed mode. – Arbitration mode can be set either fixed priority arbitration mode or round-robin arbitration mode between transfer channels. – Interrupt request: CPU interrupts from sDMAC are triggered by the following three factors. • Termination of the descriptor step • Termination of a data transfer • Occurrence of an address error
Reset Controller	7 reset functions • Power On Reset • System Reset 1 • System Reset 2 • Application Reset • DeepSTOP Reset • Module Reset • JTAG Reset External Reset output pin: $\overline{\text{RESETOUT}}$ Automatic RAM intialization after reset
Power Supply	• This product can be operated with the power management IC with three output voltages (5.0 V, 3.3 V, VDD) and with two output voltages (5.0 V/3.3 V, VDD). VDD can also be supplied by switching voltage regulator (SVR) with an external MOSFET, not by the power management IC directly. • Power on/off sequence has no constraints.
Clock Controller	• Main Oscillator (Main OSC) with an oscillation frequency of 16/20/24/25/40 MHz • High and low speed internal oscillators • PLL to generate high speed internal clocks by multiplying Main OSC input • Software-configurable external clock outputs

Item	Features
Standby Controller	• RUN mode – RUN mode is a normal operation mode where the CPU is operation and all of other modules can operate. The CPU can enter “HALT” state by executing the “HALT” instruction to stop its operation in this mode. • STOP mode: – STOP mode is a chip-level stand-by mode in which the clock supply to a certain clock domain can be stopped. • DeepSTOP mode – DeepSTOP mode is a chip-level stand-by mode in which further reduction of power consumption is possible than STOP mode. In addition to the clock supply stop, the power supply to the isolated area is switch off. • Cyclic RUN mode – Cyclic RUN mode is a low-power operation mode in which limited modules can operate at low speed. • Cyclic STOP mode – Cyclic STOP mode is a STOP mode in cyclic operation, and one CPU halts its operation. • Power off standby mode – Only RAM retention is possible in this mode. Low leakage is realized by turning off the power outside of the MCU and by stopping the oscillation. – The transition occurs by asserting the external reset pin and turning off the power. • HALT mode: – When the HALT instruction is executed, CPU transits to HALT mode and stops instruction execution. – Each CPU can be controlled individually. – CPU returns from this state by the occurrence of a reset, interrupt, or exception. • Module standby: – This function stops the clock for peripheral modules to reduce the power consumption in accordance with register settings.
Low-Power Sampler	Support checking the digital input ports and analog input ports to monitor the external input without consuming CPU resources.
Serial Flash Memory Interface	• Connection to one serial flash memory device is possible. • Selectable data bus sizes are 1 bit, 2 bits, or 4 bits. • 4-Gbyte address space • A built-in read cache enables efficient data reception (64-bit line × 16 entries). • Arbitrary bit rate settable by the on-chip baud rate generator
Multimedia Card Interface	• Compliant with JEDEC STANDARD JESD84-A441 (neither DDR mode nor 1.8-V operation is supported). • Supports 1-/4-/8-bit MMC bus widths. • Supports the backward-compatible mode. • High-speed mode is supported. • MMC Clock frequency = MMCA module clock frequency/2k (k = 1 to 10). • Supports block transfer. • Supports boot operation. • Supports high priority interrupts (HPI). • Supports background operation. • Interrupt requests: normal operation and error/timeout. • DMA transfer requests: buffer write and buffer read.
MSPI	• Chip select: Up to 12 for each channel • Master mode and slave mode selectable • Clock polarity, clock / data phase: Phase of clock and data selectablettable chip select output signals with phase of clock and data settable. • Transmission rate: Up to 40Mbps

Item	Features

RLIN3	• Protocol: LIN Specification Package Revision 1.3, 2.0, 2.1, 2.2, and SAEJ2602
	• Three operating modes
	  – LIN master mode
	  – LIN slave mode
	  – UART mode
	• Arbitary bit rate settable by the on-chip baud rate generator
	• LIN self-test mode with internal data loopback

RS-CANFD	The following are the specifications of CAN (RS-CANFD):
	• Classical CAN mode
	  – Conforming to CAN-FD ISO-11898-1 (2015)
	  – Transfer speed: Up to 1 Mbps
	  – Reception filtering
	• CAN-FD mode
	  – Conforming to CAN-FD ISO 11898-1 (2015)
	  – Transfer speed: Up to 8 Mbps
	  – Reception filtering

FlexRay	• Conforming to Protocol Specification 2.1 • Buffer size: 8 KB (up to 128 message buffers) • Message filter: ID filter, channel filter, cycle counter filter • Bit rate: Up to 10 Mbps
Renesas High Speed Bus Plus (RHSB)	The RHSB has the following two modes. • MSC-Plus mode: For communications conforming to the MSC-Plus standard • MSC mode: For communications conforming to the MSC standard The RHSB-Plus conforms to the following standards. • Micro Second Channel (MSC) Interface Description Data Sheet, V1.0 (BOSCH/Infineon, July 2003) • Micro Second Channel Interface Protocol Specification Version 1.0.0 (IPextreme, June 27, 2003)
RSENT	• Conforming to the SENT (single edge nibble transmission) protocol specified in the SAE J2716_201604 standard and the SPC (short PWM code) extension to the SENT specification • Unidirectional or bidirectional transfer through a single pin • Bidirectional transfer through two pins • Data transmission protected with CRC
RIIC	• I2C bus format with master mode or slave mode selectable • Transfer rate up to 400 kbps
RHSIF	The RHSIF module provides point-to-point communication. • Asynchronous high-speed LVDS interface based on ANSI/TIA/EIA-644 • Asynchronous high-speed LVDS interface supporting a maximum data rates of 320 Mbps • Four channels, including one channel with data streaming capability • Bus master interface used by target node to access shared memory
PSI5	• Conformance with PSI5 protocol specification V2.0 • Bit rates: Low speed(125 kbps), High speed(189 kbps), PAS compatibility mode(250 kbps) • Communication mode selectable
PSI5-S	• Supports the UART-based communication for PSI5 transceiver. • Conforming to PSI5 protocol specification V2.2 • Generates a PSI5 message from the UART transfer data. • The bit rate of the UART can be set by the built-in baud rate generator.

Item	Features
Ethernet Controller	Integrated Ethernet Communication Controller (Ethernet MAC)- according to standard IEEE 802.3 - 2008. • The controller supports these operation modes - 10/100/1000 Mbps • Full-duplex operation • PHY Interface: MII, RMII, SGMII • Supports Receive filtering (L2~L4 including VLAN(up to 10)) • Supports Jumbo Frame(up to 16KB) • Supports IEEE 802.1AS-rev • Supports IEEE 802.1Qci Ingress Policing • Supports IEEE 802.1Qav - 2009 Credit-based shaper • Supports IEEE 802.1Qbv Time Aware Shaper • Supports IEEE 802.3br Interspersed Express Traffic • Supports IEEE 802.1Qbu Frame Preemption • Automatic Tx/Rx CRC computed by Ethernet Controller • Interrupt-free data transfer between system memory and network • Integrated DMA for data transfers from/to system memory • Integrated FIFO in Ethernet controller • Test modes - Internal loopback - External loopback • Magic PacketTM*1 detection • Generation of gPTP and AVTP timers *1 Magic Packet is a trademark of Advanced Micro Devices, inc
TAUD	• 16 channels • 16-bit counter and 16-bit data register per channel • Independent channel operation • Synchronous channel operation (master and slave operation) • Generation of different types of output signal • Real-time output • Counter which can be triggered by external signal • Interrupt generation
TAUJ	• Independent channel operation function (operated using a single channel) • Synchronous channel operation function (operated using a master channel and multiple slave channels) The TAUJ can operate independently or synchronously (combine with other channels)
Watchdog Timer	Can generate a signal to the ECM when a counter overflows (timer expires). Can generate an interrupt at 75% of the counter overflow value. An interrupt request can be generated at any function of the counter value. A window open period can be set to any function of the counter value.
Secure WDT	Can generate an error signal for the Interrupt Controller in response to an error. Confirmation of matching with a specified Program Counter (PC) value of the CPU0.
OS Timer	• A 32-bit timer assuming use by an OS • Interval timer mode or free-running timer mode selectable • Synchronous start between units available
Long-Term System Counter (LTSC)	• Free-run up counting • Atomic read/write access to all registers • Anytime read access to counter registers • Application reset (SW reset) can be masked. When masked, counter keeps running on reset occurrence and counter register will not be initialized.

Item	Features
Motor Control Timer (TSG3)	The TSG3n is an 18-bit timer counter with various motor control functions. • Count clock resolution: Minimum 12.5 ns (count clock = 80 MHz) • Operating mode corresponding to various motor control methods • Compare registers with reload buffer • 10-bit dead time counter • A/D conversion trigger signal generation • Interrupt skipping – Skipping rate: 1/1 to 1/32 • Forced output stop function • Compare value setting – Reload (simultaneous rewrite) or anytime rewrite can be selected. • Various reload mode • HT-PWM mode • 120-DC contorl • Three-phase encoder function (Hall sensor signals can be input). • Active level of the output pins TSG3nO1 to TSG3nO6 can be set individually. • Fail-safe function (Warning interrupt or error interrupt can be generated) • Direct transfer of carrier-cycle setting and PWM duty setting from EMU3 • Output selectable between rectangular waveform output from EMU3 and PWM generated by TSG3.
Advanced Timer Unit VI (ATU-VI)	• ATU-VI consists of seven timer blocks (timer A to timer G). Each timer block has the following features. – Timer A has a free run counter and can capture the counter values by an assertion of the external event inputs. – Timer B can generate a multiplied-and-corrected clock signal based on an external event input and can also supply the generated signal to other timer blocks. – Timer C is a general-purpose timer; input capture or output compare is selectable. – Timer D can output one-shot pulses. The MIN guard function and the multi-shot pulse function are implemented in the limited sub-blocks. – Timer E is a PWM output timer. – Timer F can be used as a measurement timer for external inputs. 7 operation modes are available. – Timer G has a counter and can be used as an interval timer. • Provided with on-chip 4-channel prescaler provided, which generates four types of clocks by dividing on-chip peripheral clock (PCLK) by 1/1 to 1/1024 • Each channel for a timer can select a count source from among four divided clocks generated by prescaler, two external clocks, and angle clock generated by timer B. • Pin input level after noise cancellation can be read in timer A, timer C and timer F. • Pin output values of timers C, timer D and timer E can be selected by the RHSB cross bar 'RHSB_XBAR' and can be output to the RHSB for downstream communication. • The RHSB received data on upstream communication in MSC-Plus mode can be input to the timer A, timer C and timer F as external input signals via the virtual port function. • Can output pulses to SAR-ADC, DSADC, CADC, FCMP, DFE, DFP, MSPI, TAUD, PSI5 and PSI5-S.

Item	Features
Generic Timer Module (GTM)	GTM is a modular timer unit and consists of the following submodules : • Advanced Routing Unit (ARU) • Broadcast Module (BRC) • Parameter Storage Module (PSM) • Clock Management Unit (CMU) • Cluster Configuration Module (CCM) • Time Base Unit (TBU) • Timer Input Module (TIM) • Timer Output Module (TOM) • ARU-connected Timer Output Module (ATOM) • Timer Input Output Module (TIO) • Dead Time Module (DTM) • Multi-Channel Sequencer (MCS) • Memory Configuration (MCFG) • TIM0 Input Mapping Module (MAP) • Digital PLL (DPLL) • Sensor Pattern Evaluation (SPE) • Interrupt Concentrator Module (ICM) • Output Compare Unit (CMP) • Monitoring Unit (MON)
High-Resolution PWM (HRPWM)	HRPWM can control a timer output timing with a finer resolution than the timer operating clock period by adding delay for positive and negative edges. • Up to 16 channels of timer outputs supported • Generating PWM at resolution of 32-divided clock period
Real-Time Clock (RTCA)	• Counters for years, months, day of the month, day of the week, hours, minutes, seconds, and a subcounter. • One Hz pulse output function • Fixed interval interrupt function • Alarm interrupt function
Encoder Timer A (ENCA)	The Encoder Timer A (ENCA) has the following functions: • Generation of the counter control signal from the encoder input signal, and counter operation in synchronization with peripheral high-speed clock. • Capture function for capturing the counter value with an external trigger signal • Compare function for compare match judgment with the counter value • Two capture compare registers that can be set separately for capture operation and for compare operation • Interrupt mask function for masking the interrupt request signal output as a result of compare match judgment during compare operation • Function for loading the value of the capture compare register to the counter upon underflow occurrence • Encoder input signal can be applied to the timer counter clearing condition • Edge or level for clearing the encoder input signal of the timer counter clearing condition can be selected • Detection of counter overflow and underflow and output of error flags and error interrupts • Five interrupts: two capture compare interrupts, one counter clear interrupt, one overflow interrupt, and one underflow interrupt.
Timer Option (TAPA)	Timer Option (TAPA) is for use with Timer Array Unit D (TAUD), Motor Control Timer (TSG3), High-Resolution PWM (HRPWM) and SAR Analog to Digital Converter(SAR-ADC). • Asynchronous Hi-Z control to each TAUD, TSG3 and HRPWM output is enabled by TAPA input signals. • A peak and valley interrupt based on interrupt signals fromTAUD can be output. • Two conversion-trigger signals for SAR-ADC based on interrupt signals from TAUD can be output.

Item	Features
Peripheral Interconnect	The peripheral interconnection handles synchronous operations using multiple timers. • Peripheral timer control • ADC trigger selection
Analog to Digital Converter (ADCK)	• A/D conversion method: Successive approximation • Configuration of analog input pins • Resolution: 12/10-bit • Conversion speed: 1.0 μs • Scan groups for five systems for each converter • Two scan modes (multicycle scan mode and continuous scan mode) • ADCKn: Up to 64 virtual channels (n = 0 to 4 and A) • Addition mode A/D conversion functions incorporated • Can enter data directly to the Generic Timer Module. • Safety functions • Supporting an upper / lower-limit-excess-notice-function for the ADC Voltage Monitor Secondary Error Generator in each virtual channel • Track & Hold (T&H) input channels Several channel inputs can select T&H circuit for synchronize conversion.
Delta-Sigma A/D Converter	10 channels incorporated • A/D conversion method: Delta-Sigma modulation method • Configuration of analog input pins 38 analog input pins in total • Input type: Single-ended input and differential input • Input gain function (PGA): ×1, ×2, ×4, or ×8 selectable • Calibration function • Can enter data directly to the digital filter and the Generic Timer Module. • Timestamp function • Outputs a boundary flag generating signal to the ADC Boundary Flag Generator. • Safety functions
Cyclic A/D Converter	8 channels incorporated • A/D conversion method: Cyclic conversion method • Configuration of analog input pins CADC00: 8 single-ended inputs / 4 differential inputs • Input type: Single-ended input and differential input • Calibration function • Can enter data directly to the digital filter and the Generic Timer Module. • Outputs a boundary flag generating signal to the ADC Boundary Flag Generator. • Safety functions
DSMIF	4 channels incorporated • Support sampling point: at falling edge / at rising edge / at both edge (master only) • Clock input/output: Supports master mode and slave mode. • Sampling clock frequency: 10, 16.7, 20MHz • Input Filter: SINC number of stages is selectable (1 - 3), Decimation ratio 4-256, Output data 16bit. • Limit check: High limit / Low limit / inbound / outbound (collaborate with ABFG) • 2 post filter (FIR0: 8 TAPs, FIR1: 24 TAPs) • Result output: Connected to GTM and DFE. • Supports module stand-by

Item	Features
Fast comparator	10 channels incorporated • Incorporates Comparator and DAC • DAC can output the reference voltage for the comparator • Resolution: 12 bits • Conversion time: 0.2 us • Self-diagnosis function
Digital Filter Engine (DFE)	DFE contributes the offload of CPU processing and enables the high-precision detection of the powertrain states (e.g. knocking) & its change positions by the following features. • Incorporates a 16- or 12- or 4-channel FIR and IIR digital filter unit in one DFE unit. (A 64-tap FIR filter can be selected only with even-numbered channels.) • IIR filtering performance will be two times higher than that of the E2x DFE. • Filter coefficients and data are stored in the DFE-dedicated RAM. • Incorporates an accumulation circuit (for accumulation processing or decimation processing for filtered data) and a peak-hold circuit (for peak-hold processing or comparison processing). • Number of accumulation/decimation count setting registers: up to 16 • Number of comparison value setting registers: up to 16 • Subtraction circuit is incorporated for subtraction on output data from two channels. • Peak-hold circuit is incorporated for peak holding and comparison • With the peak-hold function, either upper peaks or lower peaks can be selected. • On a single arbitrary channel in each DFE unit, three different peak values can be detected and held during a single Peak-hold process. • Generates an interrupt request if the comparison calculation result is true. • Generates the output data update interrupt and the filter processing end interrupt in parallel. • Inputs converted data from the on-chip A/D converter via private bus. • Enables enhanced mapping of SAR-ADC/DS-ADC/C-ADC conversion results to more DFE channels when compared with that of the E2x DFE and a parallel input of DS-ADC conversion results to both DFE units. • Flexible cascade function enables the input of output data to channels in the same and/or different DFE unit. (Output data of channels in DFE0, for instance, can be input to channels in DFE0 and/or DFE1.) • Two types of FIFOs can be used: 8-input 8-stage buffer and 1-input 8-stage buffer.
Data Flow Processor (DFP)	• Execution Core – 512-bit vector data pipeline with large vector register file for in-place calculation – Scalar multi-threading processors to feed multiple threads to vector data pipeline for utilizing pipeline efficiently – Scalable clustering capability for heavy-load applications • Memory System – L0 sub-banked vector register file (VRF) – L1 instruction caches (Inst.$) – L1 high bandwidth vector local RAM (VLRAM) – System bus interface – NVM bus interface – Data transfer unit for prevention of vector unit interference (DTU) • Thread Scheduling – Hardware assisted thread scheduling by the control core – Multiple sources interrupt handling capability (INT)
Enhanced Motor Control Unit 3 (EMU3)	The Enhanced Motor Control Unit 3 (EMU3) is a set of motor control accelerator engines that calculate the 3-phase PWM compare value using a vector control algorithm and generate rectangle wave patterns based on the current value measured by an A/D converter and the motor's angle value obtained through an R/D converter.
Resolver Digital Converter (RDC3A)	RDC3AL is R/D converter of successive-approximation-register analog-to-digital type. And RDC3AL works by using dedicated SAR-ADC for RDC3AL. RDC3AS is R/D converter of delta-sigma analog-to-digital type. And RDC3AS works by cooperats with DSADC.

Item	Features
Timer Pattern Buffer (TPBA)	• Count clock resolution: Minimum 12.5 ns (count clock = 80 MHz) • 16-bit counter • 16-bit duty register • 16-bit period setting register • 7-bit address counter register • 7-bit pattern number setting register • Interrupt request signals – Period-matched detection interrupt – Duty-cycle-matched detection interrupt – Number-of-patterns matched detection interrupt • Number of duty patterns – 64 patterns (16 bits) or 128 patterns (8 bits) • Automatic duty generation according to the number of patterns • Output control by software • The count clock can be selected from PCLK, PCLK/2, PCLK/4, and PCLK/8 according to the prescaler set value. • Synchronous start with another timer
Functional Safety	• Development compliant with ISO26262 (ASIL-D) functional safety standard • Major safety mechanisms – Flash memory ECC error detection function – RAM ECC error detection function – Peripheral module RAM ECC error detection function (e.g. FlexRay, CAN, GTM) – Clock monitor – Error Control Module (ECM) – Duplexing of modules (e.g. CPUs, ECM, error output pins) – Automatic Power-on BIST execution after reset – Standby Resume BIST (SR-BIST) execution selection after wake-up from DeepSTOP mode.
Error Control Module (ECM)	The ECM collects error signals coming from different error sources and monitoring circuits. When an error is detected, ECM: • Generates FE or EI interrupt. • Generates internal reset. • Changes port state to Hi-Z. • Outputs errors via a dedicated pin. And the error flags which are captured in the ECM can be only cleared by Power Up Reset or Deep Stop Reset.
Data CRC Function (KCRC)	Supports major CRC polynomials 64-bit CRC64ECMA: 42F0E1EB A9EA3693$_H$ 32-bit Ethernet CRC: 04C11DB7$_H$ 32-bit CRC32C: 1EDC6F41$_H$ 32-bit CRC32P4: F4ACFB13$_H$ 16-bit CCITT: 1021$_H$ 8-bit SAE J1850: 1D$_H$ 8-bit 0x2F: 2F$_H$
Power Supply Voltage Monitor	• The power supply voltage monitor is used for monitoring power domains. • The power supply voltage monitor has H-side (HDET) and L-side (LDET) voltage detectors, which detect if the monitored voltage is higher or lower than the specified voltage. • If the power supply voltage monitor detects an error, it gives the error notification to external devices outside of the MCU.
Clock Monitor	• Up to 17 clock monitors depending on the device configuration. • Detects clock disturbances that results in lower or higher frequency than target frequency, and sends an error notification to the ECM. Supports the self-diagnosis function.

Item	Features
Temperature Sensor	• Out of range detection of temperature. • Operating modes – Single measurement mode – Continuous measurement mode • Interrupt requests – Temperature measurment end interrupt (INTOTSOTI) – Temperature rise/fall interrupt (INTOTSOTULI) – Temperature alarm error interrupt (OTABE) – Temperature sensor error interrupt (INTOTSOTE) • Support self-diagnosis function
Debugging and Calibration	• Debug interface – Nexus JTAG interface – Low Pin Debug Interface (4-pin) – External event triggers (input and output) – RHSIF Debug interface – Debug over CAN – DFP JTAG interface – Aurora Trade Interface
Flash Memory	• Code Flash – Program unit: 512 bytes, Erase unit: 16 KB for the first 8 blocks and 64 KB for remaining blocks in each Bank – Per-block OTP (One Time Programmable) support • Data Flash – Program unit: 4, 8, 16, 32, 64, 128 bytes, Erase unit:4-Kbyte • ECC support for error detection and correction • Programming method – Serial programming: Programming of flash memory by external flash memory programmer – Self programming: Programming of flash memory by a user program • BGO (Back Ground Operation) support – Code Flash read is possible during Data Flash programming/erasure. – Code Flash read from other banks is possible during programming/erasure of a bank of Code Flash. – Data Flash read from other Data Areas is possible during programming/erasure of a Data Area of Data Flash. – Data Flash read is possible during Code Flash programming/erasure. • Multi FPSYS Operation support • Suspend/Resume supported • Hardware Property Area is separated from Code Flash and Data Area to store the system configuration parameters. • OTA (Over-the-Air) update support
Basic Security	• This product includes the following security functions: – Hardware-level domain separation – Flash protection – Mode entry protection – Debugger authentication – Serial programing authentication – Device degradation
ICUM	The ICUMHB is an on-chip Hardware Security Module (HSM). The ICUMHB supports user-defined security services to the overall system based on cryptographic primitives.
ACEU	• The ACEU supports encryption and decryption based on AES algorithm.

Item	Features
RAM Modules	• Local RAM for each CPU – Any bus masters can access the Local RAM of CPUs. • Cluster RAM – Any bus masters can access the Cluster RAM. • Local RAM, Cluster RAM, DFP RAM and peripheral module RAMs are protected by SEC-DED ECC on data for functional safety. – Local RAM and Cluster RAM also include address feedback.
Boundary Scan	• Compliant with IEEE 1149.1 • JTAG interface (TDI, TDO, TMS, TCK and TRST) • TAP controller • Supports the following instructions – BYPASS – EXTEST – SAMPLE/PRELOAD – CLAMP – HIGHZ – IDCODE

1.3 Application Fields

Automotive field (including body control, chassis & safety, engine control system and transmission control system)

1.4 Ordering Information

Table 1.1 Product Name List

Product Name	Package	On-Chip ROM	Operating Temperature (Tj)	External Oscillator	Maximum Operating Frequency	ADCK restriction*1 RAM restriction*2	Note
R7F70251*EABG (RH850/U2B24)	FCBGA-468 0.8-mm ball pitch 25 mm × 25mm	24 MB	max. 150°C	16/20/24/25/40 MHz	400 MHz	No	—
R7F70251*EABA (RH850/U2B24)	FCBGA-373 0.8-mm ball pitch 21 mm × 21 mm	24 MB	max. 150°C	16/20/24/25/40 MHz	400 MHz	No	—
R7F70252*FABG-C (RH850/U2B20)	Plastic FBGA-468 0.8-mm ball pitch 25 mm × 25 mm	20 MB	max. 160°C	16/20/24/25/40 MHz	400 MHz	No	—
R7F70252*FABA-C (RH850/U2B20)	Plastic FBGA-373 0.8-mm ball pitch 21 mm × 21 mm	20 MB	max. 160°C	16/20/24/25/40 MHz	400 MHz	No	—
R7F70254*FABG-C (RH850/U2B10)	Plastic FBGA-468 0.8-mm ball pitch 25 mm × 25 mm	10 MB	max. 160°C	16/20/24/25/40 MHz	400 MHz	No	—
R7F70254*FABA-C (RH850/U2B10)	Plastic FBGA-373 0.8-mm ball pitch 21 mm × 21 mm	10 MB	max. 160°C	16/20/24/25/40 MHz	400 MHz	No	—
R7F70254*FABB-C (RH850/U2B10)	Plastic FBGA-292 0.8-mm ball pitch 17 mm × 17 mm	10 MB	max. 160°C	16/20/24/25/40 MHz	400 MHz	No	—
R7F70255*FABB-C (RH850/U2B6)	Plastic FBGA-292 0.8-mm ball pitch 17 mm × 17 mm	6 MB	max. 160°C	16/20/24/25/40 MHz	400 MHz	Yes	*3
R7F70255*AFABB-C (RH850/U2B6)	Plastic FBGA-292 0.8-mm ball pitch 17 mm × 17 mm	6 MB	max. 160°C	16/20/24/25/40 MHz	400 MHz	No	—

Note 1. Refer to **Section 50.12, ADCK restriction**.
Note 2. Refer to **Section 64.5, RAM restriction**.
Note 3. As sales of these products is limited, please contact your sales representative first when considering these products for purchasing and project development.

FBGA is hereafter referred to as BGA unless the complete abbreviation is required.

The RH850/U2B-FCC has 8 variants. Select the proper product according to the table below.

Table 1.2 RH850/U2B-FCC, EVA List

Frequency	Package			
	FCBGA-600	FCBGA-468	FCBGA-373	FCBGA-292
400 MHz	R7F702Z2*ADBG (For EVA)	R7F702Z2*EDBG (For U2B24) R7F702Z2*EDBG (For U2B20) R7F702Z2*EDBG (For U2B10)	R7F702Z2*EDBA (For U2B24) R7F702Z2*EDBA (For U2B20) R7F702Z2*EDBA (For U2B10)	R7F702Z2*EDBB (For U2B10) R7F702Z2*EDBB (For U2B6)

1.5 Differences in the Specifications of RH850/U2B

The table below lists the differences in the specifications of RH850/U2B.

Table 1.3 Device Overview Table (1/3)

Feature		RH850 U2B24	RH850 U2B20	RH850 U2B10	RH850 U2B6
Core	Main Core/Lockstep in performance config	6/4	6/2	4/2	3/2
	Main Core/Lockstep in safety config	5/5	4/4	3/3	3/2
	FPU performence config / safety config	6/5 (each main core)	6/4 (each main core)	4/3 (each main core)	3 (each main core)
	FXU (FP-SIMD)	2	2	FCC/EVA:2 MP:1	1
	MPU Region	32	32	32	32
	Frequency	400MHz	400MHz	400MHz	400MHz
System	Virtualization	Yes	Yes	Yes	Yes
	QoS	Yes	Yes	Yes	-
Flash	Code Flash	24Mbyte	20Mbyte	10Mbyte	6Mbyte
	Data Flash (EEPROM emulation)	512Kbyte	512Kbyte	256Kbyte	128Kbyte
	Data Flash (HSM)	160Kbyte	96Kbyte	64Kbyte	64Kbyte
SRAM	Total (CRAM + LRAM)	4096Kbyte	2560Kbyte	1280Kbyte	576Kbyte
	Local Data (LRAM)	64Kbyte/core	64Kbyte/core	64Kbyte/core	64Kbyte/core
	Cluster (CRAM)	3712Kbyte	2176Kbyte	1024Kbyte	384Kbyte
	Standby RAM (included in CRAM)	128Kbyte	128Kbyte	128Kbyte	32Kbyte
	Instruction Cache	16Kbyte/core	16Kbyte/core	16Kbyte/core	16Kbyte/core
		(PBS 4way)	(PBS 4way)	(PBS 4way)	(PBS 4way)
	Data Cache (Flash only)	4lines (256bit/line)	4lines (256bit/line)	4lines (256bit/line)	4lines (256bit/line)
	Emulation RAM	(7MByte*1)	(7MByte*1)	(6MByte*1)	(5MByte*1)
	Instrumentation RAM	- (96KByte*1)	- (96KByte*1)	- (96KByte*1)	- (96KByte*1)
	Trace RAM	- (64KByte*1)	- (64KByte*1)	- (64KByte*1)	- (64KByte*1)
DMA	Channels (sDMAC/DTS)	32/128	32/128	32/128	16/128
SAR ADC	Modules	5	5	4	3
	Total inputs	96	96	96	64
	Virtual channels per module	64	64	64	64
Delta Sigma ADC	Modules	10	10	10	4
	Total inputs	38	38	38	18
Cyclic ADC	Modules	1	1	1	-
	Total inputs	8	8	8	-
DSMIF*3	Unit	2	2	2	-
	Channel	4	4	4	-

Table 1.3 Device Overview Table (2/3)

Feature		RH850 U2B24	RH850 U2B20	RH850 U2B10	RH850 U2B6
RDC	RDC3AL	-	-	-	2
	RDC3AS	2	2	2	-
Fast comparator	Modules	1	1	1	1
	Comparator	10	8	4	4
	DAC	10	8	4	4
Timer (GTM)	GTM	1	1	1	1
Timer (ATU-VI)	Timer A	8	8	8	6
	Timer B	1	1	1	1
	Timer C	External: 44 Internal: 16	External: 44 Internal: 16	External: 36 Internal: 8	External: 16 Internal: 4
	Timer D	24 w/MSF* 24 w/o MSF* *MSF: Multi Shot Pulse Function	24 w/MSF* 24 w/o MSF* *MSF: Multi Shot Pulse Function	20 w/MSF* 12 w/o MSF* *MSF: Multi Shot Pulse Function	8 w/MSF* 8 w/o MSF* *MSF: Multi Shot Pulse Function
	Timer E	40	40	36	20
	Timer F	20	20	16	7
	Timer G	14	14	14	10
Timer	HRPWM	16	16	16	16
	TAPA	6	6	6	6
	TAUD	4	4	4	4
	TAUJ	2(AWO)	2(AWO)	-	-
	TPBA	2	2	2	2
	TSG3	3	3	3	3
	ENCA	2	2	2	2
	RTCA	1(AWO	1(AWO)	-	-
	Window Watchdog Timer (WDTB)	6+1(AWO)	6+1(AWO)	4	3
	Secure Watchdog Timer (SWDT)	1	1	1	1
	OS Timer (OSTM)	6	6	4	3
	Time Protection Timer (TPTM)	6	6	4	3
	Long-Term System Counter (LTSC)	1	1	1	1
Accelerator	Data Flow Processor (DFP)	DR1000C	DR1000C	DR1000C	-
	Digital Filter Engine (DFE) (units/channels)	2/20	2/20	2/20	2/16
	Enhanced Motor Control Unit (EMU)	-	-	-	2

Table 1.3 Device Overview Table (3/3)

Feature		RH850 U2B24	RH850 U2B20	RH850 U2B10	RH850 U2B6
Security	ICUMHB	Yes	Yes	Yes	Yes
	Secure RAM	64 KB	64 KB	64 KB	64 KB
	Secure Data Flash	160 KB	96 KB	64 KB	64 KB
	Code Flash Protection	Yes	Yes	Yes	Yes
	ACEU	2	2	-	-
Interfaces	FlexRay Nodes [Channels]	1 [2ch]	1 [2ch]	1 [2ch]	1 [2ch]
	RS-CANFD	10	8	8	8
	MSPI/MSPI(LVDS)/RLIN3/ RLIN3(25Mpbs)	8/2/23/1	8/2/6/2	8/2/6/1	6/1/6/1
	RIIC	2	2	2	-
	RSENT/PSI5/PSI5S	30/4/2	30/4/2	30/4/2	10/-/-
	RHSIF/RHSB	2/4	1/3	1/3	-/2
	Ethernet (100Mb(TSN)/ 1Gb(TSN))	-/2	1/1	1/-	1/-
Safety	CRC	6	6	4	4
	Voltage monitor	Yes	Yes	Yes	Yes
	Clock monitor	Yes	Yes	Yes	Yes
	Temperature Sensor	Yes	Yes	Yes	Yes
	ECM	1	1	1	1
Power Management	LPS	Yes	Yes	-	-
	STBC	Yes	Yes	Yes	Yes
	Stop mode	Yes	Yes	Yes	Yes
	Deep stop mode	Yes	Yes	-	-
	Cyclic run/cyclic stop	Yes	Yes	-	-
	Power off standby	Yes	Yes	Yes	Yes
External Memory Interfaces	MMCA	1	1	1	-
	SFMA	1	1	1	1
Debug	Nexus-JTAG	Yes	Yes	Yes	Yes
	Trace I/F (Aurora)	(No*1)	(No*1)	(No*1)	(No*1)
	Low Pin Debug I/F(4-pin)	Yes	Yes	Yes	Yes
	RHSIF Debug I/F	(No*1)	(No*1)	(No*1)	(No*1)
	Boundary Scan	Yes	Yes	Yes	Yes
Package	FCBGA600*2	No	No	No	No
	FCBGA468	Yes	No	No	No
	FCBGA373	Yes	No	No	No
	BGA468	No	Yes	Yes	No
	BGA373	No	Yes	Yes	No
	BGA292	No	No	Yes	Yes

Note 1. Only FCC and EVA device is supported
Note 2. Only EVA device is supported
Note 3. FCC and EVA device is not supported

1.6 Pin Connection Diagram (Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	
A	VSS(NC)	VSS	P31_14	P33_13	P33_12	P33_11	P33_10	P33_9	P33_8	P33_7	P33_6	P33_5	P33_4	P33_3	P33_2	P33_1	P33_0	VSS	AN120	AN121	AN100	AN051	AN052	AN033	AN032	AN021	AN020	AN003	AN001	ADSVSS	ADSVSS(NC)	A
B	VSS	VSS	P31_15	P31_13	P31_12	P31_11	P31_10	P31_9	P31_8	P31_7	P31_6	P31_5	P31_4	P31_3	P31_2	P31_1	P31_0	VSS	AN111	AN113	AN101	AN053	AN050	AN042	AN040	AN010	AN011	AN002	AN000	ADSVSS	ADSVSS	B
C	P23_5	P23_6	VSS	P32_6	P32_4	P32_2	P32_0	P34_4	P34_2	P34_0	P30_12	P30_11	P30_10	P30_9	P30_8	P30_7	VSS	VSS	AN112	AN122	AN103	AN041	AN043	AN031	AN022	AN013	AN012	ADSVSS	ADSVSS	AN200	AN201	C
D	P23_4	P23_7	P23_1	VSS	P32_5	P32_3	P32_1	P34_3	P34_1	P30_6	P30_5	P30_4	P30_3	P30_2	P30_1	P30_0	VSS	AN123	A1VREFH	AN110	AN102	AN061	AN060	A0VREFH	A0VSS	ADSVREFH	ADSVCL	ADSVREFL	AN203	AN202	AN221	D
E	P23_8	P23_9	P23_3	P23_2	VSS	P23_0	E0VCC	VSS	VSS	VSS	VSS	VSS	VSS	VSS	E0VCC	VSS	VSS	A1VSS	A1VCC	AN062	AN063	AN030	AN023	A0VCC	A0VSS	ADSVCC	ADSVSS	AN220	AN211	AN213	AN231	E
F	P23_10	P23_11	P23_12	P24_1	P24_0																						AN210	AN212	AN233	AN230	AN223	F
G	P24_12	P24_13	P24_14	P24_15	P24_2																						A2VCC	AN222	AN240	AN232	AN243	G
H	VSS	VSS	VCC	P24_8	P24_9																						A2VSS	A2VREFH	AN242	AN241	AN260	H
J	ETH1_SG_TXD_N	ETH1_SG_TXD_P	VSS	GETH1VCL	P24_10																						AN253	AN252	AN251	AN250	AN263	J
K	ETH1_SG_RXDN	ETH1_SG_RXDP	GETH0PVCC	GETH0BVCC	P24_11																						AFCVCC	AN270	AN261	AN262	AN273	K
L	ETH0_SG_TXD_N	ETH0_SG_TXD_P	VSS	GETH0VCL	VSS																						AFCVSS	AN302	AN271	AN272	AN301	L
M	ETH0_SG_RXDN	ETH0_SG_RXDP	VSS	ETH_SG_REFCLK	VSS							VDD	VDD	VDD	VSS	VSS	VSS	VDD	VDD	VDD							AN300	AN303	AN310	AN313	AN311	M
N	OSCVCC	VSS	P25_2	P25_10	VSS							VDD	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD							A3VCC	A3VREFH	AN312	AN363	AN360	N
P	X1	X2	P25_11	P25_7	VSS							VDD	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD							A3VSS	AN370	AN362	AN383	AN373	P
R	LVDVCC	VSS	P25_9	P25_8	VSS							VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS							AN371	AN372	AN393	AN380	AN390	R
T	P21_3	P21_2	P22_3	P22_0	E0VCC							VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS							AN361	AN382	AN381	AN392	AN391	T
U	P21_5	P21_4	P22_1	P22_2	VSS							VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS							VSS	VSS	VSS	VSS	VSS	U
V	P24_5	P24_4	P24_3	P22_4	VSS							VDD	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD							CICREFN	VSS	VSS	TODN3	TODP3	V
W	P24_7	P24_6	P22_5	P22_6	VSS							VDD	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD							CICREFP	VSS	VSS	TODN2	TODP2	W
Y	P25_3	P25_4	P22_8	P22_7	AWOVCL							VDD	VDD	VDD	VSS	VSS	VSS	VDD	VDD	VDD							VSS	VSS	VSS	VSS	VSS	Y
AA	P25_5	P25_6	P22_10	P22_9	RAMSVCL																						EMUVDD	EMUVDD	VSS	TODN1	TODP1	AA
AB	P25_13	P25_12	P22_12	P22_11	VSS																						VSS	VSS	VSS	TODN0	TODP0	AB
AC	P25_15	P25_14	P22_13	P25_0	P25_1																						EMUVCC	EMUVCC	VSS	VSS	VSS	AC
AD	$\overline{\mathrm{TRST}}$	FLMD0	$\overline{\mathrm{RESETOUT}}$	SBMD	$\overline{\mathrm{MSYN}}$																						$\overline{\mathrm{AURORES}}$	EMUVDD	P00_11	P00_5	P02_10	AD
AE	$\overline{\text{ERROROUT\_M}}$	JP0_2	$\overline{\mathrm{RESET}}$	PWRCTL	JP1_5																						EMUVDD	EMUVDD	P02_8	P00_4	P02_11	AE
AF	JP0_3	JP0_0	$\overline{\mathrm{VMONOUT}}$	JP1_3	$\overline{\mathrm{EVTI}}$																						E0VCC	VSS	P00_9	P02_7	P00_10	AF
AG	JP0_1	JP0_5	JP1_2	$\overline{\mathrm{EVTO0}}$	VSS	VSS	VSS	VSS	VSS	E2VCC	P15_8	P15_11	P15_12	P14_10	P14_12	P12_6	P13_8	E2VCC	VSS	E1VCC	P11_8	P10_10	E1VCC	VSS	P01_7	E0VCC	VSS	P02_4	P00_8	P02_6	P00_3	AG
AH	SYSVCC	JP1_1	JP1_0	VSS	VSS	VSS	PEMD0	P20_5	P20_7	P15_2	P15_7	P15_9	P14_6	P14_9	P14_11	P12_1	P12_5	P13_9	P13_11	P11_0	P11_4	P11_10	P10_12	P10_14	P01_11	P00_6	P02_0	P00_1	P02_3	P02_5	P02_9	AH
AJ	J0VCC	J1VCC	SVRNGATE	SVRPGATE	VSS	DBGSEL0	DBGSEL1	P20_4	P20_6	P15_3	P15_6	P15_10	P14_7	P14_8	P12_0	P12_2	P12_7	P13_10	P13_12	P11_1	P11_5	P11_9	P10_11	P10_13	P01_10	P01_13	P01_4	P01_5	P02_2	P00_2	P02_1	AJ
AK	VSS	VSS	SVRDRVSS	SVRAVSS	VSS	PEMD1	P20_1	P20_3	P15_0	P15_4	P14_0	P14_2	P13_0	P13_2	P14_5	P12_3	P12_8	P13_13	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	P01_12	P01_14	P01_3	P00_0	P01_6	VSS	AK
AL	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	ICE	PEMD2	P20_2	P20_0	P15_1	P15_5	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_9	P13_14	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_8	P01_9	P01_15	P00_7	VSS		AL
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	

Figure 1.1 Pin Connection Diagram U2B-EVA(FCBGA600)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_10	P33_9	P33_8	P33_7	P33_6	P33_5	P33_4	P33_3	P33_2	P33_1	P33_0	AN111	AN110	AN101	AN100	AN051	AN050	AN033	AN032	AN021	AN020	AN003	AN001	ADSVSS	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P31_12	P31_11	P31_10	P31_9	P31_8	P31_7	P31_6	P31_5	P31_4	P31_3	P31_2	P31_1	P31_0	AN112	AN113	AN102	AN053	AN052	AN041	AN043	AN031	AN022	AN010	AN011	AN002	AN000	ADSVSS	B
C	P31_14	E0VCC	VSS	P32_3	P32_2	P32_0	P34_4	P34_2	P34_0	P30_12	P30_11	P30_10	P30_9	P30_8	P30_7	AN121	AN120	AN103	A1VREFH	AN061	AN060	AN042	AN040	A0VREFH	AN013	AN012	ADSVCC	ADSVSS	ADSVCL	AN200	C
D	P31_15	P31_13	P32_5			P32_1	P34_3	P34_1	P30_6	P30_5	P30_4	P30_3	P30_2	P30_1	P30_0	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN202	AN201	D
E	P23_1	P23_0	P32_6																									ADSVREFL	AN203	AN221	E
F	P23_5	P23_4	P23_3	P23_2																							AN210	AN211	AN213	AN231	F
G	P23_7	P23_6	P24_1	P24_0																							AN220	AN233	AN230	AN223	G
H	VSS	VSS	VCC	P24_2																							AN212	AN222	AN240	AN232	H
J	ETH1_SG_TXD_N	ETH1_SG_TXD_P	VSS	GETH1VCL																							A2VCC	AN242	AN241	AN243	J
K	ETH1_SG_RXD_N	ETH1_SG_RXD_P	GETH0PVCC	GETH0BVCC																							A2VSS	A2VREFH	AN250	AN251	K
L	ETH0_SG_TXD_N	ETH0_SG_TXD_P	VSS	GETH0VCL																							AN252	AN260	AN253	AN261	L
M	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	ETH_SG_REFCLK								VSS	VSS	VDD	VDD	CICREFN	CICREFP	EMUVDD	VSS								AFCVCC	AN262	AN263	AN270	M
N	OSCVCC	VSS	P25_2	P25_10								VDD	VSS	VSS	VSS	VSS	VSS	VSS	EMUVDD								AFCVSS	AN271	AN272	AN273	N
P	X1	X2	P25_11	P25_7								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD								AN300	AN301	AN302	AN303	P
R	LVDVCC	VSS	P25_9	P25_8								TODN0	VSS	VSS	VSS	VSS	VSS	EMUVCC	VDD								A3VCC	A3VREFH	AN310	AN311	R
T	P21_3	P21_2	P22_3	P22_0								TODP0	VSS	VSS	VSS	VSS	VSS	AURORES	VDD								A3VSS	AN312	AN313	AN360	T
U	P21_5	P21_4	P22_1	P22_2								VDD	MSYN	VSS	VSS	VSS	VSS	VSS	VDD								AN361	AN362	AN363	AN370	U
V	P24_5	P24_4	P24_3	P22_4								VDD	VSS	EVTO0	VSS	VSS	VSS	VSS	VDD								AN371	AN372	AN373	AN380	V
W	P24_7	P24_6	P22_5	P22_6								VSS	AWOVCL	VDD	VDD	VDD	VDD	VSS	VSS								AN381	AN382	AN383	AN390	W
Y	P25_3	P25_4	P22_8	P22_7																							P00_5	AN391	AN392	AN393	Y
AA	P25_5	P25_6	P22_10	P22_9																							P00_4	P00_11	P02_10	P02_11	AA
AB	P25_13	P25_12	P22_12	P22_11																							P00_3	P00_10	P02_8	P02_9	AB
AC	P25_15	P25_14	P22_13	RAMSVCL																							P00_2	P00_9	P02_6	P02_7	AC
AD	TRST	FLMD0	RESETOUT	SBMD																							P00_1	P00_8	P02_4	P02_5	AD
AE	ERROROUT_M	JP0_2	RESET	PWRCTL																							P00_0	P00_7	P02_2	P02_3	AE
AF	JP0_3	JP0_0	VMONOUT																									P00_6	P02_0	P02_1	AF
AG	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P15_3	P15_7	P15_11	P14_6	P14_8	P14_10	P14_12	P12_1	P12_5	P13_8	P13_11	P11_0	P11_4	P11_9	P10_10	P10_12	P10_14	P01_9			P01_3	P01_4	P01_5	AG
AH	SYSVCC	J1VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P15_0	P15_4	P15_8	P15_12	P14_7	P14_9	P14_11	P12_0	P12_2	P12_6	P13_9	P13_12	P11_1	P11_5	P11_8	P11_10	P10_11	P10_13	P01_8	P01_11	E1VCC	VSS	P01_6	P01_7	AH
AJ	J0VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P15_1	P15_5	P15_9	P14_0	P14_2	P13_0	P13_2	P14_5	P12_3	P12_7	P13_10	P13_13	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	P01_12	P01_13	E0VCC	VSS	VSS	AJ
AK	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P15_2	P15_6	P15_10	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P12_9	P13_14	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_10	P01_14	P01_15	E0VCC	VSS(NC)	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	

Figure 1.2 Pin Connection Diagram U2B24-FCC(FCBGA468)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_9	P33_7	P33_5	P33_3	P33_1	P33_0	AN111	AN110	AN100	AN051	AN050	AN042	AN040	AN021	AN020	AN010	AN011	AN002	AN000	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P32_2	P32_1	P33_10	P33_8	P33_6	P33_4	P33_2	AN121	AN120	AN102	AN101	AN053	AN052	AN033	AN032	AN031	AN013	AN003	AN001	ADSVCL	AN203	AN200	B
C	P23_0	E0VCC	VSS	P32_3	P32_0	P34_2	P34_0	P30_4	P30_2	P30_0	AN112	AN113	AN103	A1VREFH	AN061	AN060	AN041	AN043	AN022	A0VREFH	AN012	ADSVCC	ADSVSS	AN202	AN201	C
D	P23_6	P23_1	P32_5			P34_4	P34_3	P34_1	P30_3	P30_1	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN221	AN213	D
E	P23_7	P23_5	P32_6																				ADSVREFL	AN231	AN230	E
F	VSS	VSS	P23_3	P23_2																		AN210	AN220	AN223	AN233	F
G	ETH1_SG_TXD_N	ETH1_SG_TXD_P	VCC	P23_4																		A2VCC	A2VREFH	AN232	AN222	G
H	ETH1_SG_RXD_N	ETH1_SG_RXD_P	VSS	GETH1VCL																		A2VSS	AN211	AN240	AN250	H
J	ETH0_SG_TXD_N	ETH0_SG_TXD_P	GETH0PVCC	GETH0BVCC																		AN212	AN243	AN241	AN251	J
K	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	GETH0VCL						VSS	VSS	VDD	CICREFN	CICREFP	EMUVDD	VSS						AN242	AN252	AN260	AN253	K
L	OSCVCC	VSS	VSS	ETH_SG_REFCLK						VDD	VSS	VSS	VSS	VSS	VSS	EMUVDD						AFCVCC	AN262	AN270	AN261	L
M	X1	X2	P25_2	P22_0						TODN0	VSS	VSS	VSS	VSS	VSS	VDD						AFCVSS	AN263	AN271	AN272	M
N	LVDVCC	VSS	P22_3	P22_1						TODP0	VSS	VSS	VSS	VSS	EMUVCC	VDD						AN311	AN310	AN300	AN273	N
P	P21_3	P21_2	P22_2	P22_4						VDD	MSYN	VSS	VSS	VSS	AURORES	VDD						A3VCC	A3VREFH	AN302	AN301	P
R	P21_5	P21_4	P22_5	P22_6						VDD	VSS	EVTO0	VSS	VSS	VSS	VDD						A3VSS	AN313	AN312	AN303	R
T	P25_3	P25_4	P22_7	P22_8						VSS	AWOVCL	VDD	VDD	VDD	VSS	VSS						P00_5	P00_11	P02_10	P02_11	T
U	P25_5	P25_6	P22_9	P22_10																		P00_4	P00_10	P02_8	P02_9	U
V	P22_11	P22_12	P22_13	RAMSVCL																		P00_3	P00_9	P02_6	P02_7	V
W	TRST	FLMD0	RESETOUT	SBMD																		P00_2	P00_8	P02_4	P02_5	W
Y	ERROROUT_M	JP0_2	RESET	PWRCTL																		P00_1	P00_7	P02_2	P02_3	Y
AA	JP0_3	JP0_0	VMONOUT																				P00_0	P02_0	P02_1	AA
AB	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P14_6	P14_10	P14_11	P12_0	P12_2	P12_6	P13_8	P13_12	P11_5	P11_8	P11_10	P10_10	P10_13			P00_6	P01_4	P01_5	AB
AC	SYSVCC	J1VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P14_8	P14_7	P14_9	P14_12	P12_1	P12_3	P12_7	P13_9	P13_13	P11_0	P11_4	P11_9	P10_11	P10_12	P10_14	E1VCC	VSS	P01_6	P01_7	AC
AD	J0VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P14_0	P14_2	P13_0	P13_2	P14_5	P12_4	P12_8	P13_10	P13_14	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	E0VCC	VSS	P01_3	AD
AE	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P14_1	P14_3	P13_1	P13_3	P14_4	P12_5	P12_9	P13_11	P11_3	P11_1	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_8	E0VCC	VSS(NC)	AE
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Figure 1.3 Pin Connection Diagram U2B24-FCC(FCBGA373)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_10	P33_9	P33_8	P33_7	P33_6	P33_5	P33_4	P33_3	P33_2	P33_1	P33_0	AN111	AN110	AN101	AN100	AN051	AN050	AN033	AN032	AN021	AN020	AN003	AN001	ADSVSS	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P31_12	P31_11	P31_10	P31_9	P31_8	P31_7	P31_6	P31_5	P31_4	P31_3	P31_2	P31_1	P31_0	AN112	AN113	AN102	AN053	AN052	AN041	AN043	AN031	AN022	AN010	AN011	AN002	AN000	ADSVSS	B
C	P31_14	E0VCC	VSS	P32_3	P32_2	P32_0	P34_4	P34_2	P34_0	P30_12	P30_11	P30_10	P30_9	P30_8	P30_7	AN121	AN120	AN103	A1VREFH	AN061	AN060	AN042	AN040	A0VREFH	AN013	AN012	ADSVCC	ADSVSS	ADSVCL	AN200	C
D	P31_15	P31_13	P32_5			P32_1	P34_3	P34_1	P30_6	P30_5	P30_4	P30_3	P30_2	P30_1	P30_0	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN202	AN201	D
E	P23_1	P23_0	P32_6																									ADSVREFL	AN203	AN221	E
F	P23_5	P23_4	P23_3	P23_2																							AN210	AN211	AN213	AN231	F
G	P23_7	P23_6	P24_1	P24_0																							AN220	AN233	AN230	AN223	G
H	VSS	VSS	VCC	P24_2																							AN212	AN222	AN240	AN232	H
J	P23_9	P23_8	VSS	VDD(NC)																							A2VCC	AN242	AN241	AN243	J
K	P23_11	P23_10	GETH0PVCC	GETH0BVCC																							A2VSS	A2VREFH	AN250	AN251	K
L	ETH0_SG_TXD_N	ETH0_SG_TXD_P	VSS	GETH0VCL																							AN252	AN260	AN253	AN261	L
M	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	ETH_SG_REFCLK								VSS	VSS	VDD	VDD	CICREFN	CICREFP	EMUVDD	VSS								AFCVCC	AN262	AN263	AN270	M
N	OSCVCC	VSS	P25_2	P25_10								VDD	VSS	VSS	VSS	VSS	VSS	VSS	EMUVDD								AFCVSS	AN271	AN272	AN273	N
P	X1	X2	P25_11	P25_7								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD								AN300	AN301	AN302	AN303	P
R	LVDVCC	VSS	P25_9	P25_8								TODN0	VSS	VSS	VSS	VSS	VSS	EMUVCC	VDD								A3VCC	A3VREFH	AN310	AN311	R
T	P21_3	P21_2	P22_3	P22_0								TODP0	VSS	VSS	VSS	VSS	VSS	AURORES	VDD								A3VSS	AN312	AN313	AN360	T
U	P21_5	P21_4	P22_1	P22_2								VDD	MSYN	VSS	VSS	VSS	VSS	VSS	VDD								AN361	AN362	AN363	AN370	U
V	P24_5	P24_4	P24_3	P22_4								VDD	VSS	EVTO0	VSS	VSS	VSS	VSS	VDD								AN371	AN372	AN373	AN380	V
W	P24_7	P24_6	P22_5	P22_6								VSS	AWOVCL	VDD	VDD	VDD	VDD	VSS	VSS								AN381	AN382	AN383	AN390	W
Y	P25_3	P25_4	P22_8	P22_7																							P00_5	AN391	AN392	AN393	Y
AA	P25_5	P25_6	P22_10	P22_9																							P00_4	P00_11	P02_10	P02_11	AA
AB	P25_13	P25_12	P22_12	P22_11																							P00_3	P00_10	P02_8	P02_9	AB
AC	P25_15	P25_14	P22_13	RAMSVCL																							P00_2	P00_9	P02_6	P02_7	AC
AD	TRST	FLMD0	RESETOUT	SBMD																							P00_1	P00_8	P02_4	P02_5	AD
AE	ERROROUT_M	JP0_2	RESET	PWRCTL																							P00_0	P00_7	P02_2	P02_3	AE
AF	JP0_3	JP0_0	VMONOUT																									P00_6	P02_0	P02_1	AF
AG	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P15_3	P15_7	P15_11	P14_6	P14_8	P14_10	P14_12	P12_1	P12_5	P13_8	P13_11	P11_0	P11_4	P11_9	P10_10	P10_12	P10_14	P01_9			P01_3	P01_4	P01_5	AG
AH	SYSVCC	J1VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P15_0	P15_4	P15_8	P15_12	P14_7	P14_9	P14_11	P12_0	P12_2	P12_6	P13_9	P13_12	P11_1	P11_5	P11_8	P11_10	P10_11	P10_13	P01_8	P01_11	E1VCC	VSS	P01_6	P01_7	AH
AJ	J0VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P15_1	P15_5	P15_9	P14_0	P14_2	P13_0	P13_2	P14_5	P12_3	P12_7	P13_10	P13_13	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	P01_12	P01_13	E0VCC	VSS	VSS	AJ
AK	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P15_2	P15_6	P15_10	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P12_9	P13_14	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_10	P01_14	P01_15	E0VCC	VSS(NC)	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	

Figure 1.4 Pin Connection Diagram U2B20-FCC(FCBGA468)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_9	P33_7	P33_5	P33_3	P33_1	P33_0	AN111	AN110	AN100	AN051	AN050	AN042	AN040	AN021	AN020	AN010	AN011	AN002	AN000	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P32_2	P32_1	P33_10	P33_8	P33_6	P33_4	P33_2	AN121	AN120	AN102	AN101	AN053	AN052	AN033	AN032	AN031	AN013	AN003	AN001	ADSVCL	AN203	AN200	B
C	P23_0	E0VCC	VSS	P32_3	P32_0	P34_2	P34_0	P30_4	P30_2	P30_0	AN112	AN113	AN103	A1VREFH	AN061	AN060	AN041	AN043	AN022	A0VREFH	AN012	ADSVCC	ADSVSS	AN202	AN201	C
D	P23_6	P23_1	P32_5			P34_4	P34_3	P34_1	P30_3	P30_1	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN221	AN213	D
E	P23_7	P23_5	P32_6																				ADSVREFL	AN231	AN230	E
F	VSS	VSS	P23_3	P23_2																		AN210	AN220	AN223	AN233	F
G	P23_9	P23_8	VCC	P23_4																		A2VCC	A2VREFH	AN232	AN222	G
H	P23_11	P23_10	VSS	VDD(NC)																		A2VSS	AN211	AN240	AN250	H
J	ETH0_SG_TXD_N	ETH0_SG_TXD_P	GETH0PVCC	GETH0BVCC																		AN212	AN243	AN241	AN251	J
K	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	GETH0VCL						VSS	VSS	VDD	CICREFN	CICREFP	EMUVDD	VSS						AN242	AN252	AN260	AN253	K
L	OSCVCC	VSS	VSS	ETH_SG_REFCLK						VDD	VSS	VSS	VSS	VSS	VSS	EMUVDD						AFCVCC	AN262	AN270	AN261	L
M	X1	X2	P25_2	P22_0						TODN0	VSS	VSS	VSS	VSS	VSS	VDD						AFCVSS	AN263	AN271	AN272	M
N	LVDVCC	VSS	P22_3	P22_1						TODP0	VSS	VSS	VSS	VSS	EMUVCC	VDD						AN311	AN310	AN300	AN273	N
P	P21_3	P21_2	P22_2	P22_4						VDD	MSYN	VSS	VSS	VSS	AURORES	VDD						A3VCC	A3VREFH	AN302	AN301	P
R	P21_5	P21_4	P22_5	P22_6						VDD	VSS	EVTO0	VSS	VSS	VSS	VDD						A3VSS	AN313	AN312	AN303	R
T	P25_3	P25_4	P22_7	P22_8						VSS	AWOVCL	VDD	VDD	VDD	VSS	VSS						P00_5	P00_11	P02_10	P02_11	T
U	P25_5	P25_6	P22_9	P22_10																		P00_4	P00_10	P02_8	P02_9	U
V	P22_11	P22_12	P22_13	RAMSVCL																		P00_3	P00_9	P02_6	P02_7	V
W	TRST	FLMD0	RESETOUT	SBMD																		P00_2	P00_8	P02_4	P02_5	W
Y	ERROROUT_M	JP0_2	RESET	PWRCTL																		P00_1	P00_7	P02_2	P02_3	Y
AA	JP0_3	JP0_0	VMONOUT																				P00_0	P02_0	P02_1	AA
AB	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P14_6	P14_10	P14_11	P12_0	P12_2	P12_6	P13_8	P13_12	P11_5	P11_8	P11_10	P10_10	P10_13			P00_6	P01_4	P01_5	AB
AC	SYSVCC	J1VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P14_8	P14_7	P14_9	P14_12	P12_1	P12_3	P12_7	P13_9	P13_13	P11_0	P11_4	P11_9	P10_11	P10_12	P10_14	E1VCC	VSS	P01_6	P01_7	AC
AD	J0VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P14_0	P14_2	P13_0	P13_2	P14_5	P12_4	P12_8	P13_10	P13_14	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	E0VCC	VSS	P01_3	AD
AE	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P14_1	P14_3	P13_1	P13_3	P14_4	P12_5	P12_9	P13_11	P11_3	P11_1	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_8	E0VCC	VSS(NC)	AE
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Figure 1.5 Pin Connection Diagram U2B20-FCC(FCBGA373)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_10	P33_9	P33_8	P33_7	P33_6	P33_5	P33_4	P33_3	P33_2	P33_1	P33_0	AN111	AN110	AN101	AN100	AN051	AN050	AN033	AN032	AN021	AN020	AN003	AN001	ADSVSS	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P31_12	P31_11	P31_10	P31_9	P31_8	P31_7	P31_6	P31_5	P31_4	P31_3	P31_2	P31_1	P31_0	AN112	AN113	AN102	AN053	AN052	AN041	AN043	AN031	AN022	AN010	AN011	AN002	AN000	ADSVSS	B
C	P31_14	E0VCC	VSS	P32_3	P32_2	P32_0	P34_4	P34_2	P34_0	P30_12	P30_11	P30_10	P30_9	P30_8	P30_7	AN121	AN120	AN103	A1VREFH	AN061	AN060	AN042	AN040	A0VREFH	AN013	AN012	ADSVCC	ADSVSS	ADSVCL	AN200	C
D	P31_15	P31_13	P32_5			P32_1	P34_3	P34_1	P30_6	P30_5	P30_4	P30_3	P30_2	P30_1	P30_0	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN202	AN201	D
E	P23_1	P23_0	P32_6																									ADSVREFL	AN203	AN221	E
F	P23_5	P23_4	P23_3	P23_2																							AN210	AN211	AN213	AN231	F
G	P23_7	P23_6	P24_1	P24_0																							AN220	AN233	AN230	AN223	G
H	VSS	VSS	VCC	P24_2																							AN212	AN222	AN240	AN232	H
J	P23_9	P23_8	VSS	VDD(NC)																							A2VCC	AN242	AN241	AN243	J
K	P23_11	P23_10	E0VCC(NC)	E0VCC(NC)																							A2VSS	A2VREFH	AN250	AN251	K
L	P24_9	P24_8	VSS	VDD(NC)																							AN252	AN260	AN253	AN261	L
M	P24_11	P24_10	VSS	P24_12								VSS	VSS	VDD	VDD	CICREFN	CICREFP	EMUVDD	VSS								AFCVCC	AN262	AN263	AN270	M
N	OSCVCC	VSS	P25_2	P25_10								VDD	VSS	VSS	VSS	VSS	VSS	VSS	EMUVDD								AFCVSS	AN271	AN272	AN273	N
P	X1	X2	P25_11	P25_7								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD								AN300	AN301	AN302	AN303	P
R	LVDVCC	VSS	P25_9	P25_8								TODN0	VSS	VSS	VSS	VSS	VSS	EMUVCC	VDD								A3VCC	A3VREFH	AN310	AN311	R
T	P21_3	P21_2	P22_3	P22_0								TODP0	VSS	VSS	VSS	VSS	VSS	AURORES	VDD								A3VSS	AN312	AN313	AN360	T
U	P21_5	P21_4	P22_1	P22_2								VDD	MSYN	VSS	VSS	VSS	VSS	VSS	VDD								AN361	AN362	AN363	AN370	U
V	P24_5	P24_4	P24_3	P22_4								VDD	VSS	EVTO0	VSS	VSS	VSS	VSS	VDD								AN371	AN372	AN373	AN380	V
W	P24_7	P24_6	P22_5	P22_6								VSS	VDD	VDD	VDD	VDD	VDD	VSS	VSS								AN381	AN382	AN383	AN390	W
Y	P25_3	P25_4	P22_8	P22_7																							P00_5	AN391	AN392	AN393	Y
AA	P25_5	P25_6	P22_10	P22_9																							P00_4	P00_11	P02_10	P02_11	AA
AB	P25_13	P25_12	P22_12	P22_11																							P00_3	P00_10	P02_8	P02_9	AB
AC	P25_15	P25_14	P22_13	RAMSVCL																							P00_2	P00_9	P02_6	P02_7	AC
AD	TRST	FLMD0	RESETOUT	P25_0																							P00_1	P00_8	P02_4	P02_5	AD
AE	ERROROUT_M	JP0_2	RESET	P25_1																							P00_0	P00_7	P02_2	P02_3	AE
AF	JP0_3	JP0_0	VMONOUT																									P00_6	P02_0	P02_1	AF
AG	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P15_3	P15_7	P15_11	P14_6	P14_8	P14_10	P14_12	P12_1	P12_5	P13_8	P13_11	P11_0	P11_4	P11_9	P10_10	P10_12	P10_14	P01_9			P01_3	P01_4	P01_5	AG
AH	SYSVCC	J1VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P15_0	P15_4	P15_8	P15_12	P14_7	P14_9	P14_11	P12_0	P12_2	P12_6	P13_9	P13_12	P11_1	P11_5	P11_8	P11_10	P10_11	P10_13	P01_8	P01_11	E1VCC	VSS	P01_6	P01_7	AH
AJ	J0VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P15_1	P15_5	P15_9	P14_0	P14_2	P13_0	P13_2	P14_5	P12_3	P12_7	P13_10	P13_13	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	P01_12	P01_13	E0VCC	VSS	VSS	AJ
AK	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P15_2	P15_6	P15_10	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P12_9	P13_14	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_10	P01_14	P01_15	E0VCC	VSS(NC)	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	

Figure 1.6 Pin Connection Diagram U2B10-FCC(FCBGA468)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_9	P33_7	P33_5	P33_3	P33_1	P33_0	AN111	AN110	AN100	AN051	AN050	AN042	AN040	AN021	AN020	AN010	AN011	AN002	AN000	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P32_2	P32_1	P33_10	P33_8	P33_6	P33_4	P33_2	AN121	AN120	AN102	AN101	AN053	AN052	AN033	AN032	AN031	AN013	AN003	AN001	ADSVCL	AN203	AN200	B
C	P23_0	E0VCC	VSS	P32_3	P32_0	P34_2	P34_0	P30_4	P30_2	P30_0	AN112	AN113	AN103	A1VREFH	AN061	AN060	AN041	AN043	AN022	A0VREFH	AN012	ADSVCC	ADSVSS	AN202	AN201	C
D	P23_6	P23_1	P32_5			P34_4	P34_3	P34_1	P30_3	P30_1	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN221	AN213	D
E	P23_7	P23_5	P32_6																				ADSVREFL	AN231	AN230	E
F	VSS	VSS	P23_3	P23_2																		AN210	AN220	AN223	AN233	F
G	P23_9	P23_8	VCC	P23_4																		A2VCC	A2VREFH	AN232	AN222	G
H	P23_11	P23_10	VSS	VDD(NC)																		A2VSS	AN211	AN240	AN250	H
J	P24_9	P24_8	E0VCC(NC)	E0VCC(NC)																		AN212	AN243	AN241	AN251	J
K	P24_11	P24_10	VSS	VDD(NC)						VSS	VSS	VDD	CICREFN	CICREFP	EMUVDD	VSS						AN242	AN252	AN260	AN253	K
L	OSCVCC	VSS	VSS	P24_12						VDD	VSS	VSS	VSS	VSS	VSS	EMUVDD						AFCVCC	AN262	AN270	AN261	L
M	X1	X2	P25_2	P22_0						TODN0	VSS	VSS	VSS	VSS	VSS	VDD						AFCVSS	AN263	AN271	AN272	M
N	LVDVCC	VSS	P22_3	P22_1						TODP0	VSS	VSS	VSS	VSS	EMUVCC	VDD						AN311	AN310	AN300	AN273	N
P	P21_3	P21_2	P22_2	P22_4						VDD	$\overline{\text{MSYN}}$	VSS	VSS	VSS	$\overline{\text{AURORES}}$	VDD						A3VCC	A3VREFH	AN302	AN301	P
R	P21_5	P21_4	P22_5	P22_6						VDD	VSS	$\overline{\text{EVTO0}}$	VSS	VSS	VSS	VDD						A3VSS	AN313	AN312	AN303	R
T	P25_3	P25_4	P22_7	P22_8						VSS	VDD	VDD	VDD	VDD	VSS	VSS						P00_5	P00_11	P02_10	P02_11	T
U	P25_5	P25_6	P22_9	P22_10																		P00_4	P00_10	P02_8	P02_9	U
V	P22_11	P22_12	P22_13	RAMSVCL																		P00_3	P00_9	P02_6	P02_7	V
W	$\overline{\text{TRST}}$	FLMD0	$\overline{\text{RESETOUT}}$	P25_0																		P00_2	P00_8	P02_4	P02_5	W
Y	$\overline{\text{ERROROUT\_M}}$	JP0_2	$\overline{\text{RESET}}$	P25_1																		P00_1	P00_7	P02_2	P02_3	Y
AA	JP0_3	JP0_0	VMONOUT																				P00_0	P02_0	P02_1	AA
AB	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P14_6	P14_10	P14_11	P12_0	P12_2	P12_6	P13_8	P13_12	P11_5	P11_8	P11_10	P10_10	P10_13			P00_6	P01_4	P01_5	AB
AC	SYSVCC	J1VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P14_8	P14_7	P14_9	P14_12	P12_1	P12_3	P12_7	P13_9	P13_13	P11_0	P11_4	P11_9	P10_11	P10_12	P10_14	E1VCC	VSS	P01_6	P01_7	AC
AD	J0VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P14_0	P14_2	P13_0	P13_2	P14_5	P12_4	P12_8	P13_10	P13_14	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	E0VCC	VSS	P01_3	AD
AE	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P14_1	P14_3	P13_1	P13_3	P14_4	P12_5	P12_9	P13_11	P11_3	P11_1	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_8	E0VCC	VSS(NC)	AE
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Figure 1.7 Pin Connection Diagram U2B10-FCC(FCBGA373)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
A	VSS(NC)	P32_2	P33_7	P33_5	P33_2	P33_1	P33_0	AN120	AN110	AN103	AN101	AN042	AN040	AN033	AN032	AN010	AN011	AN003	AN001	ADSVSS(NC)	A
B	P32_4	VSS	P32_3	P33_8	P33_6	P33_4	P33_3	AN112	AN121	AN111	AN102	AN100	AN030	AN023	AN021	AN020	AN002	AN000	ADSVSS	AN221	B
C	P23_1	P32_5																	ADSVCC	AN231	C
D	P23_5	P32_6		VSS(NC)	P32_1	P34_4	P34_2	P34_0	AN122	AN113	A1VREFH	AN043	A0VREFH	AN031	AN022	AN013	AN012		AN200	AN213	D
E	P23_6	P23_3		P23_0	VSS	P32_0	P34_3	P34_1	AN123	A1VSS	A1VCC	AN041	A0VCC	A0VSS	ADSVCL	ADSVREFL	AN202		AN201	AN230	E
F	P23_7	P23_4		P23_2	E0VCC											ADSVREFH	AN203		AN222	AN223	F
G	OSCVCC	P25_2		P22_1	P22_0			VDD	VDD	CICREFN	CICREFP	VSS	EMUVDD			AN210	AN211		AN233	AN232	G
H	X1	X2		P22_3	P22_2		VDD		VSS	VSS	VSS	VSS		EMUVDD		AN220	AN212		AN240	AN242	H
J	LVDVCC	VSS		P22_5	P22_4		VDD	VSS		VSS	VSS		VSS	VSS		A2VCC	A2VREFH		AN241	AN243	J
K	P21_3	P21_2		P22_7	P22_6		TODN0	VSS	VSS	VSS	VSS	VSS	VSS	EMUVCC		A2VSS	AN252		AFCVCC	AFCVSS	K
L	P21_5	P21_4		P22_9	P22_8		TODP0	VSS	VSS	VSS	VSS	VSS	VSS	AURORES		AN260	AN261		AN250	AN251	L
M	P25_3	P25_4		P22_11	P22_10		VDD	MSYN		VSS	VSS		VSS	VDD		AN262	AN263		AN270	AN253	M
N	P25_5	P25_6		P22_12	RAMSVCL		VDD		EVTO0	VSS	VSS	VSS		VDD		AN271	AN273		P02_9	P02_10	N
P	TRST	FLMD0		P22_13	RESETOUT			VDD	VDD	VSS	VSS	VDD	VDD			AN272	P00_5		P02_7	P02_8	P
R	ERROROUT_M	JP0_2		RESET	P20_0											E0VCC	P00_4		P02_5	P02_6	R
T	JP0_3	JP0_0		VMONOUT	P20_1	P20_3	E2VCC	P12_0	P12_2	P12_5	P11_0	P11_8	P11_9	E1VCC	P00_2	VSS	P00_3		P02_3	P02_4	T
U	JP0_1	JP0_5		SYSVCC	P20_4	P20_6	P20_7	P12_1	P12_6	P12_3	P11_1	P11_5	P11_4	P11_10	P00_1	P00_0	VSS(NC)		P02_1	P02_2	U
V	VCC	SVRDRVCC																	P00_6	P02_0	V
W	J0VCC	SVRDRVSS	SVRAVSS	SVRNGATE	P20_2	P14_0	P14_2	P13_0	P13_2	P14_5	P12_7	P12_9	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	VSS	P00_7	W
Y	VSS(NC)	SVRDRVSS	SVRAVCC	SVRPGATE	P20_5	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	VSS(NC)	Y
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	

Figure 1.8 Pin Connection Diagram U2B10-FCC(FCBGA292)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
A	VSS(NC)	P32_2	P33_7	P33_5	P33_2	P33_1	P33_0	AN120	AN110	AN103	AN101	AN042	AN040	AN033	AN032	AN010	AN011	AN003	AN001	ADSVSS(NC)	A
B	P32_4	VSS	P32_3	P33_8	P33_6	P33_4	P33_3	AN112	AN121	AN111	AN102	AN100	AN030	AN023	AN021	AN020	AN002	AN000	ADSVSS	AN221	B
C	P23_1	P32_5																	ADSVCC	AN231	C
D	P23_5	P32_6		VSS(NC)	P32_1	P34_4	P34_2	P34_0	AN122	AN113	A1VREFH	AN043	A0VREFH	AN031	AN022	AN013	AN012		AN200	AN213	D
E	P23_6	P23_3		P23_0	VSS	P32_0	P34_3	P34_1	AN123	A1VSS	A1VCC	AN041	A0VCC	A0VSS	ADSVCL	ADSVREFL	AN202		AN201	AN230	E
F	P23_7	P23_4		P23_2	E0VCC											ADSVREFH	AN203		AN222	AN223	F
G	OSCVCC	P25_2		P22_1	P22_0			VDD	VDD	CICREFN	CICREFP	VSS	EMUVDD			AN210	AN211		AN233	AN232	G
H	X1	X2		P22_3	P22_2		VDD		VSS	VSS	VSS	VSS		EMUVDD		AN220	AN212		AN240	AN242	H
J	LVDVCC	VSS		P22_5	P22_4		VDD	VSS		VSS	VSS		VSS	VSS		A2VCC	A2VREFH		AN241	AN243	J
K	P21_3	P21_2		P22_7	P22_6		TODN0	VSS	VSS	VSS	VSS	VSS	VSS	EMUVCC		A2VSS	AN252		AFCVCC	AFCVSS	K
L	P21_5	P21_4		P22_9	P22_8		TODP0	VSS	VSS	VSS	VSS	VSS	VSS	AURORES		AN260	AN261		AN250	AN251	L
M	P25_3	P25_4		P22_11	P22_10		VDD	MSYN		VSS	VSS		VSS	VDD		AN262	AN263		AN270	AN253	M
N	P25_5	P25_6		P22_12	RAMSVCL		VDD		EVTO0	VSS	VSS	VSS		VDD		AN271	AN273		P02_9	P02_10	N
P	TRST	FLMD0		P22_13	RESETOUT			VDD	VDD	VSS	VSS	VDD	VDD			AN272	P00_5		P02_7	P02_8	P
R	ERROROUT_M	JP0_2		RESET	P20_0											E0VCC	P00_4		P02_5	P02_6	R
T	JP0_3	JP0_0		VMONOUT	P20_1	P20_3	E2VCC	P12_0	P12_2	P12_5	P11_0	P11_8	P11_9	E1VCC	P00_2	VSS	P00_3		P02_3	P02_4	T
U	JP0_1	JP0_5		SYSVCC	P20_4	P20_6	P20_7	P12_1	P12_6	P12_3	P11_1	P11_5	P11_4	P11_10	P00_1	P00_0	VSS(NC)		P02_1	P02_2	U
V	VCC	SVRDRVCC																	P00_6	P02_0	V
W	J0VCC	SVRDRVSS	SVRAVSS	SVRNGATE	P20_2	P14_0	P14_2	P13_0	P13_2	P14_5	P12_7	P12_9	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	VSS	P00_7	W
Y	VSS(NC)	SVRDRVSS	SVRAVCC	SVRPGATE	P20_5	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	VSS(NC)	Y
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	

Figure 1.9 Pin Connection Diagram U2B6-FCC(FCBGA292)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_10	P33_9	P33_8	P33_7	P33_6	P33_5	P33_4	P33_3	P33_2	P33_1	P33_0	AN111	AN110	AN101	AN100	AN051	AN050	AN033	AN032	AN021	AN020	AN003	AN001	ADSVSS	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P31_12	P31_11	P31_10	P31_9	P31_8	P31_7	P31_6	P31_5	P31_4	P31_3	P31_2	P31_1	P31_0	AN112	AN113	AN102	AN053	AN052	AN041	AN043	AN031	AN022	AN010	AN011	AN002	AN000	ADSVSS	B
C	P31_14	E0VCC	VSS	P32_3	P32_2	P32_0	P34_4	P34_2	P34_0	P30_12	P30_11	P30_10	P30_9	P30_8	P30_7	AN121	AN120	AN103	A1VREFH	AN061	AN060	AN042	AN040	A0VREFH	AN013	AN012	ADSVCC	ADSVSS	ADSVCL	AN200	C
D	P31_15	P31_13	P32_5			P32_1	P34_3	P34_1	P30_6	P30_5	P30_4	P30_3	P30_2	P30_1	P30_0	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN202	AN201	D
E	P23_1	P23_0	P32_6																									ADSVREFL	AN203	AN221	E
F	P23_5	P23_4	P23_3	P23_2																							AN210	AN211	AN213	AN231	F
G	P23_7	P23_6	P24_1	P24_0																							AN220	AN233	AN230	AN223	G
H	VSS	VSS	VCC	P24_2																							AN212	AN222	AN240	AN232	H
J	ETH1_SG_TXD_N	ETH1_SG_TXD_P	VSS	GETH1VCL																							A2VCC	AN242	AN241	AN243	J
K	ETH1_SG_RXD_N	ETH1_SG_RXD_P	GETH0PVCC	GETH0BVCC																							A2VSS	A2VREFH	AN250	AN251	K
L	ETH0_SG_TXD_N	ETH0_SG_TXD_P	VSS	GETH0VCL																							AN252	AN260	AN253	AN261	L
M	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	ETH_SG_REFCLK								VSS	VSS	VDD	VDD	VDD(NC)	VDD(NC)	VDD(NC)	VSS								AFCVCC	AN262	AN263	AN270	M
N	OSCVCC	VSS	P25_2	P25_10								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD(NC)								AFCVSS	AN271	AN272	AN273	N
P	X1	X2	P25_11	P25_7								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD								AN300	AN301	AN302	AN303	P
R	LVDVCC	VSS	P25_9	P25_8								VDD(NC)	VSS	VSS	VSS	VSS	VSS	VSS(NC)	VDD								A3VCC	A3VREFH	AN310	AN311	R
T	P21_3	P21_2	P22_3	P22_0								VDD(NC)	VSS	VSS	VSS	VSS	VSS	VSS(NC)	VDD								A3VSS	AN312	AN313	AN360	T
U	P21_5	P21_4	P22_1	P22_2								VDD	VSS(NC)	VSS	VSS	VSS	VSS	VSS	VDD								AN361	AN362	AN363	AN370	U
V	P24_5	P24_4	P24_3	P22_4								VDD	VSS	VSS(NC)	VSS	VSS	VSS	VSS	VDD								AN371	AN372	AN373	AN380	V
W	P24_7	P24_6	P22_5	P22_6								VSS	AWOVCL	VDD	VDD	VDD	VDD	VSS	VSS								AN381	AN382	AN383	AN390	W
Y	P25_3	P25_4	P22_8	P22_7																							P00_5	AN391	AN392	AN393	Y
AA	P25_5	P25_6	P22_10	P22_9																							P00_4	P00_11	P02_10	P02_11	AA
AB	P25_13	P25_12	P22_12	P22_11																							P00_3	P00_10	P02_8	P02_9	AB
AC	P25_15	P25_14	P22_13	RAMSVCL																							P00_2	P00_9	P02_6	P02_7	AC
AD	TRST	FLMD0	RESETOUT	SBMD																							P00_1	P00_8	P02_4	P02_5	AD
AE	ERROROUT_M	JP0_2	RESET	PWRCTL																							P00_0	P00_7	P02_2	P02_3	AE
AF	JP0_3	JP0_0	VMONOUT																									P00_6	P02_0	P02_1	AF
AG	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P15_3	P15_7	P15_11	P14_6	P14_8	P14_10	P14_12	P12_1	P12_5	P13_8	P13_11	P11_0	P11_4	P11_9	P10_10	P10_12	P10_14	P01_9			P01_3	P01_4	P01_5	AG
AH	SYSVCC	VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P15_0	P15_4	P15_8	P15_12	P14_7	P14_9	P14_11	P12_0	P12_2	P12_6	P13_9	P13_12	P11_1	P11_5	P11_8	P11_10	P10_11	P10_13	P01_8	P01_11	E1VCC	VSS	P01_6	P01_7	AH
AJ	VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P15_1	P15_5	P15_9	P14_0	P14_2	P13_0	P13_2	P14_5	P12_3	P12_7	P13_10	P13_13	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	P01_12	P01_13	E0VCC	VSS	VSS	AJ
AK	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P15_2	P15_6	P15_10	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P12_9	P13_14	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_10	P01_14	P01_15	E0VCC	VSS(NC)	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	

Figure 1.10 Pin Connection Diagram U2B24(FCBGA468)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_9	P33_7	P33_5	P33_3	P33_1	P33_0	AN111	AN110	AN100	AN051	AN050	AN042	AN040	AN021	AN020	AN010	AN011	AN002	AN000	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P32_2	P32_1	P33_10	P33_8	P33_6	P33_4	P33_2	AN121	AN120	AN102	AN101	AN053	AN052	AN033	AN032	AN031	AN013	AN003	AN001	ADSVCL	AN203	AN200	B
C	P23_0	E0VCC	VSS	P32_3	P32_0	P34_2	P34_0	P30_4	P30_2	P30_0	AN112	AN113	AN103	A1VREFH	AN061	AN060	AN041	AN043	AN022	A0VREFH	AN012	ADSVCC	ADSVSS	AN202	AN201	C
D	P23_6	P23_1	P32_5			P34_4	P34_3	P34_1	P30_3	P30_1	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN221	AN213	D
E	P23_7	P23_5	P32_6																				ADSVREFL	AN231	AN230	E
F	VSS	VSS	P23_3	P23_2																		AN210	AN220	AN223	AN233	F
G	ETH1_SG_TXD_N	ETH1_SG_TXD_P	VCC	P23_4																		A2VCC	A2VREFH	AN232	AN222	G
H	ETH1_SG_RXD_N	ETH1_SG_RXD_P	VSS	GETH1VCL																		A2VSS	AN211	AN240	AN250	H
J	ETH0_SG_TXD_N	ETH0_SG_TXD_P	GETH0PVCC	GETH0BVCC																		AN212	AN243	AN241	AN251	J
K	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	GETH0VCL						VSS	VSS	VDD	VDD(NC)	VDD(NC)	VDD(NC)	VSS						AN242	AN252	AN260	AN253	K
L	OSCVCC	VSS	VSS	ETH_SG_REFCLK						VDD	VSS	VSS	VSS	VSS	VSS	VDD(NC)						AFCVCC	AN262	AN270	AN261	L
M	X1	X2	P25_2	P22_0						VDD(NC)	VSS	VSS	VSS	VSS	VSS	VDD						AFCVSS	AN263	AN271	AN272	M
N	LVDVCC	VSS	P22_3	P22_1						VDD(NC)	VSS	VSS	VSS	VSS	VSS(NC)	VDD						AN311	AN310	AN300	AN273	N
P	P21_3	P21_2	P22_2	P22_4						VDD	VSS(NC)	VSS	VSS	VSS	VSS(NC)	VDD						A3VCC	A3VREFH	AN302	AN301	P
R	P21_5	P21_4	P22_5	P22_6						VDD	VSS	VSS(NC)	VSS	VSS	VSS	VDD						A3VSS	AN313	AN312	AN303	R
T	P25_3	P25_4	P22_7	P22_8						VSS	AWOVCL	VDD	VDD	VDD	VSS	VSS						P00_5	P00_11	P02_10	P02_11	T
U	P25_5	P25_6	P22_9	P22_10																		P00_4	P00_10	P02_8	P02_9	U
V	P22_11	P22_12	P22_13	RAMSVCL																		P00_3	P00_9	P02_6	P02_7	V
W	TRST	FLMD0	RESETOUT	SBMD																		P00_2	P00_8	P02_4	P02_5	W
Y	ERROROUT_M	JP0_2	RESET	PWRCTL																		P00_1	P00_7	P02_2	P02_3	Y
AA	JP0_3	JP0_0	VMONOUT																				P00_0	P02_0	P02_1	AA
AB	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P14_6	P14_10	P14_11	P12_0	P12_2	P12_6	P13_8	P13_12	P11_5	P11_8	P11_10	P10_10	P10_13			P00_6	P01_4	P01_5	AB
AC	SYSVCC	VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P14_8	P14_7	P14_9	P14_12	P12_1	P12_3	P12_7	P13_9	P13_13	P11_0	P11_4	P11_9	P10_11	P10_12	P10_14	E1VCC	VSS	P01_6	P01_7	AC
AD	VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P14_0	P14_2	P13_0	P13_2	P14_5	P12_4	P12_8	P13_10	P13_14	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	E0VCC	VSS	P01_3	AD
AE	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P14_1	P14_3	P13_1	P13_3	P14_4	P12_5	P12_9	P13_11	P11_3	P11_1	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_8	E0VCC	VSS(NC)	AE
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Figure 1.11 Pin Connection Diagram U2B24(FCBGA373)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_10	P33_9	P33_8	P33_7	P33_6	P33_5	P33_4	P33_3	P33_2	P33_1	P33_0	AN111	AN110	AN101	AN100	AN051	AN050	AN033	AN032	AN021	AN020	AN003	AN001	ADSVSS	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P31_12	P31_11	P31_10	P31_9	P31_8	P31_7	P31_6	P31_5	P31_4	P31_3	P31_2	P31_1	P31_0	AN112	AN113	AN102	AN053	AN052	AN041	AN043	AN031	AN022	AN010	AN011	AN002	AN000	ADSVSS	B
C	P31_14	E0VCC	VSS	P32_3	P32_2	P32_0	P34_4	P34_2	P34_0	P30_12	P30_11	P30_10	P30_9	P30_8	P30_7	AN121	AN120	AN103	A1VREFH	AN061	AN060	AN042	AN040	A0VREFH	AN013	AN012	ADSVCC	ADSVSS	ADSVCL	AN200	C
D	P31_15	P31_13	P32_5			P32_1	P34_3	P34_1	P30_6	P30_5	P30_4	P30_3	P30_2	P30_1	P30_0	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN202	AN201	D
E	P23_1	P23_0	P32_6																									ADSVREFL	AN203	AN221	E
F	P23_5	P23_4	P23_3	P23_2																							AN210	AN211	AN213	AN231	F
G	P23_7	P23_6	P24_1	P24_0																							AN220	AN233	AN230	AN223	G
H	VSS	VSS	VCC	P24_2																							AN212	AN222	AN240	AN232	H
J	P23_9	P23_8	VSS	VDD(NC)																							A2VCC	AN242	AN241	AN243	J
K	P23_11	P23_10	GETH0PVCC	GETH0BVCC																							A2VSS	A2VREFH	AN250	AN251	K
L	ETH0_SG_TXD_N	ETH0_SG_TXD_P	VSS	GETH0VCL																							AN252	AN260	AN253	AN261	L
M	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	ETH_SG_REFCLK								VSS	VSS	VDD	VDD	VDD(NC)	VDD(NC)	VDD(NC)	VSS								AFCVCC	AN262	AN263	AN270	M
N	OSCVCC	VSS	P25_2	P25_10								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD(NC)								AFCVSS	AN271	AN272	AN273	N
P	X1	X2	P25_11	P25_7								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD								AN300	AN301	AN302	AN303	P
R	LVDVCC	VSS	P25_9	P25_8								VDD(NC)	VSS	VSS	VSS	VSS	VSS	VSS(NC)	VDD								A3VCC	A3VREFH	AN310	AN311	R
T	P21_3	P21_2	P22_3	P22_0								VDD(NC)	VSS	VSS	VSS	VSS	VSS	VSS(NC)	VDD								A3VSS	AN312	AN313	AN360	T
U	P21_5	P21_4	P22_1	P22_2								VDD	VSS(NC)	VSS	VSS	VSS	VSS	VSS	VDD								AN361	AN362	AN363	AN370	U
V	P24_5	P24_4	P24_3	P22_4								VDD	VSS	VSS(NC)	VSS	VSS	VSS	VSS	VDD								AN371	AN372	AN373	AN380	V
W	P24_7	P24_6	P22_5	P22_6								VSS	AWOVCL	VDD	VDD	VDD	VDD	VSS	VSS								AN381	AN382	AN383	AN390	W
Y	P25_3	P25_4	P22_8	P22_7																							P00_5	AN391	AN392	AN393	Y
AA	P25_5	P25_6	P22_10	P22_9																							P00_4	P00_11	P02_10	P02_11	AA
AB	P25_13	P25_12	P22_12	P22_11																							P00_3	P00_10	P02_8	P02_9	AB
AC	P25_15	P25_14	P22_13	RAMSVCL																							P00_2	P00_9	P02_6	P02_7	AC
AD	TRST	FLMD0	RESETOUT	SBMD																							P00_1	P00_8	P02_4	P02_5	AD
AE	ERROROUT_M	JP0_2	RESET	PWRCTL																							P00_0	P00_7	P02_2	P02_3	AE
AF	JP0_3	JP0_0	VMONOUT																									P00_6	P02_0	P02_1	AF
AG	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P15_3	P15_7	P15_11	P14_6	P14_8	P14_10	P14_12	P12_1	P12_5	P13_8	P13_11	P11_0	P11_4	P11_9	P10_10	P10_12	P10_14	P01_9			P01_3	P01_4	P01_5	AG
AH	SYSVCC	VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P15_0	P15_4	P15_8	P15_12	P14_7	P14_9	P14_11	P12_0	P12_2	P12_6	P13_9	P13_12	P11_1	P11_5	P11_8	P11_10	P10_11	P10_13	P01_8	P01_11	E1VCC	VSS	P01_6	P01_7	AH
AJ	VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P15_1	P15_5	P15_9	P14_0	P14_2	P13_0	P13_2	P14_5	P12_3	P12_7	P13_10	P13_13	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	P01_12	P01_13	E0VCC	VSS	VSS	AJ
AK	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P15_2	P15_6	P15_10	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P12_9	P13_14	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_10	P01_14	P01_15	E0VCC	VSS(NC)	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	

Figure 1.12 Pin Connection Diagram U2B20(BGA468)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_9	P33_7	P33_5	P33_3	P33_1	P33_0	AN111	AN110	AN100	AN051	AN050	AN042	AN040	AN021	AN020	AN010	AN011	AN002	AN000	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P32_2	P32_1	P33_10	P33_8	P33_6	P33_4	P33_2	AN121	AN120	AN102	AN101	AN053	AN052	AN033	AN032	AN031	AN013	AN003	AN001	ADSVCL	AN203	AN200	B
C	P23_0	E0VCC	VSS	P32_3	P32_0	P34_2	P34_0	P30_4	P30_2	P30_0	AN112	AN113	AN103	A1VREFH	AN061	AN060	AN041	AN043	AN022	A0VREFH	AN012	ADSVCC	ADSVSS	AN202	AN201	C
D	P23_6	P23_1	P32_5			P34_4	P34_3	P34_1	P30_3	P30_1	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN221	AN213	D
E	P23_7	P23_5	P32_6																				ADSVREFL	AN231	AN230	E
F	VSS	VSS	P23_3	P23_2																		AN210	AN220	AN223	AN233	F
G	P23_9	P23_8	VCC	P23_4																		A2VCC	A2VREFH	AN232	AN222	G
H	P23_11	P23_10	VSS	VDD(NC)																		A2VSS	AN211	AN240	AN250	H
J	ETH0_SG_TXD_N	ETH0_SG_TXD_P	GETH0PVCC	GETH0BVCC																		AN212	AN243	AN241	AN251	J
K	ETH0_SG_RXD_N	ETH0_SG_RXD_P	VSS	GETH0VCL						VSS	VSS	VDD	VDD(NC)	VDD(NC)	VDD(NC)	VSS						AN242	AN252	AN260	AN253	K
L	OSCVCC	VSS	VSS	ETH_SG_REFCLK						VDD	VSS	VSS	VSS	VSS	VSS	VDD(NC)						AFCVCC	AN262	AN270	AN261	L
M	X1	X2	P25_2	P22_0						VDD(NC)	VSS	VSS	VSS	VSS	VSS	VDD						AFCVSS	AN263	AN271	AN272	M
N	LVDVCC	VSS	P22_3	P22_1						VDD(NC)	VSS	VSS	VSS	VSS	VSS(NC)	VDD						AN311	AN310	AN300	AN273	N
P	P21_3	P21_2	P22_2	P22_4						VDD	VSS(NC)	VSS	VSS	VSS	VSS(NC)	VDD						A3VCC	A3VREFH	AN302	AN301	P
R	P21_5	P21_4	P22_5	P22_6						VDD	VSS	VSS(NC)	VSS	VSS	VSS	VDD						A3VSS	AN313	AN312	AN303	R
T	P25_3	P25_4	P22_7	P22_8						VSS	AWOVCL	VDD	VDD	VDD	VSS	VSS						P00_5	P00_11	P02_10	P02_11	T
U	P25_5	P25_6	P22_9	P22_10																		P00_4	P00_10	P02_8	P02_9	U
V	P22_11	P22_12	P22_13	RAMSVCL																		P00_3	P00_9	P02_6	P02_7	V
W	$\overline{\text{TRST}}$	FLMD0	$\overline{\text{RESETOUT}}$	SBMD																		P00_2	P00_8	P02_4	P02_5	W
Y	$\overline{\text{ERROROUT\_M}}$	JP0_2	$\overline{\text{RESET}}$	PWRCTL																		P00_1	P00_7	P02_2	P02_3	Y
AA	JP0_3	JP0_0	$\overline{\text{VMONOUT}}$																				P00_0	P02_0	P02_1	AA
AB	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P14_6	P14_10	P14_11	P12_0	P12_2	P12_6	P13_8	P13_12	P11_5	P11_8	P11_10	P10_10	P10_13			P00_6	P01_4	P01_5	AB
AC	SYSVCC	VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P14_8	P14_7	P14_9	P14_12	P12_1	P12_3	P12_7	P13_9	P13_13	P11_0	P11_4	P11_9	P10_11	P10_12	P10_14	E1VCC	VSS	P01_6	P01_7	AC
AD	VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P14_0	P14_2	P13_0	P13_2	P14_5	P12_4	P12_8	P13_10	P13_14	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	E0VCC	VSS	P01_3	AD
AE	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P14_1	P14_3	P13_1	P13_3	P14_4	P12_5	P12_9	P13_11	P11_3	P11_1	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_8	E0VCC	VSS(NC)	AE
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Figure 1.13 Pin Connection Diagram U2B20(BGA373)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_10	P33_9	P33_8	P33_7	P33_6	P33_5	P33_4	P33_3	P33_2	P33_1	P33_0	AN111	AN110	AN101	AN100	AN051	AN050	AN033	AN032	AN021	AN020	AN003	AN001	ADSVSS	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P31_12	P31_11	P31_10	P31_9	P31_8	P31_7	P31_6	P31_5	P31_4	P31_3	P31_2	P31_1	P31_0	AN112	AN113	AN102	AN053	AN052	AN041	AN043	AN031	AN022	AN010	AN011	AN002	AN000	ADSVSS	B
C	P31_14	E0VCC	VSS	P32_3	P32_2	P32_0	P34_4	P34_2	P34_0	P30_12	P30_11	P30_10	P30_9	P30_8	P30_7	AN121	AN120	AN103	A1VREFH	AN061	AN060	AN042	AN040	A0VREFH	AN013	AN012	ADSVCC	ADSVSS	ADSVCL	AN200	C
D	P31_15	P31_13	P32_5			P32_1	P34_3	P34_1	P30_6	P30_5	P30_4	P30_3	P30_2	P30_1	P30_0	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN202	AN201	D
E	P23_1	P23_0	P32_6																									ADSVREFL	AN203	AN221	E
F	P23_5	P23_4	P23_3	P23_2																							AN210	AN211	AN213	AN231	F
G	P23_7	P23_6	P24_1	P24_0																							AN220	AN233	AN230	AN223	G
H	VSS	VSS	VCC	P24_2																							AN212	AN222	AN240	AN232	H
J	P23_9	P23_8	VSS	VDD(NC)																							A2VCC	AN242	AN241	AN243	J
K	P23_11	P23_10	E0VCC(NC)	E0VCC(NC)																							A2VSS	A2VREFH	AN250	AN251	K
L	P24_9	P24_8	VSS	VDD(NC)																							AN252	AN260	AN253	AN261	L
M	P24_11	P24_10	VSS	P24_12								VSS	VSS	VDD	VDD	VDD(NC)	VDD(NC)	VDD(NC)	VSS								AFCVCC	AN262	AN263	AN270	M
N	OSCVCC	VSS	P25_2	P25_10								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD(NC)								AFCVSS	AN271	AN272	AN273	N
P	X1	X2	P25_11	P25_7								VDD	VSS	VSS	VSS	VSS	VSS	VSS	VDD								AN300	AN301	AN302	AN303	P
R	LVDVCC	VSS	P25_9	P25_8								VDD(NC)	VSS	VSS	VSS	VSS	VSS	VSS(NC)	VDD								A3VCC	A3VREFH	AN310	AN311	R
T	P21_3	P21_2	P22_3	P22_0								VDD(NC)	VSS	VSS	VSS	VSS	VSS	VSS(NC)	VDD								A3VSS	AN312	AN313	AN360	T
U	P21_5	P21_4	P22_1	P22_2								VDD	VSS(NC)	VSS	VSS	VSS	VSS	VSS	VDD								AN361	AN362	AN363	AN370	U
V	P24_5	P24_4	P24_3	P22_4								VDD	VSS	VSS(NC)	VSS	VSS	VSS	VSS	VDD								AN371	AN372	AN373	AN380	V
W	P24_7	P24_6	P22_5	P22_6								VSS	VDD	VDD	VDD	VDD	VDD	VSS	VSS								AN381	AN382	AN383	AN390	W
Y	P25_3	P25_4	P22_8	P22_7																							P00_5	AN391	AN392	AN393	Y
AA	P25_5	P25_6	P22_10	P22_9																							P00_4	P00_11	P02_10	P02_11	AA
AB	P25_13	P25_12	P22_12	P22_11																							P00_3	P00_10	P02_8	P02_9	AB
AC	P25_15	P25_14	P22_13	RAMSVCL																							P00_2	P00_9	P02_6	P02_7	AC
AD	$\overline{\text{TRST}}$	FLMD0	$\overline{\text{RESETOUT}}$	P25_0																							P00_1	P00_8	P02_4	P02_5	AD
AE	$\overline{\text{ERROROUT\_M}}$	JP0_2	$\overline{\text{RESET}}$	P25_1																							P00_0	P00_7	P02_2	P02_3	AE
AF	JP0_3	JP0_0	$\overline{\text{VMONOUT}}$																									P00_6	P02_0	P02_1	AF
AG	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P15_3	P15_7	P15_11	P14_6	P14_8	P14_10	P14_12	P12_1	P12_5	P13_8	P13_11	P11_0	P11_4	P11_9	P10_10	P10_12	P10_14	P01_9			P01_3	P01_4	P01_5	AG
AH	SYSVCC	VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P15_0	P15_4	P15_8	P15_12	P14_7	P14_9	P14_11	P12_0	P12_2	P12_6	P13_9	P13_12	P11_1	P11_5	P11_8	P11_10	P10_11	P10_13	P01_8	P01_11	E1VCC	VSS	P01_6	P01_7	AH
AJ	VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P15_1	P15_5	P15_9	P14_0	P14_2	P13_0	P13_2	P14_5	P12_3	P12_7	P13_10	P13_13	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	P01_12	P01_13	E0VCC	VSS	VSS	AJ
AK	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P15_2	P15_6	P15_10	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P12_9	P13_14	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_10	P01_14	P01_15	E0VCC	VSS(NC)	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	

Figure 1.14 Pin Connection Diagram U2B10(BGA468)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	
A	VSS(NC)	VSS	P33_13	P33_12	P33_11	P33_9	P33_7	P33_5	P33_3	P33_1	P33_0	AN111	AN110	AN100	AN051	AN050	AN042	AN040	AN021	AN020	AN010	AN011	AN002	AN000	ADSVSS(NC)	A
B	E0VCC	VSS	P32_4	P32_2	P32_1	P33_10	P33_8	P33_6	P33_4	P33_2	AN121	AN120	AN102	AN101	AN053	AN052	AN033	AN032	AN031	AN013	AN003	AN001	ADSVCL	AN203	AN200	B
C	P23_0	E0VCC	VSS	P32_3	P32_0	P34_2	P34_0	P30_4	P30_2	P30_0	AN112	AN113	AN103	A1VREFH	AN061	AN060	AN041	AN043	AN022	A0VREFH	AN012	ADSVCC	ADSVSS	AN202	AN201	C
D	P23_6	P23_1	P32_5			P34_4	P34_3	P34_1	P30_3	P30_1	AN123	AN122	A1VSS	A1VCC	AN063	AN062	AN030	AN023	A0VCC	A0VSS			ADSVREFH	AN221	AN213	D
E	P23_7	P23_5	P32_6																				ADSVREFL	AN231	AN230	E
F	VSS	VSS	P23_3	P23_2																		AN210	AN220	AN223	AN233	F
G	P23_9	P23_8	VCC	P23_4																		A2VCC	A2VREFH	AN232	AN222	G
H	P23_11	P23_10	VSS	VDD(NC)																		A2VSS	AN211	AN240	AN250	H
J	P24_9	P24_8	E0VCC(NC)	E0VCC(NC)																		AN212	AN243	AN241	AN251	J
K	P24_11	P24_10	VSS	VDD(NC)						VSS	VSS	VDD	VDD(NC)	VDD(NC)	VDD(NC)	VSS						AN242	AN252	AN260	AN253	K
L	OSCVCC	VSS	VSS	P24_12						VDD	VSS	VSS	VSS	VSS	VSS	VDD(NC)						AFCVCC	AN262	AN270	AN261	L
M	X1	X2	P25_2	P22_0						VDD(NC)	VSS	VSS	VSS	VSS	VSS	VDD						AFCVSS	AN263	AN271	AN272	M
N	LVDVCC	VSS	P22_3	P22_1						VDD(NC)	VSS	VSS	VSS	VSS	VSS(NC)	VDD						AN311	AN310	AN300	AN273	N
P	P21_3	P21_2	P22_2	P22_4						VDD	VSS(NC)	VSS	VSS	VSS	VSS(NC)	VDD						A3VCC	A3VREFH	AN302	AN301	P
R	P21_5	P21_4	P22_5	P22_6						VDD	VSS	VSS(NC)	VSS	VSS	VSS	VDD						A3VSS	AN313	AN312	AN303	R
T	P25_3	P25_4	P22_7	P22_8						VSS	VDD	VDD	VDD	VDD	VSS	VSS						P00_5	P00_11	P02_10	P02_11	T
U	P25_5	P25_6	P22_9	P22_10																		P00_4	P00_10	P02_8	P02_9	U
V	P22_11	P22_12	P22_13	RAMSVCL																		P00_3	P00_9	P02_6	P02_7	V
W	TRST	FLMD0	RESETOUT	P25_0																		P00_2	P00_8	P02_4	P02_5	W
Y	ERROROUT_M	JP0_2	RESET	P25_1																		P00_1	P00_7	P02_2	P02_3	Y
AA	JP0_3	JP0_0	VMONOUT																				P00_0	P02_0	P02_1	AA
AB	JP0_1	JP0_5	E2VCC			P20_0	P20_7	P14_6	P14_10	P14_11	P12_0	P12_2	P12_6	P13_8	P13_12	P11_5	P11_8	P11_10	P10_10	P10_13			P00_6	P01_4	P01_5	AB
AC	SYSVCC	VCC	SVRNGATE	SVRPGATE	P20_1	P20_4	P14_8	P14_7	P14_9	P14_12	P12_1	P12_3	P12_7	P13_9	P13_13	P11_0	P11_4	P11_9	P10_11	P10_12	P10_14	E1VCC	VSS	P01_6	P01_7	AC
AD	VCC	VSS	SVRDRVSS	SVRAVSS	P20_2	P20_5	P14_0	P14_2	P13_0	P13_2	P14_5	P12_4	P12_8	P13_10	P13_14	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	P10_9	E0VCC	VSS	P01_3	AD
AE	VSS(NC)	SVRDRVSS	SVRDRVCC	SVRAVCC	P20_3	P20_6	P14_1	P14_3	P13_1	P13_3	P14_4	P12_5	P12_9	P13_11	P11_3	P11_1	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	P01_8	E0VCC	VSS(NC)	AE
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	

Figure 1.15 Pin Connection Diagram U2B10(BGA373)

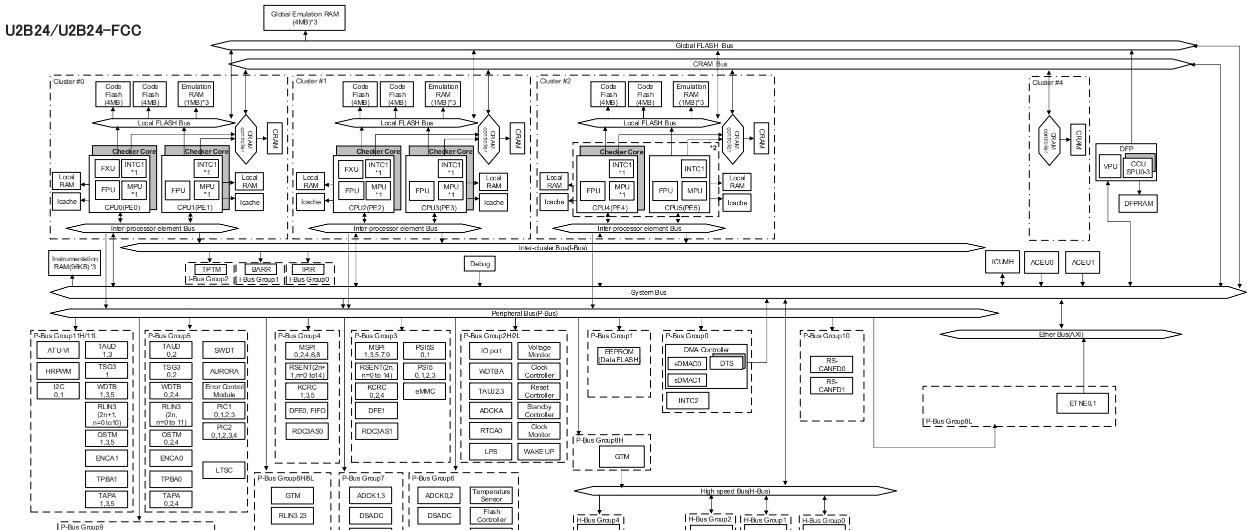
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
A	VSS(NC)	P32_2	P33_7	P33_5	P33_2	P33_1	P33_0	AN120	AN110	AN103	AN101	AN042	AN040	AN033	AN032	AN010	AN011	AN003	AN001	ADSVSS(NC)	A
B	P32_4	VSS	P32_3	P33_8	P33_6	P33_4	P33_3	AN112	AN121	AN111	AN102	AN100	AN030	AN023	AN021	AN020	AN002	AN000	ADSVSS	AN221	B
C	P23_1	P32_5																	ADSVCC	AN231	C
D	P23_5	P32_6		VSS(NC)	P32_1	P34_4	P34_2	P34_0	AN122	AN113	A1VREFH	AN043	A0VREFH	AN031	AN022	AN013	AN012		AN200	AN213	D
E	P23_6	P23_3		P23_0	VSS	P32_0	P34_3	P34_1	AN123	A1VSS	A1VCC	AN041	A0VCC	A0VSS	ADSVCL	ADSVREFL	AN202		AN201	AN230	E
F	P23_7	P23_4		P23_2	E0VCC											ADSVREFH	AN203		AN222	AN223	F
G	OSCVCC	P25_2		P22_1	P22_0			VDD	VDD	VSS(NC)	VSS(NC)	VSS	VDD(NC)			AN210	AN211		AN233	AN232	G
H	X1	X2		P22_3	P22_2		VDD		VSS	VSS	VSS	VSS		VDD(NC)		AN220	AN212		AN240	AN242	H
J	LVDVCC	VSS		P22_5	P22_4		VDD	VSS		VSS	VSS		VSS	VSS		A2VCC	A2VREFH		AN241	AN243	J
K	P21_3	P21_2		P22_7	P22_6		VSS(NC)	VSS	VSS	VSS	VSS	VSS	VSS	VSS(NC)		A2VSS	AN252		AFCVCC	AFCVSS	K
L	P21_5	P21_4		P22_9	P22_8		VSS(NC)	VSS	VSS	VSS	VSS	VSS	VSS	VSS(NC)		AN260	AN261		AN250	AN251	L
M	P25_3	P25_4		P22_11	P22_10		VDD	VSS(NC)		VSS	VSS		VSS	VDD		AN262	AN263		AN270	AN253	M
N	P25_5	P25_6		P22_12	RAMSVCL		VDD		VSS(NC)	VSS	VSS	VSS		VDD		AN271	AN273		P02_9	P02_10	N
P	TRST	FLMD0		P22_13	RESETOUT			VDD	VDD	VSS	VSS	VDD	VDD			AN272	P00_5		P02_7	P02_8	P
R	ERROROUT_M	JP0_2		RESET	P20_0											E0VCC	P00_4		P02_5	P02_6	R
T	JP0_3	JP0_0		VMONOUT	P20_1	P20_3	E2VCC	P12_0	P12_2	P12_5	P11_0	P11_8	P11_9	E1VCC	P00_2	VSS	P00_3		P02_3	P02_4	T
U	JP0_1	JP0_5		SYSVCC	P20_4	P20_6	P20_7	P12_1	P12_6	P12_3	P11_1	P11_5	P11_4	P11_10	P00_1	P00_0	VSS(NC)		P02_1	P02_2	U
V	VCC	SVRDRVCC																	P00_6	P02_0	V
W	VCC	SVRDRVSS	SVRAVSS	SVRNGATE	P20_2	P14_0	P14_2	P13_0	P13_2	P14_5	P12_7	P12_9	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	VSS	P00_7	W
Y	VSS(NC)	SVRDRVSS	SVRAVCC	SVRPGATE	P20_5	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	VSS(NC)	Y
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	

Figure 1.16 Pin Connection Diagram U2B10(BGA292)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
A	VSS(NC)	P32_2	P33_7	P33_5	P33_2	P33_1	P33_0	AN120	AN110	AN103	AN101	AN042	AN040	AN033	AN032	AN010	AN011	AN003	AN001	ADSVSS(NC)	A
B	P32_4	VSS	P32_3	P33_8	P33_6	P33_4	P33_3	AN112	AN121	AN111	AN102	AN100	AN030	AN023	AN021	AN020	AN002	AN000	ADSVSS	AN221	B
C	P23_1	P32_5																	ADSVCC	AN231	C
D	P23_5	P32_6		VSS(NC)	P32_1	P34_4	P34_2	P34_0	AN122	AN113	A1VREFH	AN043	A0VREFH	AN031	AN022	AN013	AN012		AN200	AN213	D
E	P23_6	P23_3		P23_0	VSS	P32_0	P34_3	P34_1	AN123	A1VSS	A1VCC	AN041	A0VCC	A0VSS	ADSVCL	ADSVREFL	AN202		AN201	AN230	E
F	P23_7	P23_4		P23_2	E0VCC											ADSVREFH	AN203		AN222	AN223	F
G	OSCVCC	P25_2		P22_1	P22_0			VDD	VDD	VSS(NC)	VSS(NC)	VSS	VDD(NC)			AN210	AN211		AN233	AN232	G
H	X1	X2		P22_3	P22_2		VDD		VSS	VSS	VSS	VSS		VDD(NC)		AN220	AN212		AN240	AN242	H
J	LVDVCC	VSS		P22_5	P22_4		VDD	VSS		VSS	VSS		VSS	VSS		A2VCC	A2VREFH		AN241	AN243	J
K	P21_3	P21_2		P22_7	P22_6		VSS(NC)	VSS	VSS	VSS	VSS	VSS	VSS	VSS(NC)		A2VSS	AN252		AFCVCC	AFCVSS	K
L	P21_5	P21_4		P22_9	P22_8		VSS(NC)	VSS	VSS	VSS	VSS	VSS	VSS	VSS(NC)		AN260	AN261		AN250	AN251	L
M	P25_3	P25_4		P22_11	P22_10		VDD	VSS(NC)		VSS	VSS		VSS	VDD		AN262	AN263		AN270	AN253	M
N	P25_5	P25_6		P22_12	RAMSVCL		VDD		VSS(NC)	VSS	VSS	VSS		VDD		AN271	AN273		P02_9	P02_10	N
P	TRST	FLMD0		P22_13	RESETOUT			VDD	VDD	VSS	VSS	VDD	VDD			AN272	P00_5		P02_7	P02_8	P
R	ERROROUT_M	JP0_2		RESET	P20_0											E0VCC	P00_4		P02_5	P02_6	R
T	JP0_3	JP0_0		VMONOUT	P20_1	P20_3	E2VCC	P12_0	P12_2	P12_5	P11_0	P11_8	P11_9	E1VCC	P00_2	VSS	P00_3		P02_3	P02_4	T
U	JP0_1	JP0_5		SYSVCC	P20_4	P20_6	P20_7	P12_1	P12_6	P12_3	P11_1	P11_5	P11_4	P11_10	P00_1	P00_0	VSS(NC)		P02_1	P02_2	U
V	VCC	SVRDRVCC																	P00_6	P02_0	V
W	VCC	SVRDRVSS	SVRAVSS	SVRNGATE	P20_2	P14_0	P14_2	P13_0	P13_2	P14_5	P12_7	P12_9	P11_2	P11_6	P10_1	P10_3	P10_5	P10_7	VSS	P00_7	W
Y	VSS(NC)	SVRDRVSS	SVRAVCC	SVRPGATE	P20_5	P14_1	P14_3	P13_1	P13_3	P14_4	P12_4	P12_8	P11_3	P11_7	P10_0	P10_2	P10_4	P10_6	P10_8	VSS(NC)	Y
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	

Figure 1.17 Pin Connection Diagram U2B6(BGA292)

1.7 Functional Block Configuration



Note1. This unit is only in Master Core.
Note2. CPU structure (DCLSx1 or Single Core x2) in cluster depends on device configuration(safety or performance).
Note3. This unit is only in FCC device.
Note4. This unit is not in FCC device.

Figure 1.18 Internal Block Diagram (U2B24 and U2B24-FCC)

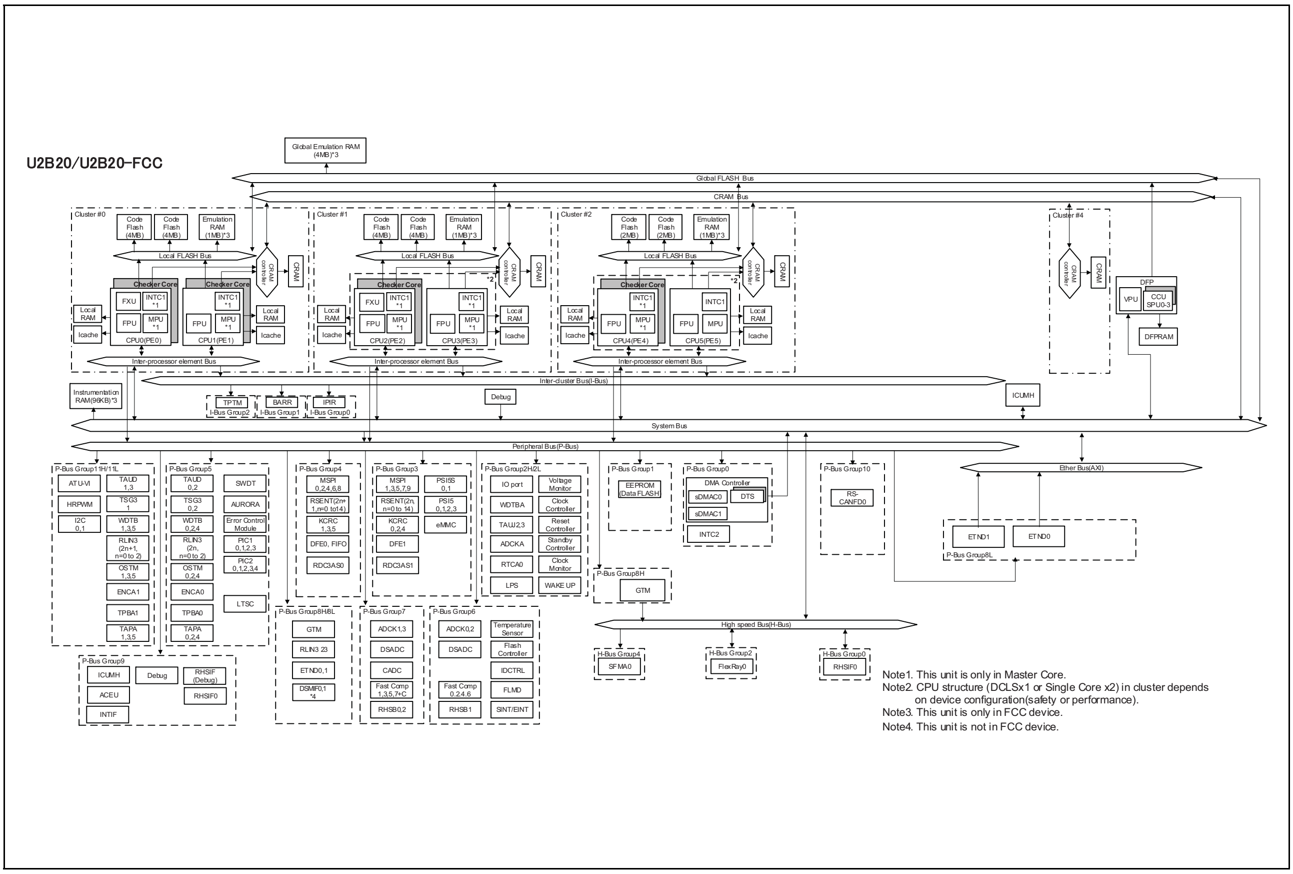


Figure 1.19 Internal Block Diagram (U2B20 and U2B20-FCC)

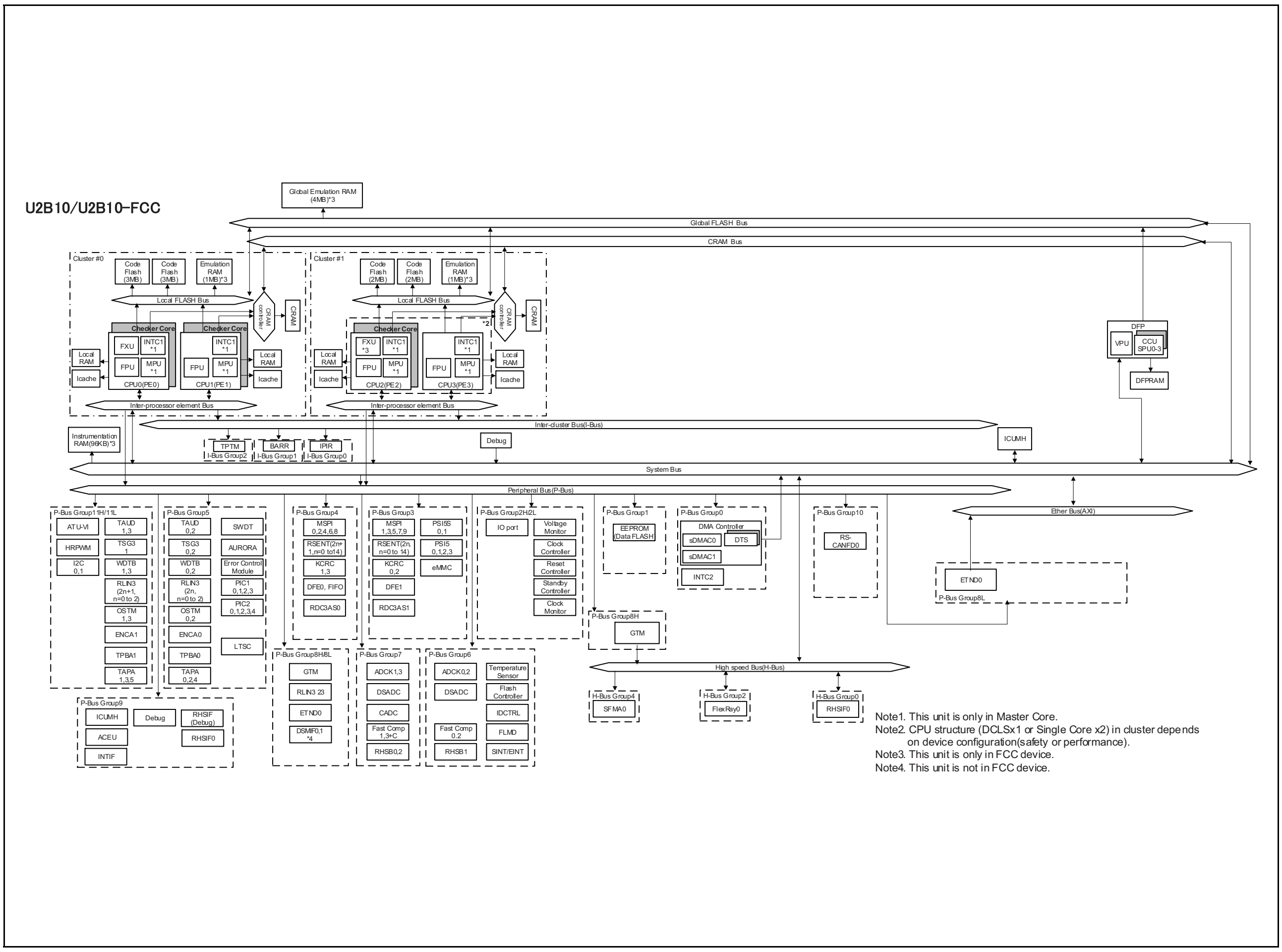


Figure 1.20 Internal Block Diagram (U2B10 and U2B10-FCC)

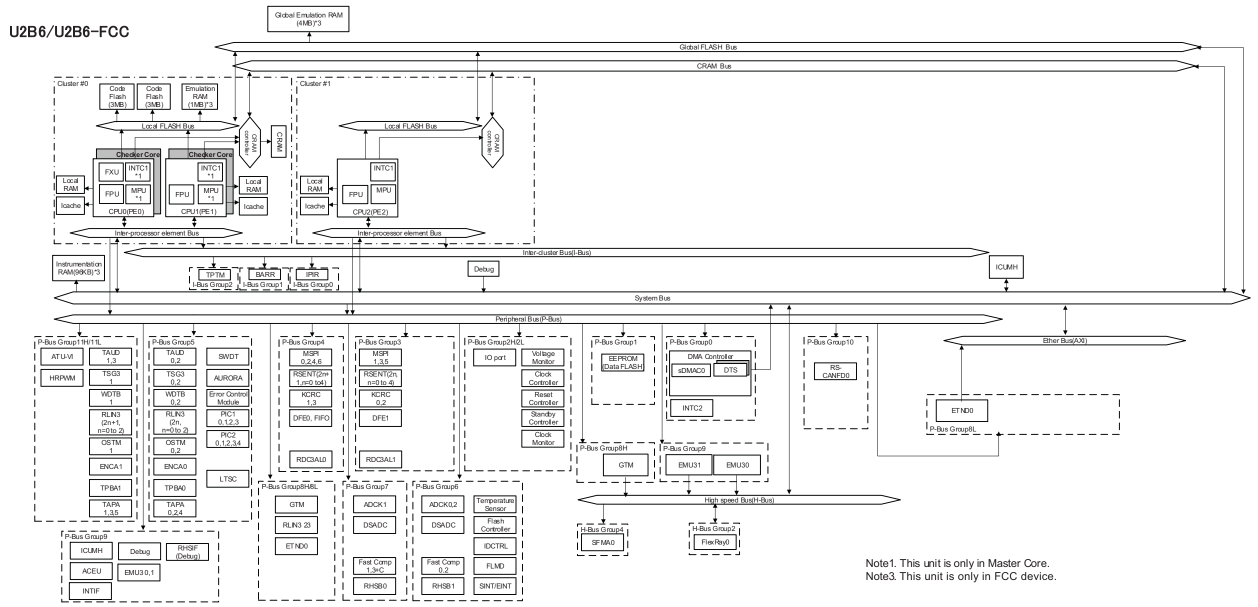


Figure 1.21 Internal Block Diagram (U2B6 and U2B-FCC for U2B6)

Section 2 Pin Functions

This section describes the pin connections and respective pin functions.

2.1 Term Definition

The following terms are used in this section:

Pin

Denotes the physical pin. Every pin is denoted by a unique pin number.

A pin can be used in several modes. Each pin is assigned a name that reflects its function, which is determined by the selected mode.

Port group

Denotes a group of pins. All the pins of a specific port group are controlled by the same port control register.

This product provides the following port groups, indicated by the numbers in the table below.

Table 2.1 Port Group

Product Name	RH850 U2B-EVA (600pins)	RH850 U2B24/ U2B24-FCC (468pins)	RH850 U2B20/ U2B20-FCC (468pins)	RH850 U2B10/ U2B10-FCC (468pins)	RH850 U2B24/ U2B24-FCC (373pins)	RH850 U2B20/ U2B20-FCC (373pins)	RH850 U2B10/ U2B10-FCC (373pins)	RH850 U2B10/ U2B10-FCC (292pins)	RH850 U2B6/ U2B6-FCC (292pins)
Number of port group	24	23			20		21	18	
Port Group Name	JP0, JP1, P00 to P02, P10 to P15, P20 to P25, P27, P30 to P34, P40	JP0, P00 to P02, P10 to P15, P20 to P25, P27, P30 to P34, P40			JP0, P00 to P02, P10 to P14, P20 to P23, P25, P27, P30, P32 to P34 P40		JP0, P00 to P02, P10 to P14, P20 to P25, P27, P30, P32 to P34 P40	JP0, P00, P02, P10 to P14, P20 to P23, P25, P27, P32 to P34, P40	

Port group index n

Each port group is identified by its own index “n” throughout this section; e.g.

PMCn for the port mode control register of the Pn port.

Port mode and ports

A pin in port mode works as a general purpose input/output pin. It is then called “port”.

The corresponding name is Pn_m. For example, P00_7 denotes port 7 of port group 00. It is referenced as “port P00_7”.

Alternative mode

In alternative mode, a pin can be used for various non-general-purpose input/output functions.

It is such as MSPI and IRQ.

SHMT1/SHMT4/SHMTMSC/TTL

Denotes input buffer types. Each types have a different DC characteristics.

For details, refer to the **Section 66, Electrical Characteristics**.

2.2 Pin List

2.2.1 Pin List and Function assignment

For detail information, refer to Appendix “E02_01_List_of_Pin_Assignment.xlsx”.

2.2.2 Pin Status

Table 2.2 Pin Status (1/4)

Pin Function		Pin Status						
			RESET = H					
Category	Pin Name	RESET = L	During Internal Reset	FBIST0 *1	After Internal Reset	DeepSTOP	FBIST1/ FBIST2 *1	Power Off Stanby
System Control	X1	I	I	I	I	I	I	Z
	X2	O	O	O	O	O	O	Z
	RESET	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	RESETOUT	O	O	O	O	O	O	Z
	FLMD0	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	P22_5 (FLMD1)	I *6	Z	Z	Z	Z *3	Z *3	Z
	SBMD	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	ICE	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	PEMD0	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	PEMD1	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	PEMD2	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	DBGSEL0	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	DBGSEL1	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
	PWRCTL*9	O	O	O	O	O	O	Z
	VMONOUT	O	O	O	O	O	O	O
	SVRPGATE	O	O	O	O	O	O	Z
	SVRNGATE	O	O	O	O	O	O	Z
PORT	Port in AWO area*7	Z	Z	Z	Z	Z	Z	Z
	Port in ISO area*8	Z	Z	Z	Z	Z *3*4	Z *3*4	Z
	ANxyz(x=0-3, A)	Z	Z	Z	Z	Z	Z	Z
ADC	AnVREFH(n=0-3)	I	I	I	I	I	I	Z
	ADSVREFH,ADSVREFL	I	I	I	I	I	I	Z
ECM	ERROROUT_M	Z	O	O	O	Z	O	Z

Table 2.2 Pin Status (2/4)

Pin Function			Pin Status						
				$\overline{\text{RESET}}$ = H					
Category		Pin Name	$\overline{\text{RESET}}$ = L	During Internal Reset	FBIST0 *1	After Internal Reset	DeepSTOP	FBIST1/ FBIST2 *1	Power Off Stanby
Aurora		CICREFP	Z	Z	Z	Z	Z	Z	Z
		CICREFN	Z	Z	Z	Z	Z	Z	Z
		TODP0	Z	Z	Z	Z	Z	Z	Z
		TODN0	Z	Z	Z	Z	Z	Z	Z
		TODP1	Z	Z	Z	Z	Z	Z	Z
		TODN1	Z	Z	Z	Z	Z	Z	Z
		TODP2	Z	Z	Z	Z	Z	Z	Z
		TODN2	Z	Z	Z	Z	Z	Z	Z
		TODP3	Z	Z	Z	Z	Z	Z	Z
		TODN3	Z	Z	Z	Z	Z	Z	Z
		$\overline{\text{AURORES}}$	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	Z
		$\overline{\text{MSYN}}$	I (pull-up)	I (pull-up)	pull-up	I (pull-up)	I (pull-up)	I (pull-up)	Z
SGMII		ETH0_SG_TXD_N	Z	Z	Z	Z	Z	Z	Z
		ETH0_SG_TXD_P	Z	Z	Z	Z	Z	Z	Z
		ETH0_SG_RXD_N	Z	Z	Z	Z	Z	Z	Z
		ETH0_SG_RXD_P	Z	Z	Z	Z	Z	Z	Z
		ETH1_SG_TXD_N	Z	Z	Z	Z	Z	Z	Z
		ETH1_SG_TXD_P	Z	Z	Z	Z	Z	Z	Z
		ETH1_SG_RXD_N	Z	Z	Z	Z	Z	Z	Z
		ETH1_SG_RXD_P	Z	Z	Z	Z	Z	Z	Z
		ETH_SG_REFCLK	Z	Z	Z	Z	Z	Z	Z
Debug	NEXUS (without RDY)	JP0_0	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP0_1	Z	Z	-	Z	Z	-	-
		JP0_2	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP0_3	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		$\overline{\text{TRST}}$	I (pull-down)	I (pull-down)	-	I (pull-down)	I (pull-down)	-	-
		JP0_5	Z	Z	-	Z	Z	-	-
		JP1_0	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_1	Z	Z	-	Z	Z	-	-
		JP1_2	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_3	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_5	Z	Z	-	Z	Z	-	-

Table 2.2 Pin Status (3/4)

Pin Function			Pin Status						
				$\overline{\text{RESET}}$ = H					
Category		Pin Name	$\overline{\text{RESET}}$ = L	During Internal Reset	FBIST0 *1	After Internal Reset	DeepSTOP	FBIST1/ FBIST2 *1	Power Off Stanby
Debug	NEXUS (with $\overline{\text{RDY}}$)	JP0_0	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP0_1	Z	Z	-	Z	Z	-	-
		JP0_2	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP0_3	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		$\overline{\text{TRST}}$	I (pull-down)	I (pull-down)	-	I (pull-down)	I (pull-down)	-	-
		JP0_5	Z *5	Z *5	-	O	O	-	-
		JP1_0	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_1	Z	Z	-	Z	Z	-	-
		JP1_2	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_3	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_5	Z *5	Z *5	-	O	O	-	-
	LPD4	JP0_0	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP0_1	Z *5	Z*5	-	O	O	-	-
		JP0_2	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP0_3	Z	Z	-	Z	Z	-	-
		$\overline{\text{TRST}}$	I (pull-down)	I (pull-down)	-	I (pull-down)	I (pull-down)	-	-
		JP0_5	Z *5	Z *5	-	O	O	-	-
		JP1_0	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_1	Z *5	Z *5	-	O	O	-	-
		JP1_2	I (pull-up)	I (pull-up)	-	I (pull-up)	I (pull-up)	-	-
		JP1_3	Z	Z	-	Z	Z	-	-
		JP1_5	Z *5	Z *5	-	O	O	-	-
	RHSIF Debug Interface*10	JP0_0	Z *5	Z *5	-	I	Z *5	-	-
		JP0_1	Z *5	Z *5	-	O	Z *5	-	-
		JP0_2	Z *5	Z *5	-	I	Z *5	-	-
		JP0_3	Z *5	Z *5	-	I	Z *5	-	-
		$\overline{\text{TRST}}$	I (pull-down)	I (pull-down)	-	I (pull-down)	I (pull-down)	-	-
		JP0_5	Z *5	Z *5	-	O	Z *5	-	-

Table 2.2 Pin Status (4/4)

Pin Function			Pin Status						
			$\overline{\text{RESET}}$ = L	$\overline{\text{RESET}}$ = H					
Category		Pin Name		During Internal Reset	FBIST0 *1	After Internal Reset	DeepSTOP	FBIST1/ FBIST2 *1	Power Off Stanby
	GPIO	JP0_0	I (pull-up)	I (pull-up)	pull-up	I (pull-up)	I (pull-up)	I (pull-up)	Z
		JP0_1	Z	Z	Z	Z	Z	Z	Z
		JP0_2	I (pull-up)	I (pull-up)	pull-up	I (pull-up)	I (pull-up)	I (pull-up)	Z
		JP0_3	Z	Z	Z	Z	Z	Z	Z
		$\overline{\text{TRST}}$	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)	I (pull-down)
		JP0_5	Z	Z	Z	Z	Z	Z	Z
		JP1_0	I (pull-up)	I (pull-up)	pull-up	I (pull-up)	I (pull-up)	I (pull-up)	Z
		JP1_1	Z	Z	Z	Z	Z	Z	Z
		JP1_2	I (pull-up)	I (pull-up)	pull-up	I (pull-up)	I (pull-up)	I (pull-up)	Z
		JP1_3	Z	Z	Z	Z	Z	Z	Z
		JP1_5	Z	Z	Z	Z	Z	Z	Z
Debug	BSCAN	JP0_0	I (pull-up)	I (pull-up)	-	I (pull-up)	-	-	-
		JP0_1	Z	Z	-	Z	-	-	-
		JP0_2	I (pull-up)	I (pull-up)	-	I (pull-up)	-	-	-
		JP0_3	I (pull-up)	I (pull-up)	-	I (pull-up)	-	-	-
		$\overline{\text{TRST}}$	I (pull-down)	I (pull-down)	-	I (pull-down)	-	-	-
		JP0_5	Z	Z	-	Z	-	-	-
	-	$\overline{\text{EVTI}}$	I (pull-up)	I (pull-up)	pull-up	I (pull-up)	I (pull-up)	pull-up	Z
		$\overline{\text{EVTO0}}$	O	O	O	O	Z *2	O	Z

Note: I: Input
O: Output
Z: Hi-Z
pull-up: On-chip pull-up resistor
pull-down: On-chip pull-down resistor

Note 1. FBIST0/FBIST1/FBIST2 are executed only when $\overline{\text{TRST}}$ = L
Note 2. If $\overline{\text{TRST}}$ = H, the pin keeps output
Note 3. The pin is in I/O buffer hold state. For detail, refer to **Section 17, Standby Controller (STBC).**
Note 4. LVDS input/output function become disable even if LVDSCTRL register is set.
Note 5. If reset mask function is enable, the pin keeps input/output. For detail, refer to **Section 62, Debugging and Calibration**.
Note 6. The pin is Hi-Z when FLMD0 = L
Note 7. P31,P33
Note 8. P00-P02,P10-P15,P20-P25,P30,P32,P34, excluding P22_5,P22_6,P22_9
Note 9. The PWRCTL pin can be used only when SBMD = H.
Note 10. RHSIF debug interface is supported only FCC device.

CAUTION

Fix the input of FLMD1 pin to low level in order to determine the Serial Programming Mode 1. For detail, refer to Section 5, Operating Mode.

2.2.3 Pin Function name

For detail information, refer to Appendix “E02_03_List_of_Pin_Function.xlsx".

2.2.4 Handling of Unused Pins

Handling of unused pins are listed in following table.

Table 2.3 Handling of Unused Pins (1/3)

Category	Pin Name	Handling of Unused Pins
System Control	X1 RESET FLMD0 SYSVCC VCC VDD RAMSVCL VSS AWOVCL	(These pins will always be used.)
	X2 ICE SBMD PEMD0/PEMD1/ PEMD2 DBGSEL0/DBGSEL1 PWRCTL VMONOUT	These pins can be left open.
	RESETOUT	
SVR	SVRPGATE SVRNGATE	These pins can be left open.
	SVRAVCC SVRDRVCC	These pins must be connected to SYSVCC.
	SVRAVSS SVRDRVSS	These pins must be connected to VSS.
ECM	ERROROUT_M	This pin can be left open.
PORT	Pn_m (excluding P27_0)	[Input mode] These pins can be left open and the settings are to disable input (when PMCn_m = 0, PMn_m = 1, PIBCn_m = 0) These pins can be left open and the settings are to enable on-chip pull-up/pull-down resistors (use PUn_m, PDn_m) These pins can be connected to power supply or ground of each pin via individual resistor. [Output mode] These pins can be left open.
	Analog input-only AN0xx, AN1xx, AN2xx, AN3xx (excluding AN012, AN013, AN02x,AN030,AN031, AN21x,AN22x)	These pins can be left open.
	Alternative analog input and digital input. AN012, AN013, AN02x,AN030,AN031, AN21x,AN22x	These pins can be left open with the settings are to disable input (values after reset are PMCn_m = 0, and PIBCn_m = 0)
	E0VCC/E1VCC/E2VCC	(These pins will always be used.)
	LVDVCC	This pin must be connected to E0VCC.
	OSCVCC J0VCC/J1VCC	This pin must be connected to VCC.

Table 2.3 Handling of Unused Pins (2/3)

Category	Pin Name	Handling of Unused Pins
Debug	JP0_0,JP1_0 JP0_1,JP1_1 JP0_2,JP1_2 JP0_3,JP1_3 JP0_5, JP1_5 $\overline{\text{EVTI}}$ $\overline{\text{EVTO0}}$	These pins can be left open.
	$\overline{\text{TRST}}$	This pin must be connected to VSS via individual resistor.
Aurora	CICREFP CICREFN TODP0 TODN0	Refer to **Section 2.8, Handling of Thermal Area Pins**
	TODP1 TODN1 TODP2 TODN2 TODP3 TODN3 $\overline{\text{MSYN}}$	These pins can be left open.
	$\overline{\text{AURORES}}$	Refer to **Section 2.8, Handling of Thermal Area Pins**
	EMUVCC EMUVDD	Refer to **Section 2.8, Handling of Thermal Area Pins**
SGMII	ETH0_SG_TXD_N ETH0_SG_TXD_P ETH0_SG_RXD_N ETH0_SG_RXD_P ETH1_SG_TXD_N ETH1_SG_TXD_P ETH1_SG_RXD_N ETH1_SG_RXD_P ETH_SG_REFCLK	These pins can be left open.
	GETH0PVCC	These pins must be connected to E0VCC. (When E0VCC = E1VCC = E2VCC then these pins can be connected to each of them.)
	GETH0BVCC	This pin must be connected to 3.0V to 3.6V voltage power supply, or VSS with 1 kΩ or more pull-down resistance.
	GETH0VCL GETH1VCL	This pin can be left open.

Table 2.3 Handling of Unused Pins (3/3)

Category	Pin Name	Handling of Unused Pins
ADC	ADSVCC ADSVSS ADSVREFH ADSVREFL ADSVCL	Connections when the DS-ADC and CADC is not in use but the ADC module is in use are listed below. ADSVCC: This pin must be connected to A0VCC. ADSVSS: This pin must be connected to A0VSS. ADSVREFH: This pin must be connected to A0VCC. ADSVREFL: This pin must be connected to A0VSS. ADSVCL: This pin can be left open. Connections when neither the DSADC nor CADC nor the ADCK module is in use are listed below. ADSVCC: This pin must be connected to E0VCC. ADSVSS: This pin must be connected to VSS. ADSVREFH: This pin must be connected to E0VCC. ADSVREFL: This pin must be connected to VSS. ADSVCL: This pin can be left open
	A0VCC A1VCC A2VCC A3VCC A0VREFH A1VREFH A2VREFH A3VREFH A0VSS A1VSS A2VSS A3VSS	Connections when the ADCK module is not in use but the DS-ADC or CADC is in use are listed below. A0VCC: This pin must be connected to ADSVCC. A0VSS: This pin must be connected to ADSVSS. A0VREFH: This pin must be connected to ADSVCC. A1VCC: This pin must be connected to ADSVCC. A1VSS: This pin must be connected to ADSVSS. A1VREFH: This pin must be connected to ADSVCC. A2VCC: This pin must be connected to ADSVCC. A2VSS: This pin must be connected to ADSVSS. A2VREFH: This pin must be connected to ADSVCC. A3VCC: This pin must be connected to ADSVCC. A3VSS: This pin must be connected to ADSVSS. A3VREFH: This pin must be connected to ADSVCC. Connections when neither the DSADC nor the ADCK nor the CADC module is in use are listed below. A0VCC: This pin must be connected to E0VCC. A0VSS: This pin must be connected to VSS. A0VREFH: This pin must be connected to E0VCC. A1VCC: This pin must be connected to E0VCC. A1VSS: This pin must be connected to VSS. A1VREFH: This pin must be connected to E0VCC. A2VCC: This pin must be connected to E0VCC. A2VSS: This pin must be connected to VSS. A2VREFH: This pin must be connected to E0VCC. A3VCC: This pin must be connected to E0VCC. A3VSS: This pin must be connected to VSS. A3VREFH: This pin must be connected to E0VCC.
	AFCVCC AFCVSS	Connections when the FastComparator and RDC3A is not in use but the ADC module is in use are listed below. AFCVCC: This pin must be connected to A0VCC. AFCVSS: This pin must be connected to A0VSS. Connections when neither the FastComparator nor RDC3A nor the ADCK module is in use are listed below. AFCVCC: This pin must be connected to E0VCC. AFCVSS: This pin must be connected to VSS.

2.3 PORT Overview

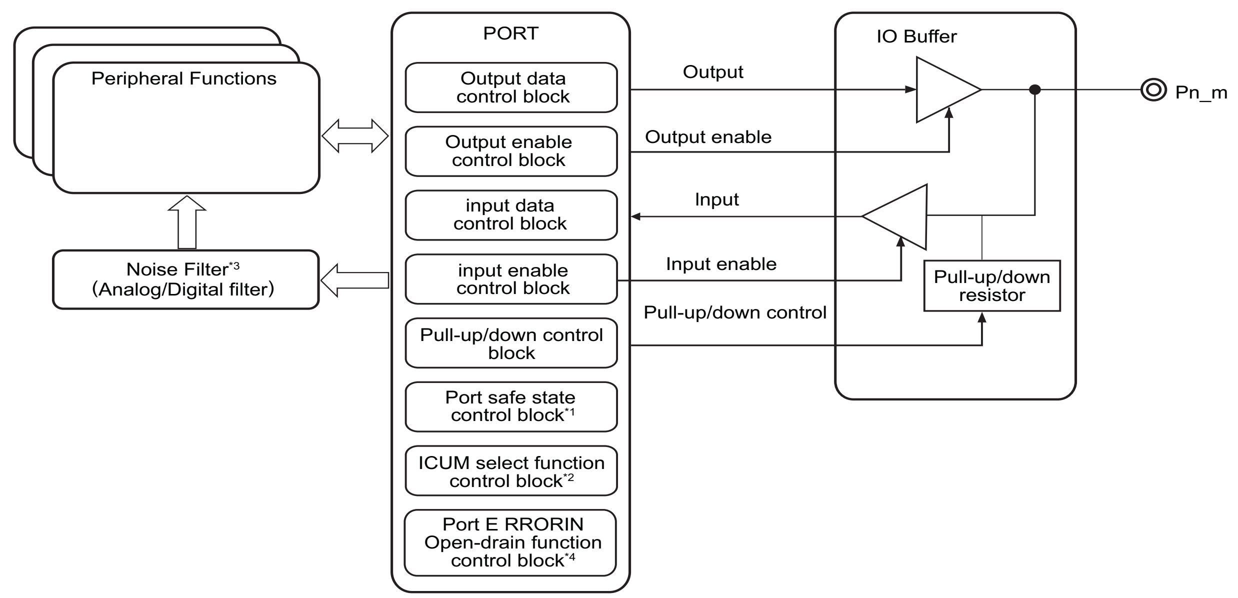
The product has various pins for input/output functions, known as ports. The ports are organized in port groups. Each pin is configurable to be used as GPIO or input/output function of internal IP.

Features summary:

- Configurable for individual pins.
- Each pin is multiplexed GPIO and many input/output functions of internal IPs.
- Electrical characteristics is configurable for most of the pins.
- Support port safe state function.
- Noise filter (Analog or Digital filter) is available for some functional signals of internal IPs.
- Supports LVDS function for high-speed interface.
- Shared analog pins with some GPIO pins are available for ADC.
- Write access protection is available for specific registers or whole port group.

2.3.1 Introduction

The port is an interface between external devices and the internal resources (IPs) and usually consists of IO buffer and the control logic as it is shown below.



Note 1. For details of port safe state function, see **Section 2.4.2, Port Safe State Function**

Note 2. For details of ICUM select function, see **Section 2.4.3, ICUM Select Function**

Note 3. For details of Noise Filter, see **Section 2.7, Noise Filter & Edge/Level Detector**

Note 4. For details of Port ERRORIN Open-drain Function, see **Section 2.4.4, Port ERRORIN Open-drain Function**

Figure 2.1 Functional Block Diagram

2.3.2 Register Base Addresses

All port addresses are given as an offset from the individual base addresses, <JPORT_base> and <PORT_base>.

Table 2.4 Port Group

Base Address Name	Base Address	Bus Group
<PORT_base>	FFD9 0000_H	P-Bus Group 2L
<JPORT_base>	FFDA 0000_H	P-Bus Group 2L

2.3.3 Clock Supply

The clock supply to ports is shown in the following table.

Table 2.5 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name
PORT	Register access clock	CLK_LSB

2.3.4 Operation Mode

Pins can operate in three operation modes.

- Port mode (PMCn.PMCn_m = 0)
  A pin in port mode operates as a general purpose input/output pin. The input / output mode is selected by setting the PMn.PMn_m bit.
- Alternative mode software I/O control (PMCn.PMCn_m bit = 1, PIPCn.PIPCn_m bit = 0)
  In this mode, the pins operate as alternative functions. The input / output mode is selected by setting the PMn.PMn_m bit.
- Alternative mode direct I/O control (PMCn.PMCn_m bit = 1, PIPCn.PIPCn_m bit = 1)
  In this mode, the pins operate as alternative functions. Unlike the alternative mode software I/O control, however, the input / output direction is selected by the alternative function.

The following is register effect related to operation modes and pin I/O direction.

- PMCn.PMCn_m bit
  This bit selects port mode (PMCn_m = 0) or alternative mode (PMCn_m = 1).
- PMn.PMn_m bit
  This bit selects input (PMn_m = 1) or output (PMn_m = 0) when the port mode (PMCn_m = 0) and alternative mode software I/O control (PMCn_m = 1, PIPCn_m = 0) have been selected.
- PIBCn.PIBCn_m bit
  This bit disables (PIBCn_m = 0) or enables (PIBCn_m = 1) the input buffer in input port mode (PMCn_m = 0, PMn_m = 1).
- PIPCn.PIPCn_m bit
  This bit selects alternative mode software I/O control or alternative mode direct I/O control.
- PBDCn.PBDCn_m bit
  In output mode or output enabled by alternative function, when this bit is set to 1, the pin enters the bidirectional mode.

The input/output direction and each modes by register setting is shown below **Table 2.6, Pin register setting**

Table 2.6 Pin register setting

PMCn_m	PMn_m	PIBCn_m	PIPCn_m	PBDCn_m	Modes	I/O Direction
0	0	X	X	0	Port mode Output mode	Output
				1	Port mode Bi-directional mode	Input/Output
	1	0		X	Port mode Input mode (input disabled)	–
		1			Port mode Input mode (input enabled)	Input
1	0	X	0	0	Alternative mode Software I/O control Output mode	Output
		X[*2]		1	Alternative mode Software I/O control Bi-directional mode	Input/Output
	1	0[*1]		X	Alternative mode Software I/O control Input mode	Input
	X		1	0	Alternative mode Direct I/O control Input or output	Controlled by the alternative function
		X[*2]		1	Alternative mode Direct I/O control Input or bi-direction	Controlled by the alternative function

Note 1. Setting PIBC 1 is prohibited.

Note 2. Setting PIBC 0 in Bi-directional mode makes propagate the pin level to alternative input function.
Setting PIBC 1 in Bi-directional mode does not make propagate the pin level to alternative input function.

Table 2.7 Alternative mode selection table

Bit					Function
PFCEAEn_m	**PFCAEn_m**	**PFCEn_m**	**PFCn_m**	**PMn_m**	
0	0	0	0	0	Alternative output mode 1 (ALT-OUT1)
				1	Alternative input mode 1 (ALT-IN1)
			1	0	Alternative output mode 2 (ALT-OUT2)
				1	Alternative input mode 2 (ALT-IN2)
		1	0	0	Alternative output mode 3 (ALT-OUT3)
				1	Alternative input mode 3 (ALT-IN3)
			1	0	Alternative output mode 4 (ALT-OUT4)
				1	Alternative input mode 4 (ALT-IN4)
	1	0	0	0	Alternative output mode 5 (ALT-OUT5)
				1	Alternative input mode 5 (ALT-IN5)
			1	0	Alternative output mode 6 (ALT-OUT6)
				1	Alternative input mode 6 (ALT-IN6)
		1	0	0	Alternative output mode 7 (ALT-OUT7)
				1	Alternative input mode 7 (ALT-IN7)
			1	0	Alternative output mode 8 (ALT-OUT8)
				1	Alternative input mode 8 (ALT-IN8)
1	0	0	0	0	Alternative output mode 9 (ALT-OUT9)
				1	Alternative input mode 9 (ALT-IN9)
			1	0	Alternative output mode 10 (ALT-OUT10)
				1	Alternative input mode 10 (ALT-IN10)
		1	0	0	Alternative output mode 11 (ALT-OUT11)
				1	Alternative input mode 11 (ALT-IN11)
			1	0	Alternative output mode 12 (ALT-OUT12)
				1	Alternative input mode 12 (ALT-IN12)
	1	0	0	0	Alternative output mode 13 (ALT-OUT13)
				1	Alternative input mode 13 (ALT-IN13)
			1	0	Alternative output mode 14 (ALT-OUT14)
				1	Alternative input mode 14 (ALT-IN14)
		1	0	0	Alternative output mode 15 (ALT-OUT15)
				1	Alternative input mode 15 (ALT-IN15)
			1	0	Alternative output mode 16 (ALT-OUT16)
				1	Alternative input mode 16 (ALT-IN16)

NOTE

Some input or output functions is assigned to more than one pin. Only activate one single pin to one given alternative input function. Do not activate a input function on multiple pins at the same time.
Do not set alternative mode selection to a place where the alternative function is not assigned.

2.3.5 Pin data input/output

The registers used for data input/output are described below.

The location that is read via the PPRn register differs depending on the pin mode.

(1) Output data

In the port mode (PMCn.PMCn_m = 0), the value of the Pn.Pn_m bit is output to the Pn_m pin.

(2) Input data

When the PPRn register is read, either the value of the Pn_m pin, the value of the corresponding bit of the port register Pn.Pn_m, or the value output by the alternative function is returned.

Which value is returned depends on the pin mode and setting of several control bits. The different PPRn read modes are shown in the **Table 2.8, PPRn_m Read Values**

Table 2.8 PPRn_m Read Values

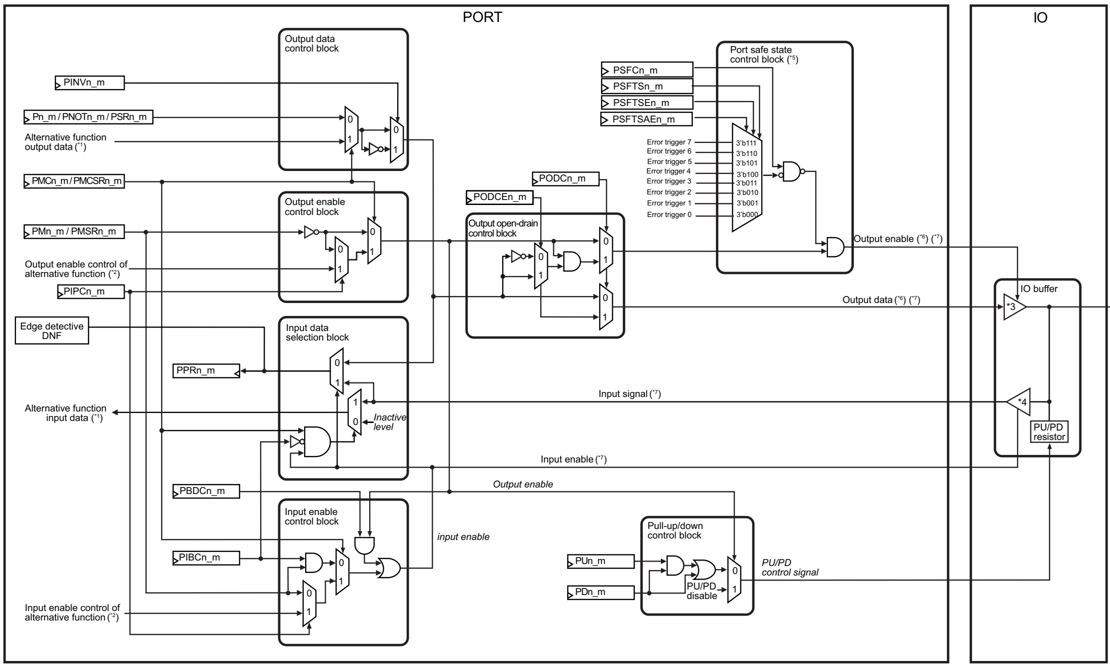
PMCn_m	PMn_m	PIBCn_m	PIPCn_m	PBDCn_m	Mode	PPRn_m Read Value
0	0	X	X	0	Port Mode Output Mode	Pn.Pn_m bit
				1	Port Mode Bi-directional Mode	Pn_m pin
	1	0		X	Port Mode Input Mode, (Input disabled)	Pn.Pn_m bit
		1			Port Mode Input Mode (Input enabled)	Pn_m pin
1	0	X	0	0	Alternative Mode Software I/O control Output Mode	Alternative-function internal output signal
				1	Alternative Mode Software I/O control Bi-directional Mode	Pn_m pin
	1			X	Alternative Mode Software I/O control Input Mode	Pn_m pin
	X		1	0	Alternative Mode Direct I/O control Input or output	I/O port in alternative mode: Input: Pn_m pin Output: Alternative-function internal output signal
				1	Alternative Mode Direct I/O control Input or bi-direction	I/O port in alternative mode: Input: Pn_m pin Bi-direction: Pn_m pin

2.4 Port Functions

This section explains port control logic functions.

2.4.1 Port Control Logic

The following figure shows the logical circuitry of the port control functions. The diagram is only a logical reference and does not show the real circuitry.



Note 1. Alternative function signal selected by PFCn_m, PFCEn_m, PFCAEn_m and PFCEAEn_m.

Note 2. Only functions exist in **Table 2.29, Alternative functions require "Direct I/O Control" (Must Set PIPCn_m=1)** are supported this configuration.

Note 3. The output drive strength can be set by PDSCn_m and PUCCn_m.

Note 4. The input buffer type can be set by PISn_m and PISAn_m.

Note 5. Detail of Port Safe State function is described in **Section 2.4.2, Port Safe State Function.**

Note 6. P22_0, P22_1, P22_2 and P22_3 are target pins for ICUM select function which requires additional higher priority control. For details, refer to **Section 2.4.3, ICUM Select Function**.

Note 7. The alternative functions which are supported LVDS buffers have additional control logic from LVDS registers. For detail, refer to **Section 2.4.6, Functions incorporated LVDS Buffers**.

Note 8. The Pull-up/down condition can be set by PULVSEL5.

Figure 2.2 Port Control Logic Block Diagram

2.4.2 Port Safe State Function

Pn_m have Port Safe State Function which can operate the output enable of each pins individually by selecting error triggers for the defined application. This product supports up to 8 error triggers named $\overline{\text{ERROROUT0}}$, $\overline{\text{ERROROUT1}}$, $\overline{\text{ERROROUT2}}$, $\overline{\text{ERROROUT3}}$, $\overline{\text{ERROROUT4}}$, $\overline{\text{ERROROUT5}}$, $\overline{\text{ERROROUT6}}$ and $\overline{\text{ERROROUT7}}$, and the trigger can be selected by setting of PSFTSn_m, PSFTSEn_m and PSFTSAEn_m registers. For details of error triggers behavior, refer to **Section 56, Error Control Module (ECM)**.

The output enable control of a pin is forcibly disabled in case that an error trigger is issued when Port Safe State Function is enabled by setting PSFCn_m register. Even Bi-directional mode output data is suppressed by Port Safe State function during an error trigger is issued.

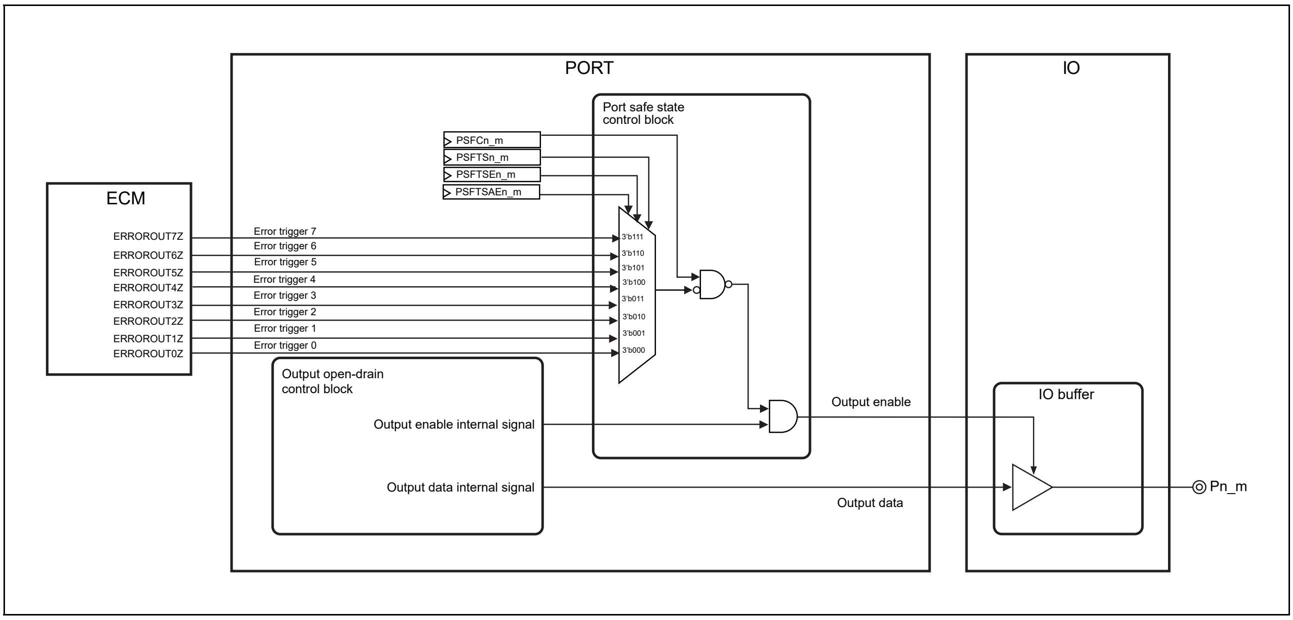


Figure 2.3 Block Diagram of Port Safe State Function

The following table shows the register configuration for Port Safe State Function

Table 2.9 Configuration for Port Safe State Function

PSFCn_m	PSFTSAEn_m	PSFTSEn_m	PSFTSn_m	Function
0	X	X	X	Port Safe State function is disabled
1	0	0	0	Port Safe State function is enabled. Error Trigger 0 is selected for Port Safe State function.
			1	Port Safe State function is enabled. Error Trigger 1 is selected for Port Safe State function.
		1	0	Port Safe State function is enabled. Error Trigger 2 is selected for Port Safe State function.
			1	Port Safe State function is enabled. Error Trigger 3 is selected for Port Safe State function.
	1	0	0	Port Safe State function is enabled. Error Trigger 4 is selected for Port Safe State function.
			1	Port Safe State function is enabled. Error Trigger 5 is selected for Port Safe State function.
		1	0	Port Safe State function is enabled. Error Trigger 6 is selected for Port Safe State function.
			1	Port Safe State function is enabled. Error Trigger 7 is selected for Port Safe State function.

CAUTION

Error triggers from ECM are triggered by DeepSTOP Reset and the status of a pin located to AWO area becomes high impedance when Port Safe State Function is enabled.

2.4.3 ICUM Select Function

Target pins: P22_0, P22_1, P22_2 and P22_3.

The ICUM Select Function takes highest priority in port functions to control output value of these target pins.

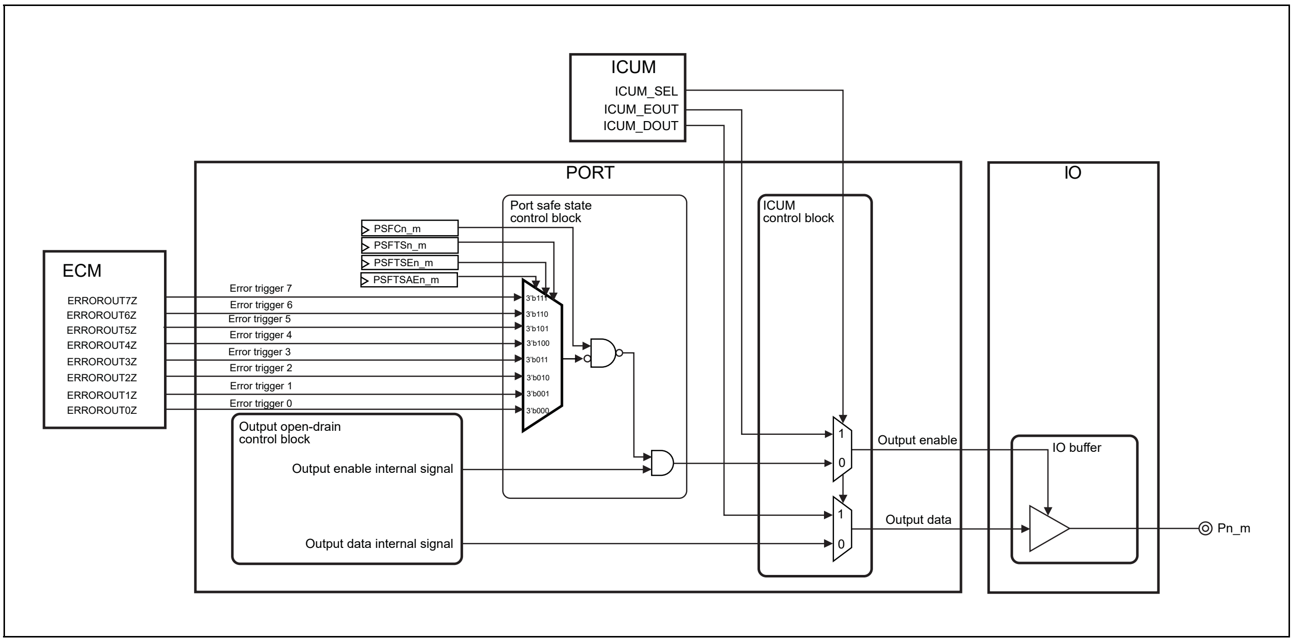


Figure 2.4 Block Diagram of ICUM Select Function

For more details how to use the ICUM function, refer to *RH850/U2B Group Security User's Manual: Hardware.*

2.4.4 Port $\overline{\text{ERRORIN}}$ Open-drain Function

Target pins: P00_2, P02_10, P11_7, P15_6, P21_2, P22_0, P23_3, P32_5, P33_1(which has $\overline{\text{ERRORIN}}$ alternatie function).

These pins have a **2.5.30** PEIODCn register to control Port $\overline{\text{ERRORIN}}$ Open-drain function.

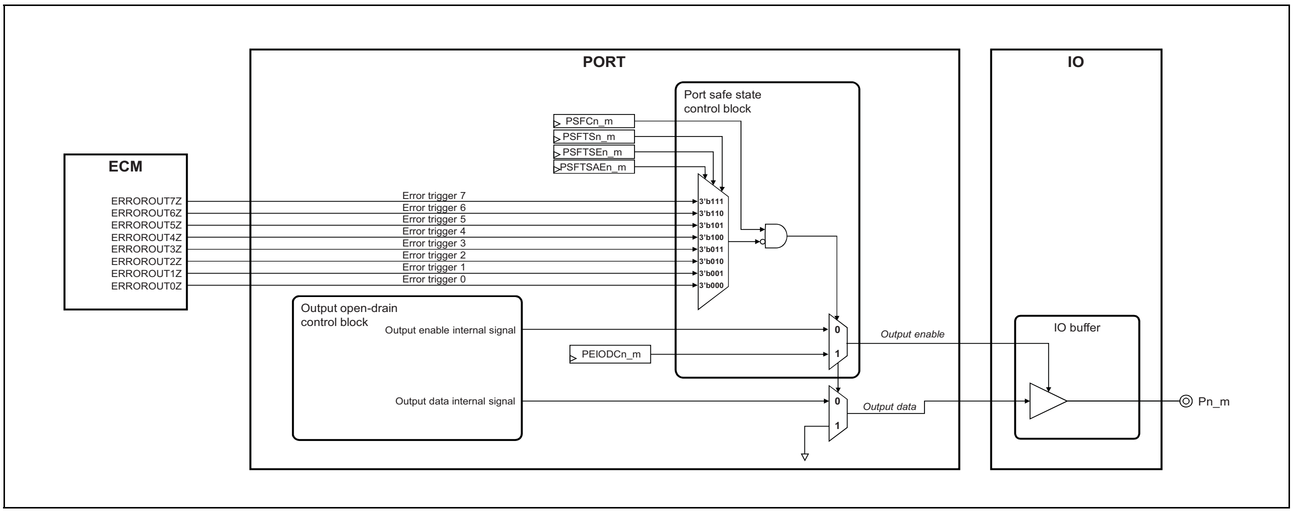


Figure 2.5 Block Diagram of Port $\overline{\text{ERRORIN}}$ Open-drain Function

2.4.5 RESETOUT Function

This product has a RESETOUT pin to handshake with a debugger. The behavior of this signal's output can be selected by using an Option Byte as follows. For detailed information about Option Bytes, refer to the **Section 63, Flash Memory**.

(1) Option Byte PFC_RESO_CFG = 1

- RESETOUT pin is assign P27 group (P27_0 only) supported with output in order to handshake to debugger only.
- This port setting is configurable as a high/low output with push-pull and open drain modes that are the same as other port. However, all of the P27 registers are protected by a key code register.
- The output signal is low when reset factor (see **Section 11.1, Features** for details) is asserted. It will hold the low output until port setting is changed by user SW command.
- This pin shows the internal reset state, so can be used as handshake for external devices like debugger.

The following figure shows the use of RESETOUT function.

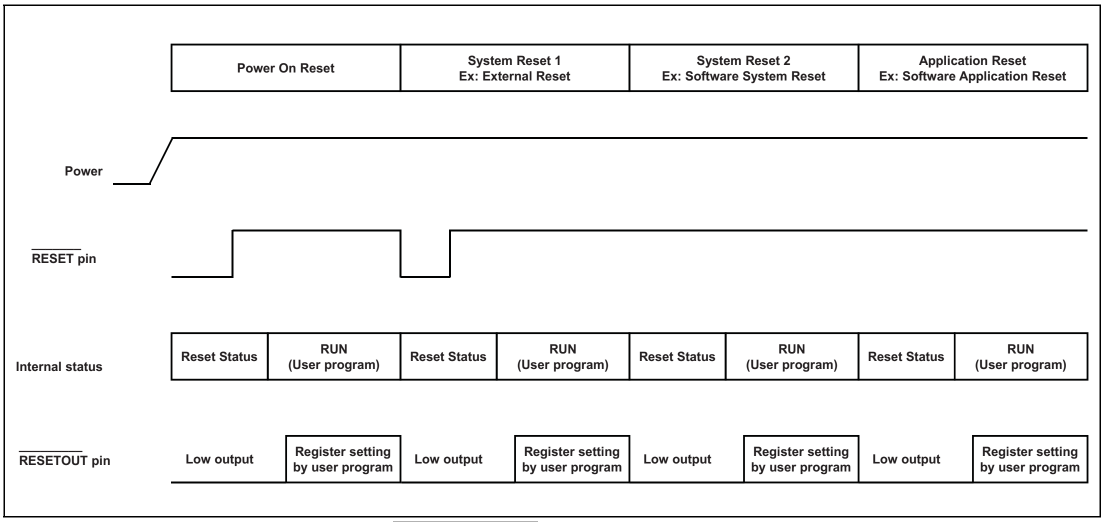


Figure 2.6 Timing Chart of RESETOUT function for each reset factor(PFC_RESO_CFG = 1)

(2) Option Byte PFC_RESO_CFG = 0

- $\overline{\text{RESETOUT}}$ pin is not assigned to a port group, so the port setting is not configurable.
- The output signal is low when reset factor (see **Section 11.1, Features** for details.) is asserted. And the terminal state is automatically changed from low to high after internal reset.

The following figure shows the use of $\overline{\text{RESETOUT}}$ function.

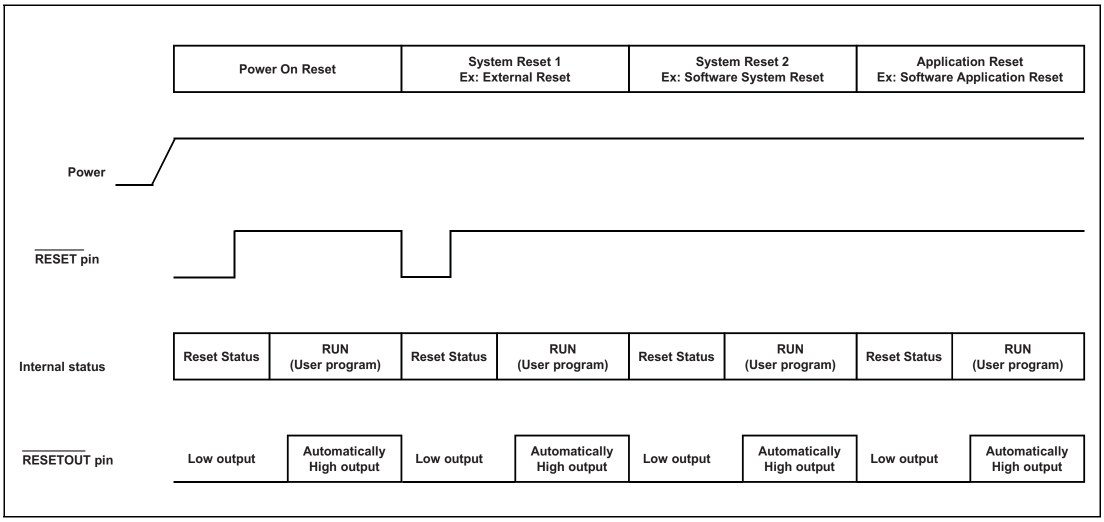


Figure 2.7 Timing Chart of $\overline{\text{RESETOUT}}$ function for each reset factor(PFC_RESO_CFG = 0)

2.4.6 Functions incorporated LVDS Buffers

When the LVDS buffer is used, set the LVDSCTRL[A,B,C,D,E,F] register to enable input/output. Also, set whether the supply voltage of LVDS buffer is 3V or 5V.

Table 2.10 LVDS Control Register Assignment (1/2)

Control Register	Register Bit	Power Domain	I/F	Pin Name	I/O	Control	Related Function
LVDSCTRLA	[3]	LVDVCC	ANSI/IEEE	P21_4/P21_5	OUT	OE	HSIF0_TXDN/HSIF0_TXDP
	[19]					SPECSEL	HSIF0_TXDN/HSIF0_TXDP
	[0]		ANSI	P21_2/P21_3	IN	IE	HSIF0_RXDN/HSIF0_RXDP
	[16]					VSEL	HSIF0_RXDN/HSIF0_RXDP
LVDSCTRLB	[5]	E2VCC	ANSI/IEEE	P13_0/P13_1	OUT	OE	MSPI9_SCKN/MSPI9_SCKP
							RHSB2FCLN/RHSB2FCLP
							RHSB2NMFCLN/RHSB2NMFCLP
	[3]			P13_2/P13_3	OUT	OE	MSPI9_SON/MSPI9_SOP
							RHSB2SON/RHSB2SOP
							RHSB2MCSON/RHSB2MCSOP
							RHSB2NMSON/RHSB2NMSOP
	[19]			P13_0/P13_1, P13_2/P13_3	OUT	SPECSEL	MSPI9_SCKN/MSPI9_SCKP
							RHSB2FCLN/RHSB2FCLP
							RHSB2NMFCLN/RHSB2NMFCLP
							MSPI9_SON/MSPI9_SOP
							RHSB2SON/RHSB2SOP
							RHSB2MCSON/RHSB2MCSOP
							RHSB2NMSON/RHSB2NMSOP
	[4]		ANSI	P13_0/P13_1	IN	IE	MSPI9_SCKN/MSPI9_SCKP
							RHSB2MCSIN/RHSB2MCSIP
	[0]			P14_4/P14_5	IN	IE	MSPI9_SIP/MSPI9_SIN
	[16]			P13_0/P13_1, P14_4/P14_5	IN	VSEL	MSPI9_SCKN/MSPI9_SCKP
							RHSB2MCSIN/RHSB2MCSIP
							MSPI9_SIP/MSPI9_SIN
LVDSCTRLC	[1]	E1VCC	ANSI/IEEE	P10_0/P10_1	OUT	OE	MSPI6_SCKP/MSPI6_SCKN
							RHSB1FCLP/RHSB1FCLN
							RHSB1NMFCLP/RHSB1NMFCLN
	[3]			P10_2/P10_3	OUT	OE	MSPI6_SOP/MSPI6_SON
							RHSB1SOP/RHSB1SON
							RHSB1MCSOP/RHSB1MCSON
							RHSB1NMSOP/RHSB1NMSON
	[17]			P10_0/P10_1, P10_2/P10_3	OUT	SPECSEL	MSPI6_SCKP/MSPI6_SCKN
							RHSB1FCLP/RHSB1FCLN
							RHSB1NMFCLP/RHSB1NMFCLN

Table 2.10 LVDS Control Register Assignment (2/2)

Control Register	Register Bit	Power Domain	I/F	Pin Name	I/O	Control	Related Function
LVDSCTRLC	[17]	E1VCC	ANSI/IEEE	P10_0/P10_1, P10_2/P10_3	OUT	SPECSEL	MSPI6_SOP/MSPI6_SON
							RHSB1SOP/RHSB1SON
							RHSB1MCSOP/RHSB1MCSON
							RHSB1NMSOP/RHSB1NMSON
	[0]		ANSI	P10_0/P10_1	IN	IE	MSPI6_SCKP/MSPI6_SCKN
							RHSB1MCSIP/RHSB1MCSIN
	[4]			P10_4/P10_5	IN	IE	MSPI6_SIP/MSPI6_SIN
	[16]			P10_0/P10_1, P10_4/P10_5	IN	VSEL	MSPI6_SCKP/MSPI6_SCKN
							RHSB1MCSIP/RHSB1MCSIN
							MSPI6_SIP/MSPI6_SIN
LVDSCTRLD	[1]	LVDVCC	ANSI	P25_3/P25_4	OUT	OE	RHSB0FCLN/RHSB0FCLP
							RHSB0NMFCLN/RHSB0NMFCLP
	[3]			P25_5/P25_6	OUT	OE	RHSB0MCSON/RHSB0MCSOP
							RHSB0SON/RHSB0SOP
							RHSB0NMSON/RHSB0NMSOP
	[0]			P25_3/P25_4	IN	IE	RHSB0MCSIN/RHSB0MCSIP
	[16]			P25_3/P25_4	IN	VSEL	RHSB0MCSIN/RHSB0MCSIP
LVDSCTRLE	[3]	LVDVCC	ANSI/IEEE	P24_6/P24_7	OUT	OE	HSIF1_TXDN/HSIF1_TXDP
	[19]			P24_6/P24_7	OUT	SPECSEL	HSIF1_TXDN/HSIF1_TXDP
	[0]		ANSI	P24_4/P24_5	IN	IE	HSIF1_RXDN/HSIF1_RXDP
	[16]			P24_4/P24_5	IN	VSEL	HSIF1_RXDN/HSIF1_RXDP
LVDSCTRLF	[5]	LVDVCC	ANSI	P25_12/ P25_13	OUT	OE	RHSB3FCLP/RHSB3FCLN
							RHSB3NMFCLP/RHSB3NMFCLN
	[3]			P25_14/ P25_15	OUT	OE	RHSB3SOP/RHSB3SON
							RHSB3MCSOP/RHSB3MCSON
							RHSB3NMSOP/RHSB3NMSON
	[4]			P25_12/ P25_13	IN	IE	RHSB3MCSIP/RHSB3MCSIN
	[20]			P25_12/ P25_13	IN	VSEL	RHSB3MCSIP/RHSB3MCSIN

CAUTION

For pins have not only functions incorporated LVDS buffers but also GPIO and alternative functions. These cannot be used at the same time. Refer to 2.6.4, LVDS Function Setting in case LVDS buffers are used.

Input/output from the LVDS buffer actually being enabled/disable takes at least 10 µs after setting LVDSCTRL[A~F], and IOHOLD0.IOHOLD_Pn(n=10,13,14,21,24,25).

2.4.7 Debug Interface

The switch between debug interface and port mode is controlled by a combination of mode pins and option byte settings. For details, see **Section 62, Debugging and Calibration**.

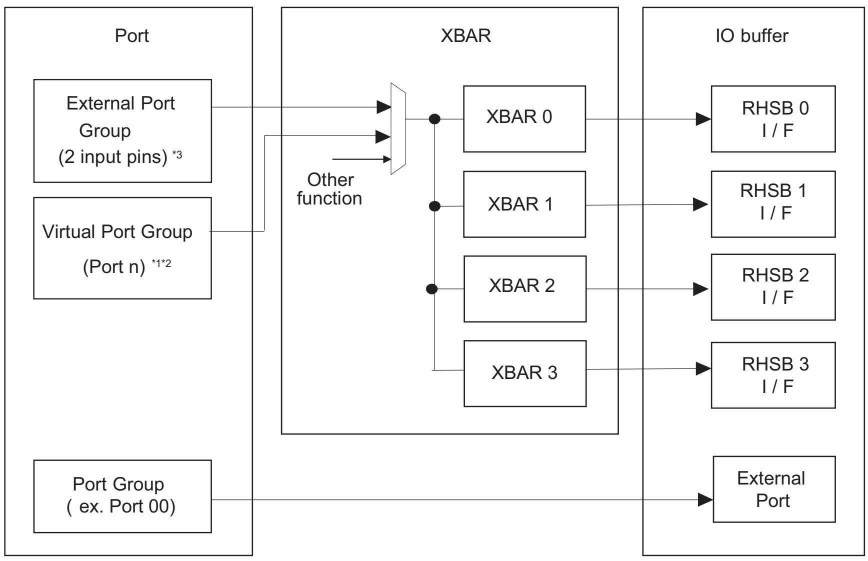
2.4.8 Virtual Port Function via RHSB (XBAR)

Virtual ports are port groups that are not connected to external pins but are instead connected to internal functions. Port groups P41, P42, P43 and P44 are virtual ports for the XBAR function of RHSB. Configurable output signals are 64 bits in total (each port group has 16 bits x 4 groups), and it is possible to transfer these data serially to other devices via XBAR in the same way as with a port register setting (XBAR function, see **Section 27.6, Cross Bar (XBAR)**).

- The output data (64 bits) of P41, P42, P43 and P44 and RHSB_XBAR_TDE0, RHSB_XBAR_TDE1 are transferring to input of XBAR function.
- Output pins of P41, P42, P43 and P44 are connected to XBAR function instead of IO output buffer.
- The data from XBAR function is transferred to RHSB.

Block Diagram

The following figure shows the use of coordination with XBAR Function.



Note 1. n: Virtual Port Group number. (n = 41, 42, 43, 44).

Note 2. Virtual Port can be read the Pn_m value as the same as Port function

Note 3. RHSB_XBAR_TDE0, RHSB_XBAR_TDE1.

Figure 2.8 Block Diagram of Virtual Port Function via RHSB

2.4.9 Virtual Port Function via Analog

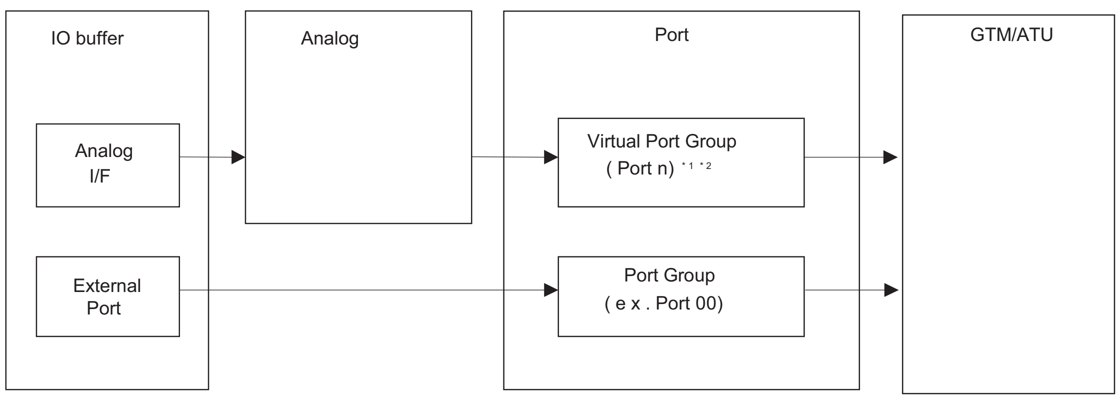
Virtual ports are port groups that are not connected to external pins but are instead connected to internal functions. The target port groups are P37, P38 and P39. For support product of virtual ports is refer to Appendix "E02_01_List_of_Pin_Assignment.xlsx". This function is to transfer the received serial data from Analog Boundary Flag generator to GTM/ATU, as the inverted function of delivering the serial data to XBAR described in **Section 2.4.8, Virtual Port Function via RHSB (XBAR)**.

About Analog Boundary Flag generator, see **Section 50, Analog to Digital Converter (ADCK)**, **Section 51, Delta-Sigma Analog to Digital Converter (DSADC)** and **Section 52, Cyclic Analog to Digital Converter (CADC)**.

- The data (max 48 bits) from Analog Boundary Flag generator is transferring to input pins of P37, P38 and P39.
- Input pins of virtual ports are connected to Analog Boundary Flag generator instead of IO input buffer.
- The data from Analog Boundary Flag generator is transferred to GTM/ATU via port function (pin multiplexed table, see **Section 2.2.1, Pin List and Function assignment**.

Block Diagram

The following figure shows the use of virtual ports. It may look like the data for GTM/ATU is from a virtual port, but actually, the data is from the analog module.



Note 1. n: Virtual Port Group number. (n = 37 ~ 39).
For support product of virtual ports is refer to Appendix "E02_01_List_of_Pin_Assignment.xlsx".

Note 2. Virtual Port can read the Pn_m value as the same as Port function.

Figure 2.9 Block Diagram of Virtual Port Function via Analog

2.4.10 Virtual Port Function via RHSB+

Virtual ports are port groups that are not connected to external pins but are instead connected to internal functions. Port groups of P45~P60 are virtual port groups for RHSB+ I/F. For support product of virtual ports is refer to Appendix "E02_01_List_of_Pin_Assignment.xlsx". This function is to transfer the received serial data from RHSB+ I/F to GTM/ATU, as the inverted function of delivering the serial data to XBAR described in **Section 2.4.8, Virtual Port Function via RHSB (XBAR)**.

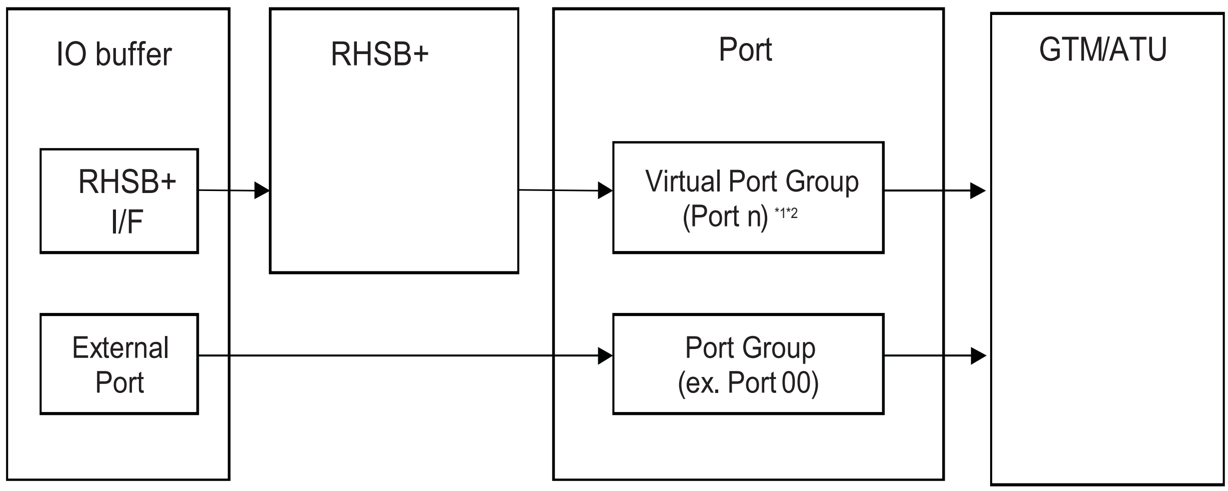
About RHSB+ function, see **Section 27, Renesas High Speed Bus (RHSB).**

- The data (max 64 bits) from RHSB+ is transferred to the input pins of P45~P60.
- The input pins of P45~P60 are connected to RHSB+ instead of the IO input buffer.
- The data from RHSB+ is transferred to GTM/ATU via the port function (pin multiplexed table, see **Section 2.2.1, Pin List and Function assignment**).
- The 256 bits of data from RHSB+ can be divided by 8 bits per frame.
- For example, the received data is 8*n bits (n = 1, 2, 3, …, 32)

Block Diagram

The following figure shows the use of P45~P60. It may look like that the data for GTM/ATU is from P45~P60, but in actually, the data is from RHSB+.

P45~P48 is used for RHSB0, P49~P52 is used for RHSB1, P53~P56 is used for RHSB2, P57~P60 is used for RHSB3.



Note 1. n: Virtual Port Group number. (n = 45~60).
For support product of virtyal ports is refer to Appendix "E02_01_List_of_Pin_Assignment.xlsx".

Note 2. Virtual Port can read the Pn_m value as the same as Port function.

Figure 2.10 Block Diagram of Virtual Port Function via RHSB+

2.5 Port Register Description

For detail list control registers of each port group, refer to Appendix “E02-02_List of Port Functions.xlsx”.

In the bitmap field, “x” means an effective bit and “—” means a reserved bit. Reserved areas are always read as the values after reset. The write value also should be the value after reset. The registers of unimplemented pins should not be modified after reset.

The list of port control register types are described below.

Table 2.11 Port Register Overview (1/2)

			Access Protection		
				Other	
Port Register Name	**Port Symbol**	**Address**	**PBG**	**Category1 *1**	**Category2 *2**
Port register	Pn	<PORT_base> + 0000_H + 40_H*n	Refer to Appendix “E02-02_List of Port Functions.xlsx”	—	PWE
	JPn	<JPORT_base> + 0000_H + 40_H*n		—	—
Port Set Reset register	PSRn	<PORT_base> + 0004_H + 40_H*n		—	PWE
	JPSRn	<JPORT_base> + 0004_H + 40_H*n		—	—
Port NOT register	PNOTn	<PORT_base> + 0008_H + 40_H*n		—	PWE
	JPNOTn	<JPORT_base> + 0008_H + 40_H*n		—	—
Port Pin Read register	PPRn	<PORT_base> + $000C_H$ + 40_H*n		—	—
	JPPRn	<JPORT_base> + $000C_H$ + 40_H*n		—	—
Port Mode register	PMn	<PORT_base> + 0010_H + 40_H*n		—	PWE
	JPMn	<JPORT_base> + 0010_H + 40_H*n		—	—
Port Mode Control register	PMCn	<PORT_base> + 0014_H + 40_H*n		—	PWE
	JPMCn	<JPORT_base> + 0014_H + 40_H*n		—	—
Port Function Control register	PFCn	<PORT_base> + 0018_H + 40_H*n		—	PWE
Port Function Control Expansion register	PFCEn	<PORT_base> + $001C_H$ + 40_H*n		—	PWE
Port Mode Set Reset register	PMSRn	<PORT_base> + 0020_H + 40_H*n		—	PWE
	JPMSRn	<JPORT_base> + 0020_H + 40_H*n		—	—
Port Mode Control Set Reset register	PMCSRn	<PORT_base> + 0024_H + 40_H*n		—	PWE
	JPMCSRn	<JPORT_base> + 0024_H + 40_H*n		—	—
Port Function Control Additional Expansion register	PFCAEn	<PORT_base> + 0028_H + 40_H*n		—	PWE
Port function control extra additional expansion register	PFCEAEn	<PORT_base> + $002C_H$ + 40_H*n		—	PWE
Port output value Inversion register	PINVn	<PORT_base> + 0030_H + 40_H*n		PWE	PWE
	JPINVn	<JPORT_base> + 0030_H + 40_H*n		—	—
Port Input Buffer Control register	PIBCn	<PORT_base> + 4000_H + 40_H*n		—	PWE
	JPIBCn	<JPORT_base> + 4000_H + 40_H*n		—	—
Port Bi-Direction Control register	PBDCn	<PORT_base> + 4004_H + 40_H*n		—	PWE
	JPBDCn	<JPORT_base> + 4004_H + 40_H*n		—	—
Port IP Control register	PIPCn	<PORT_base> + 4008_H + 40_H*n		—	PWE
	JPIPCn	<JPORT_base> + 4008_H + 40_H*n		—	—

Table 2.11 Port Register Overview (2/2)

			Access Protection		
				Other	
Port Register Name	**Port Symbol**	**Address**	**PBG**	**Category1 *1**	**Category2 *2**
Pull-Up option register	PUn	<PORT_base> + $400C_H$ + 40_H*n	Refer to Appendix “E02-02_List of Port Functions.xlsx”	—	PWE
	JPUn	<JPORT_base> + $400C_H$ + 40_H*n		—	—
Pull-Down option register	PDn	<PORT_base> + 4010_H + 40_H*n		—	PWE
	JPDn	<JPORT_base> + 4010_H + 40_H*n		—	—
Port Open Drain Control register	PODCn	<PORT_base> + 4014_H + 40_H*n		PWE	PWE
	JPODCn	<JPORT_base> + 4014_H + 40_H*n		—	—
Port Drive Strength Control register	PDSCn	<PORT_base> + 4018_H + 40_H*n		PWE	PWE
	JPDSCn	<JPORT_base> + 4018_H + 40_H*n		—	—
Port Input buffer Selection register	PISn	<PORT_base> + $401C_H$ + 40_H*n		—	PWE
	JPISn	<JPORT_base> + $401C_H$ + 40_H*n		—	—
Port Input buffer Selection Advanced register	PISAn	<PORT_base> + 4024_H + 40_H*n		—	PWE
	JPISAn	<JPORT_base> + 4024_H + 40_H*n		—	—
Port Universal Characteristic Control register	PUCCn	<PORT_base> + 4028_H + 40_H*n		PWE	PWE
	JPUCCn	<JPORT_base> + 4028_H + 40_H*n		—	—
Port Open Drain Control Expansion register	PODCEn	<PORT_base> + 4038_H + 40_H*n		PWE	PWE
	JPODCEn	<JPORT_base> + 4038_H + 40_H*n		—	—
Port Control register	PCRn_m	<PORT_base> + 2000_H + 40_H*n + 4_H*m		PWE	PWE
	JPCRn_m	<JPORT_base> + 2000_H + 40_H*n + 4_H*m		—	—
Port Safe State Control register	PSFCn	<PORT_base> + 6000_H + 40_H*n		PWE	PWE
Port ERRORIN Open-drain Control register	PEIODCn	<PORT_base> + 6008_H + 40_H*n		PWE	PWE
Port Safe State Trigger Selection register	PSFTSn	<PORT_base> + 6010_H + 40_H*n		PWE	PWE
Port Safe State Trigger Selection Expansion register	PSFTSEn	<PORT_base> + 6014_H + 40_H*n		PWE	PWE
Port Safe State Trigger Selection Additional Expansion register	PSFTSAEn	<PORT_base> + 6018_H + 40_H*n		PWE	PWE
Pull-up Pull-down Voltage Option Register	PULVSEL5	<PORT_base> + $2F80_H$	PBG20#12	—	—
Port Keycode Protection Register	PKCPROT	<PORT_base> + $2F40_H$	PBG20#12	—	—
Port Write Enable register	PWE	<PORT_base> + $2F44_H$	PBG20#12	PKCPROT	PKCPROT
LVDS control A register	LVDSCTRLA	<PORT_base> + $2F50_H$	PBG20#12	PWE	—
LVDS control B register	LVDSCTRLB	<PORT_base> + $2F54_H$	PBG20#12	PWE	—
LVDS control C register	LVDSCTRLC	<PORT_base> + $2F58_H$	PBG20#12	PWE	—
LVDS control D register	LVDSCTRLD	<PORT_base> + $2F60_H$	PBG20#12	PWE	—
LVDS control E register	LVDSCTRLE	<PORT_base> + $2F64_H$	PBG20#12	PWE	—
LVDS control F register	LVDSCTRLF	<PORT_base> + $2F68_H$	PBG20#12	PWE	—

Note 1. Category1: Port excluding P21, P24_4~P24_7, P25_3~P25_6, P25_12~P25_15, P27.

Note 2. Category2: P21, P24_4~P24_7, P25_3~P25_6, P25_12~P25_15, P27, all registers are protected. P21, P24_4~P24_7, P25_3~P25_6, P25_12~P25_15 are intended to protect the LVDS outputs which

belong to LVDVCC.
P27 is intended to protect the $\overline{\text{RESETOUT}}$ output.

Note 3. LVDSCTRLA, LVDSCTRLB are not supported in U2B6.
LVDSCTRLE, LVDSCTRLF are not supported in U2B20, U2B10 and U2B6.

2.5.1 Pn/JP0,1 — Port Register

This register defines port pins output levels for port output mode.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	Pn_15	Pn_14	Pn_13	Pn_12	Pn_11	Pn_10	Pn_9	Pn_8	Pn_7	Pn_6	Pn_5	Pn_4	Pn_3	Pn_2	Pn_1	Pn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.12 Pn Register Contents

Bit Position	Bit Name	Function
15 to 0	Pn_[15:0]	Sets the output level of pin m (m = 0 to 15): 0: Port pin drives low level 1: Port pin drives high level

NOTES

1. Reading Pn returns the register value independent from other register settings.
2. The value on this register bit is reflected to a pin level in the following conditions.
   Case : Port Mode (PMCn_m=0) & Output Mode (PMn_m=0)

2.5.2 PPRn/JPPR0,1 — Port Pin Read Register

This register reflects the actual pin level when the input buffer is active.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PPRn_15	PPRn_14	PPRn_13	PPRn_12	PPRn_11	PPRn_10	PPRn_9	PPRn_8	PPRn_7	PPRn_6	PPRn_5	PPRn_4	PPRn_3	PPRn_2	PPRn_1	PPRn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.13 PPRn Register Contents

Bit Position	Bit Name	Function
15 to 0	PPRn_[15:0]	This register reflects the actual pin level when the input buffer is active 0: Port pin is at low level 1: Port pin is at high level

2.5.3 PMn/JPM0,1 — Port Mode Register

This register selects the pin direction as input or output.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PMn_15	PMn_14	PMn_13	PMn_12	PMn_11	PMn_10	PMn_9	PMn_8	PMn_7	PMn_6	PMn_5	PMn_4	PMn_3	PMn_2	PMn_1	PMn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.14 PMn Register Contents

Bit Position	Bit Name	Function
15 to 0	PMn_[15:0]	Specifies input/output mode of the corresponding pin. 0: Output mode (output enabled) 1: Input mode (output disabled)

2.5.4 PMCn/JPMC0,1 — Port Mode Control Register

This register specifies whether the individual pins of port group n are in port mode or in alternative mode.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PMCn_15	PMCn_14	PMCn_13	PMCn_12	PMCn_11	PMCn_10	PMCn_9	PMCn_8	PMCn_7	PMCn_6	PMCn_5	PMCn_4	PMCn_3	PMCn_2	PMCn_1	PMCn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.15 PMCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PMCn_[15:0]	Specifies the operation mode of the corresponding pin. 0: Port mode 1: Alternative mode

2.5.5 PIBCn/JPIBC0,1 — Port Input Buffer Control Register

When a pin is being used in input port mode (PMCn.PMCn_m = 0 and PMn.PMn_m = 1), this register enables or disables the input buffer. However, when the pin is used as an input pin in S/W I/O control alternative mode (PMCn.PMCn_m = 1 and PIPCn.PIPCn_m = 0) or direct I/O control alternative mode (PMCn.PMCn_m = 1 and PIPCn.PIPCn_m = 1), set PIBCn.PIBCn_m = 0.

When pins are in bidirectional mode (PBDCn.PBDCn_m = 1), the shared output level loop-back function and pin output level-read function can be selected by setting PIBCn.PIBCn_m. Refer to **Section 2.5.15, PBDCn/JPBDC0,1 — Port Bi-Direction Control Register.**

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PIBCn_15	PIBCn_14	PIBCn_13	PIBCn_12	PIBCn_11	PIBCn_10	PIBCn_9	PIBCn_8	PIBCn_7	PIBCn_6	PIBCn_5	PIBCn_4	PIBCn_3	PIBCn_2	PIBCn_1	PIBCn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.16 PIBCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PIBCn_[15:0]	Enables/disables the input buffer: 0: Input buffer is disabled. 1: Input buffer is enabled.

NOTE

1. To enable port pin's input buffer, the IO direction must be set as input mode by PM=1 during Port Mode (PMC=0).
2. By keeping this register at a reset value of 0, port pin's input buffer does not consumes current even when the pin level is at an intermediate voltage. When the input buffer is disabled, in port mode, through current does not flow even when the pin level is Hi-Z. Thus the pin does not need to be fixed to a high or low level externally.
3. During this register set 1 with Port Mode (PMC=0), the values for peripheral macro are fixed.
4. During "Software I/O control alternative-function input" Mode (PMC=1, PM=1, PIPC=0), this register bit must be set to 0.

2.5.6 PFCn — Port Function Control Register

This register selects the alternative peripheral functions together with PFCEn, PFCAEn, PFCEAEn and PMn in Control Mode (PMCn = 1).

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PFCn_15	PFCn_14	PFCn_13	PFCn_12	PFCn_11	PFCn_10	PFCn_9	PFCn_8	PFCn_7	PFCn_6	PFCn_5	PFCn_4	PFCn_3	PFCn_2	PFCn_1	PFCn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.17 PFCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PFCn_[15:0]	Specifies the alternative function of a pin. See **Table 2.21, Setting Alternative Functions**

2.5.7 PFCEn — Port Function Control Expansion Register

This register selects the alternative peripheral functions together with PFCn, PFCAEn, PFCEAEn and PMn in Control Mode (PMCn = 1).

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PFCEn_15	PFCEn_14	PFCEn_13	PFCEn_12	PFCEn_11	PFCEn_10	PFCEn_9	PFCEn_8	PFCEn_7	PFCEn_6	PFCEn_5	PFCEn_4	PFCEn_3	PFCEn_2	PFCEn_1	PFCEn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix “*E02-02_List of Port Functions.xlsx*”

Table 2.18 PFCEn Register Contents

Bit Position	Bit Name	Function
15 to 0	PFCEn_[15:0]	Specifies an alternative function of a pin. See **Table 2.21, Setting Alternative Functions**

2.5.8 PFCAEn — Port Function Control Additional Expansion Register

This register selects the alternative peripheral functions together with PFCn, PFCEn, PFCEAEn and PMn in Control Mode (PMCn = 1).

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PFCAEn_15	PFCAEn_14	PFCAEn_13	PFCAEn_12	PFCAEn_11	PFCAEn_10	PFCAEn_9	PFCAEn_8	PFCAEn_7	PFCAEn_6	PFCAEn_5	PFCAEn_4	PFCAEn_3	PFCAEn_2	PFCAEn_1	PFCAEn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix “*E02-02_List of Port Functions.xlsx*”

Table 2.19 PFCAEn Register Contents

Bit Position	Bit Name	Function
15 to 0	PFCAEn_[15:0]	Specifies an alternative function of a pin. See **Table 2.21, Setting Alternative Functions**.

2.5.9 PFCEAEn — Port Function Control Extra Additional Expansion Register

This register, together with the PFCn, PFCEn, and PFCAEn registers, specifies an alternative function of the pins.Some alternative functions directly control input/output of pin Pn_m. For such alternative functions PIPCn.PIPCn_m must be set to 1 as well.For other alternative functions, input/output must be specified by PMn.PMn_m.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PFCEAEn_15	PFCEAEn_14	PFCEAEn_13	PFCEAEn_12	PFCEAEn_11	PFCEAEn_10	PFCEAEn_9	PFCEAEn_8	PFCEAEn_7	PFCEAEn_6	PFCEAEn_5	PFCEAEn_4	PFCEAEn_3	PFCEAEn_2	PFCEAEn_1	PFCEAEn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.20 PFCEAEn Register Contents

Bit Position	Bit Name	Function
15 to 0	PFCEAEn_[15:0]	Specifies an alternative function of a pin. See **Table 2.21, Setting Alternative Functions**.

Table 2.21 Setting Alternative Functions

Bit					Function
PFCEAEn_m	**PFCAEn_m**	**PFCEn_m**	**PFCn_m**	**PMn_m**	
0	0	0	0	0	Alternative output mode 1 (ALT-OUT1)
				1	Alternative input mode 1 (ALT-IN1)
			1	0	Alternative output mode 2 (ALT-OUT2)
				1	Alternative input mode 2 (ALT-IN2)
		1	0	0	Alternative output mode 3 (ALT-OUT3)
				1	Alternative input mode 3 (ALT-IN3)
			1	0	Alternative output mode 4 (ALT-OUT4)
				1	Alternative input mode 4 (ALT-IN4)
	1	0	0	0	Alternative output mode 5 (ALT-OUT5)
				1	Alternative input mode 5 (ALT-IN5)
			1	0	Alternative output mode 6 (ALT-OUT6)
				1	Alternative input mode 6 (ALT-IN6)
		1	0	0	Alternative output mode 7 (ALT-OUT7)
				1	Alternative input mode 7 (ALT-IN7)
			1	0	Alternative output mode 8 (ALT-OUT8)
				1	Alternative input mode 8 (ALT-IN8)
1	0	0	0	0	Alternative output mode 9 (ALT-OUT9)
				1	Alternative input mode 9 (ALT-IN9)
			1	0	Alternative output mode 10 (ALT-OUT10)
				1	Alternative input mode 10 (ALT-IN10)
		1	0	0	Alternative output mode 11 (ALT-OUT11)
				1	Alternative input mode 11 (ALT-IN11)
			1	0	Alternative output mode 12 (ALT-OUT12)
				1	Alternative input mode 12 (ALT-IN12)
	1	0	0	0	Alternative output mode 13 (ALT-OUT13)
				1	Alternative input mode 13 (ALT-IN13)
			1	0	Alternative output mode 14 (ALT-OUT14)
				1	Alternative input mode 14 (ALT-IN14)
		1	0	0	Alternative output mode 15 (ALT-OUT15)
				1	Alternative input mode 15 (ALT-IN15)
			1	0	Alternative output mode 16 (ALT-OUT16)
				1	Alternative input mode 16 (ALT-IN16)

Note: For JP0, ALT_IN1 or ALT_OUT1 will be selected when alternative mode is active (JPMC0_n=1).

2.5.10 PNOTn/JPNOT0,1 — Port NOT Register

This register provided a method to flip the bit values of Pn register. The bits of Pn register are flipped if the PNOTn register is written with the corresponding bit values being 1.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PNOTn_15	PNOTn_14	PNOTn_13	PNOTn_12	PNOTn_11	PNOTn_10	PNOTn_9	PNOTn_8	PNOTn_7	PNOTn_6	PNOTn_5	PNOTn_4	PNOTn_3	PNOTn_2	PNOTn_1	PNOTn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.22 PNOTn Register Contents

Bit Position	Bit Name	Function
15 to 0	PNOTn_[15:0]	0: No effect on the value of Pn_m bit 1: The value of Pn_m bit is flipped

2.5.11 PSRn/JPSR0,1 — Port Set/Reset Register

This register provides an alternative method to write/read data on Pn register.

The upper 16 bits of PSRn act as a mask which specifies whether or not the value Pn.Pn_m is set by the corresponding bit in the lower 16 bits of PSRn.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PSRn_31	PSRn_30	PSRn_29	PSRn_28	PSRn_27	PSRn_26	PSRn_25	PSRn_24	PSRn_23	PSRn_22	PSRn_21	PSRn_20	PSRn_19	PSRn_18	PSRn_17	PSRn_16
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PSRn_15	PSRn_14	PSRn_13	PSRn_12	PSRn_11	PSRn_10	PSRn_9	PSRn_8	PSRn_7	PSRn_6	PSRn_5	PSRn_4	PSRn_3	PSRn_2	PSRn_1	PSRn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix “*E02-02_List of Port Functions.xlsx*”

Table 2.23 PSRn Register Contents

Bit Position	Bit Name	Function
31 to 16	PSRn_[31:16]	Specifies whether the value of the corresponding lower bit PSRn_m value is written to Pn_m. 0: Pn_m is not affected by PSRn_m 1: Pn_m is PSRn_m Example: When PSRn.PSRn_31 = 1, the value of PSRn.PSRn_15 is written to bit Pn.Pn_15 and output. When read, 0000_H is always returned.
15 to 0	PSRn_[15:0]	Sets the output level of pin Pn_m (m = 0 to 15). 0: Low level is written on Pn_m when it is enabled by PSRn_(m+16) 1: High level is written on Pn_m when it is enabled by PSRn_(m+16) When read, Pn register value is returned.

2.5.12 PMSRn/JPMSR0,1 — Port Mode Set/Reset Register

This register provides an alternative method to write/read data on PMn register.

The upper 16 bits of PMSRn act as a mask which specifies whether or not the value PMn.PMn_m is set by the corresponding bit in the lower 16 bits of PMSRn.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PMSRn_31	PMSRn_30	PMSRn_29	PMSRn_28	PMSRn_27	PMSRn_26	PMSRn_25	PMSRn_24	PMSRn_23	PMSRn_22	PMSRn_21	PMSRn_20	PMSRn_19	PMSRn_18	PMSRn_17	PMSRn_16
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PMSRn_15	PMSRn_14	PMSRn_13	PMSRn_12	PMSRn_11	PMSRn_10	PMSRn_9	PMSRn_8	PMSRn_7	PMSRn_6	PMSRn_5	PMSRn_4	PMSRn_3	PMSRn_2	PMSRn_1	PMSRn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.24 PMSRn Register Contents

Bit Position	Bit Name	Function
31 to 16	PMSRn_[31:16]	Specifies whether the value of the corresponding lower bit PMSRn_m value is written to PMn_m. 0: PMn_m is not affected by PMSRn_m. 1: PMn_m is PMSRn_m Example: When PMSRn.PMSRn_31 = 1, the value of bit PMSRn.PMSRn_15 is written to bit PMn.PMn_15. When read, 0000_H is always returned.
15 to 0	PMSRn_[15:0]	Data bits that specify the PMn_m value if the corresponding upper bit (PMSRn_[31:16]) PMSRn_m is 1. 0: PMn_m = 0 1: PMn_m = 1 When read, PMn register value is returned.

2.5.13 PMCSRn/JPMCSR0,1 — Port Mode Control Set/Reset Register

This register provides an alternative method to write data to the PMCn register.

The upper 16 bits of PMCSRn act as a mask which specifies whether or not the value PMCn.PMCn_m is set by the corresponding bit in the lower 16 bits of PMCSRn.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PMCSRn_31	PMCSRn_30	PMCSRn_29	PMCSRn_28	PMCSRn_27	PMCSRn_26	PMCSRn_25	PMCSRn_24	PMCSRn_23	PMCSRn_22	PMCSRn_21	PMCSRn_20	PMCSRn_19	PMCSRn_18	PMCSRn_17	PMCSRn_16
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PMCSRn_15	PMCSRn_14	PMCSRn_13	PMCSRn_12	PMCSRn_11	PMCSRn_10	PMCSRn_9	PMCSRn_8	PMCSRn_7	PMCSRn_6	PMCSRn_5	PMCSRn_4	PMCSRn_3	PMCSRn_2	PMCSRn_1	PMCSRn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.25 PMCSRn Register Contents

Bit Position	Bit Name	Function
31 to 16	PMCSRn_ [31:16]	Specifies whether the value of the corresponding lower bit PMCSRn_m value is written to PMCn_m. 0: PMCn_m is not affected by PMCSRn_m 1: PMCn_m is PMCSRn_m Example: When PMCSRn.PMCSRn_31 = 1, the value of bit PMCSRn.PMCSRn_15 is written to bit PMCn.PMCn_15. When read, 0000_H is always returned.
15 to 0	PMCSRn_ [15:0]	Data bits that specify the PMCn_m value if the corresponding upper bit (PMCSRn_[31:16]) PMCSRn_m is 1. 0: PMCn_m = 0 1: PMCn_m = 1 When read, PMCn register value is returned.

2.5.14 PINVn/JPINV0,1 — Port Output value Inversion Register

This register inverts the output value of the port.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"
Address: Refer to **Table 2.11, Port Register Overview**
Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PINVn_15	PINVn_14	PINVn_13	PINVn_12	PINVn_11	PINVn_10	PINVn_9	PINVn_8	PINVn_7	PINVn_6	PINVn_5	PINVn_4	PINVn_3	PINVn_2	PINVn_1	PINVn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.26 PINVn Register Contents

Bit Position	Bit Name	Function
15 to 0	PINVn_[15:0]	This register inverts the output value of the port. 0: No effect 1: Inverted value is output

2.5.15 PBDCn/JPBDC0,1 — Port Bi-Direction Control Register

This register enables the input buffer when a pin is used in output mode, and permits bidirectional mode. The Pn_m pin level is read via PPRn.PPRn_m in bidirectional mode.

- Alternative output level loopback function
  When the Pn_m pin is used as the alternative output function, the actual pin output level based on the alternative output function can be looped back to the alternative input side by setting PBDCn.PBDCn_m = 1 and PIBCn.PIBCn_m = 0. For example, the pin output level based on the first alternative function can be looped back to the same alternative input side. Also the pin output level can be read via PPRn.PPRn_m.
- Pin output level read function
  When the Pn_m pin is used as the general output port function or the alternative output function, the actual pin output level can be read via PPRn.PPRn_m by setting PBDCn.PBDCn_m = 1 and PIBCn.PIBCn_m = 1. Under this setting, the pin output level will never be looped back to the alternative input side even in alternative output mode.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PBDCn_15	PBDCn_14	PBDCn_13	PBDCn_12	PBDCn_11	PBDCn_10	PBDCn_9	PBDCn_8	PBDCn_7	PBDCn_6	PBDCn_5	PBDCn_4	PBDCn_3	PBDCn_2	PBDCn_1	PBDCn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.27 PBDCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PBDCn_[15:0]	Enables/disables bi-direction mode of the corresponding pin. 0: Bi-direction mode disabled 1: Bi-direction mode enabled

NOTE

Loopback is enabled after four cycles of APB Low freq clock (CLK_LSB) since this register is written.

2.5.16 PIPCn/JPIPC0 — Port IP Control Register

This register specifies whether the I/O direction of pin Pn_m is controlled by the port mode register PMn.PMn_m or by an alternative function. If pin Pn_m is operated in alternative mode (PMCn.PMCn_m = 1) and the alternative function requires direct control of the I/O direction, then PIPCn.PIPCn_m must be set to 1 as well. This transfers I/O control to the alternative function and overrules the PMn.PMn_m setting.

The list of alternative functions that require direct control of the I/O direction is **Table 2.29, Alternative functions require "Direct I/O Control" (Must Set PIPCn_m=1)**.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PIPCn_15	PIPCn_14	PIPCn_13	PIPCn_12	PIPCn_11	PIPCn_10	PIPCn_9	PIPCn_8	PIPCn_7	PIPCn_6	PIPCn_5	PIPCn_4	PIPCn_3	PIPCn_2	PIPCn_1	PIPCn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.28 PIPCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PIPCn_[15:0]	Specifies the I/O control mode. 0: I/O mode is selected by PMn.PMn_m (software I/O control). 1: I/O mode is selected by the peripheral function (direct I/O control).

Table 2.29 Alternative functions require "Direct I/O Control" (Must Set PIPCn_m=1) (1/3)

Category	Pin Name	I/O	Function description
PIC	TAPA0UN	O	Motor control output U phase (negative)
	TAPA0UP	O	Motor control output U phase (positive)
	TAPA0VN	O	Motor control output V phase (negative)
	TAPA0VP	O	Motor control output V phase (positive)
	TAPA0WN	O	Motor control output W phase (negative)
	TAPA0WP	O	Motor control output W phase (positive)
TAUD	TAUD1O10	O	TAUD1 output for channel 10
	TAUD1O11	O	TAUD1 output for channel 11
	TAUD1O12	O	TAUD1 output for channel 12
	TAUD1O13	O	TAUD1 output for channel 13
	TAUD1O14	O	TAUD1 output for channel 14
	TAUD1O15	O	TAUD1 output for channel 15
	TAUD2O10	O	TAUD2 output for channel 10
	TAUD2O11	O	TAUD2 output for channel 11
	TAUD2O12	O	TAUD2 output for channel 12
	TAUD2O13	O	TAUD2 output for channel 13
	TAUD2O14	O	TAUD2 output for channel 14
	TAUD2O15	O	TAUD2 output for channel 15

Table 2.29 Alternative functions require “Direct I/O Control” (Must Set PIPCn_m=1) (2/3)

Category	Pin Name	I/O	Function description
TSG3	TSG30O1	O	TSG30 timer output 1
	TSG30O2	O	TSG30 timer output 2
	TSG30O3	O	TSG30 timer output 3
	TSG30O4	O	TSG30 timer output 4
	TSG30O5	O	TSG30 timer output 5
	TSG30O6	O	TSG30 timer output 6
	TSG31O1	O	TSG31 timer output 1
	TSG31O2	O	TSG31 timer output 2
	TSG31O3	O	TSG31 timer output 3
	TSG31O4	O	TSG31 timer output 4
	TSG31O5	O	TSG31 timer output 5
	TSG31O6	O	TSG31 timer output 6
	TSG32O1	O	TSG32 timer output 1
	TSG32O2	O	TSG32 timer output 2
	TSG32O3	O	TSG32 timer output 3
	TSG32O4	O	TSG32 timer output 4
	TSG32O5	O	TSG32 timer output 5
	TSG32O6	O	TSG32 timer output 6

Table 2.29 Alternative functions require “Direct I/O Control” (Must Set PIPCn_m=1) (3/3)

Category	Pin Name	I/O	Function description
HRPWM	TSG31O1_H	O	TSG31 timer output 1
	TSG31O2_H	O	TSG31 timer output 2
	TSG31O3_H	O	TSG31 timer output 3
	TSG31O4_H	O	TSG31 timer output 4
	TSG31O5_H	O	TSG31 timer output 5
	TSG31O6_H	O	TSG31 timer output 6
	TSG32O1_H	O	TSG32 timer output 1
	TSG32O2_H	O	TSG32 timer output 2
	TSG32O3_H	O	TSG32 timer output 3
	TSG32O4_H	O	TSG32 timer output 4
	TSG32O5_H	O	TSG32 timer output 5
	TSG32O6_H	O	TSG32 timer output 6
	ATOM1_0_H	O	GTM output signals for ATOM
	ATOM1_1_H	O	GTM output signals for ATOM
	ATOM1_2_H	O	GTM output signals for ATOM
	ATOM1_3_H	O	GTM output signals for ATOM
	ATOM2_0_H	O	GTM output signals for ATOM
	ATOM2_1_H	O	GTM output signals for ATOM
	ATOM2_2_H	O	GTM output signals for ATOM
	ATOM2_3_H	O	GTM output signals for ATOM
	ATOM1_0N_H	O	GTM inverted timer output signals for ATOM
	ATOM1_1N_H	O	GTM inverted timer output signals for ATOM
	ATOM1_2N_H	O	GTM inverted timer output signals for ATOM
	ATOM1_3N_H	O	GTM inverted timer output signals for ATOM
	ATOM2_0N_H	O	GTM inverted timer output signals for ATOM
	ATOM2_1N_H	O	GTM inverted timer output signals for ATOM
	ATOM2_2N_H	O	GTM inverted timer output signals for ATOM
	ATOM2_3N_H	O	GTM inverted timer output signals for ATOM
MSPI	MSPInSO	O	MSPI serial data output
	MSPInDCS	I	MSPI data consistency check
GTM	ATOMn_m	O	GTM output signals for ATOM
	ATOMn_mN	O	GTM inverted timer output signals for ATOM
MMCA	MMCAnCMD	I/O	MMCA command/response
	MMCAnDATm	I/O	MMCA data
SFMA	SFMAnIOm	I/O	SFMA transmit/receive data
ETH	ETHn_MDIO	I/O	PHY management transfer data signal
ATU Timer C	TIOCn0 to TIOCn3	I/O	ATU timer C Input: capture trigger input Output: compare output
DSMIF	DSMIFn_DSMCLK_INm	I	Input clock from Delta-Sigma Modulator CHm of DSMIFn (slave mode)
DSMIF	DSMIFn_DSMCLK_OUTm	O	Output clock for Delta-Sigma Modulator CHm of DSMIFn (master mode)

2.5.17 PUn/JPU0,1 — Pull-up Option Register

This register specifies whether pull-up resistor is connected to an input pin. For the electrical characteristics for each setting, refer to **Section 66, Electrical Characteristics**.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PUn_15	PUn_14	PUn_13	PUn_12	PUn_11	PUn_10	PUn_9	PUn_8	PUn_7	PUn_6	PUn_5	PUn_4	PUn_3	PUn_2	PUn_1	PUn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.30 PUn Register Contents

Bit Position	Bit Name	Function
15 to 0	PUn_[15:0]	Specifies whether a pull-up resistor is connected to the corresponding pin. 0: No pull-up resistor connected 1: Pull-up resistor connected

NOTES

1. If a pin is configured such that both an internal pull-up resistor (PUn.PUn_m = 1) and pull-down resistor (PDn.PDn_m = 1) are connected, the pull-down resistor is automatically selected and the pull-up resistor is not connected.
2. The pull-up resistor has no effect when the pin is operated in output mode.

2.5.18 PDn/JPD0,1 — Pull-down Option Register

This register specifies whether to connect an internal pull-down resistor to an input pin. For the electrical characteristics for each setting, refer to **Section 66, Electrical Characteristics**.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PDn_15	PDn_14	PDn_13	PDn_12	PDn_11	PDn_10	PDn_9	PDn_8	PDn_7	PDn_6	PDn_5	PDn_4	PDn_3	PDn_2	PDn_1	PDn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.31 PDn Register Contents

Bit Position	Bit Name	Function
15 to 0	PDn_[15:0]	Specifies whether to connect an internal pull-down resistor to the corresponding pin: 0: No internal pull-down resistor connected 1: An internal pull-down resistor connected

NOTES

1. If a pin is configured such that both an internal pull-up resistor (PUn.PUn_m = 1) and pull-down resistor (PDn.PDn_m = 1) are connected, the pull-down resistor is automatically selected and the pull-up resistor is not connected.
2. The internal pull-down resistor has no effect when the pin is operated in output mode.

2.5.19 PODCn/JPODC0,1 — Port Open-drain Control Register

This register selects push-pull or open-drain as the output buffer function.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PODC n_15	PODC n_14	PODC n_13	PODC n_12	PODC n_11	PODC n_10	PODC n_9	PODC n_8	PODC n_7	PODC n_6	PODC n_5	PODC n_4	PODC n_3	PODC n_2	PODC n_1	PODC n_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.32 PODCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PODCn_[15:0]	Specifies the output buffer function. 0: Push-pull 1: Open-drain

2.5.20 PODCEn/JPODCE0,1 — Port Open-drain Control Expansion Register

This register selects the emulated P-channel Open-drain together with PODCn.

Access: Refer to Appendix “*E02-02_List of Port Functions.xlsx*”

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix “*E02-02_List of Port Functions.xlsx*”

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PODCEn_15	PODCEn_14	PODCEn_13	PODCEn_12	PODCEn_11	PODCEn_10	PODCEn_9	PODCEn_8	PODCEn_7	PODCEn_6	PODCEn_5	PODCEn_4	PODCEn_3	PODCEn_2	PODCEn_1	PODCEn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix “*E02-02_List of Port Functions.xlsx*”

Table 2.33 PODCEn Register Contents

Bit Position	Bit Name	Function
15 to 0	PODCEn_[15:0]	Refer to **Table 2.34, Port Open Drain Control Expansion** for the detailed operation on PODCE.

Table 2.34 Port Open Drain Control Expansion

PODCEn_m	PODCn_m	Function
0	0	Push-pull
0	1	Emulated N-channel Open Drain
1	0	Push-pull
1	1	Emulated P-channel Open Drain

2.5.21 PDSCn/JPDSC0,1 — Port Drive Strength Control Register

This register, together with the PUCCn registers, specifies the output buffer drive strength of a pin. Specific driving ability settings may be required, depending on the pin function to be used. For details, refer to **Section 66, Electrical Characteristics**.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PDSCn_15	PDSCn_14	PDSCn_13	PDSCn_12	PDSCn_11	PDSCn_10	PDSCn_9	PDSCn_8	PDSCn_7	PDSCn_6	PDSCn_5	PDSCn_4	PDSCn_3	PDSCn_2	PDSCn_1	PDSCn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.35 PDSCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PDSCn_[15:0]	Specifies the port drive strength of the output buffer of the port pin. See **Table 2.37, Port Output Buffer drive Strength Selection**.

2.5.22 PUCCn/JPUCC0,1 — Port Universal Characteristic Control Register

This register, together with the PDSCn registers, specifies the output buffer drive strength of a pin. Specific driving ability settings may be required, depending on the pin function to be used. For details, refer to **Section 66, Electrical Characteristics**.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PUCCn_15	PUCCn_14	PUCCn_13	PUCCn_12	PUCCn_11	PUCCn_10	PUCCn_9	PUCCn_8	PUCCn_7	PUCCn_6	PUCCn_5	PUCCn_4	PUCCn_3	PUCCn_2	PUCCn_1	PUCCn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.36 PUCCn Register Contents

Bit Position	Bit Name	Function
15 to 0	PUCCn_[15:0]	Specifies the port drive strength of the output buffer of the port pin. See **Table 2.37, Port Output Buffer drive Strength Selection**.

Table 2.37 Port Output Buffer drive Strength Selection

PUCCn_m	PDSCn_m	Output Buffer drive Strength
0	0	Drive strength = 5 (very low)
	1	Drive strength = 4 (low)
1	0	Drive strength = 3 (medium)
	1	Drive strength = 2 (high) Drive strength = 1*1 (very high)

Note 1. Only P11_2, P11_5, P11_7, P11_9, P14_0, P14_3.

NOTES

1. For details of selectable input characteristics, refer to the appendix "E02_01_List_of_Pin_Assignment.xlsx".
2. JP0_1 is selected to "Drive strength = 3" when the pin is used as LPD4 interface and "Drive strength = 2" when the pin is used as Nexus and Boundary SCAN interface.
3. JP0_5 is selected to "Drive strength = 3" when the pin is used as LPD4 interface, and "Drive strength = 2" when the pin is used as Nexus interface.
4. JP1_1, JP1_5 is selected to "Drive strength = 3" when the pin is used as LPD4 interface and "Drive strength = 2" when the pin is used as Nexus interface.
[For U2B-EVA Only]

2.5.23 PISn/JPIS0,1 — Port input buffer selection register

This register and the PISAn specifies the input buffer characteristics. Specific input characteristic settings may be required, depending on the pin function to be used. For details, refer to **Section 66, Electrical Characteristics**.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PISn_15	PISn_14	PISn_13	PISn_12	PISn_11	PISn_10	PISn_9	PISn_8	PISn_7	PISn_6	PISn_5	PISn_4	PISn_3	PISn_2	PISn_1	PISn_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.38 PISn Register Contents

Bit Position	Bit Name	Function
15 to 0	PISn_[15:0]	Specifies the input buffer characteristic. See **Table 2.40, Port Input buffer Characteristics Selection**.

2.5.24 PISAn/JPISA0,1 – Port Input buffer Selection Advanced register

This register and the PISn specifies the input buffer characteristics. Specific input characteristic settings may be required, depending on the pin function to be used. For details, refer to **Section 66, Electrical Characteristics**.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PISA n_15	PISA n_14	PISA n_13	PISA n_12	PISA n_11	PISA n_10	PISA n_9	PISA n_8	PISA n_7	PISA n_6	PISA n_5	PISA n_4	PISA n_3	PISA n_2	PISA n_1	PISA n_0
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.39 PISAn Register Contents

Bit Position	Bit Name	Function
15 to 0	PISAn_[15:0]	Specifies the input buffer characteristic: See **Table 2.40**.

Table 2.40 Port Input buffer Characteristics Selection

PISAn_m	PISn_m	Function
0	0	SHMT1 input buffer is selected
	1	SHMT4 input buffer is selected
1	0	TTL input buffer is selected
	1	TTL input buffer is selected SHMTMSC input buffer is selected*1

Note 1. The support pins for each products are as follows.
All product: P10_5, P11_3, P11_4, P11_6, P22_1, P22_2, P22_4, P22_8
U2B-EVA (600pins): P12_1, P12_3, P12_7, P12_9, P22_11, P22_13
U2B24 (468pins): P12_1, P12_3, P12_7, P12_9, P22_11, P22_13
U2B24 (373pins): P12_1, P12_3, P12_7, P12_9
U2B20 (468pins): P12_1, P12_3, P12_7, P12_9
U2B20 (373pins): P12_1, P12_3, P12_7, P12_9
U2B10 (468pins): P12_1, P12_3, P12_7, P12_9
U2B10 (373pins): P12_1, P12_3, P12_7, P12_9
U2B10 (292pins): P12_1, P12_3, P12_7, P12_9
U2B6 (292pins): Same as "All product".

NOTES

1. For details of selectable input characteristics, refer to the appendix "E02_01_List_of_Pin_Assignment.xlsx".
2. Input buffer type of JP0_0, JP0_2 and JP0_3 is selected to TTL type when these pins are used as Nexus, LPD4 and Boundary SCAN interface.
3. Input buffer type of JP1_0, JP1_2 and JP1_3 is selected to TTL type when these pins are used as Nexus, LPD4 and Boundary SCAN interface [For U2B-EVA Only].

2.5.25 PCRn_m/JPCR0,1_m — Port Control Register

By going through this register, it is possible to have access to the registers of each port group, and the individual pins is specified all functions by 1 PCR register setting.

Access: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Address: Refer to **Table 2.11, Port Register Overview**

Value after reset: Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	PINV	—	PODC	PODCE	—	PUCC	PDSC	—	PISA	—	PIS	PU	PD	PBDC	PIBC
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R	R/W	R	R/W	R/W	R	R/W	R/W	R	R/W	R	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	P	—	—	—	PPR	—	PMC	PIPC	PM	PFCEAE	PFCAE	PFCE	PFC
Value after reset	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1	*1
R/W	R	R	R	R/W	R	R	R	R	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Note 1. Refer to Appendix "*E02-02_List of Port Functions.xlsx*"

Table 2.41 PCRn_m Register Contents (1/2)

Bit Position	Bit Name	Function
31	Reserved	When read, the value after reset is read. When writing, write the value after reset.
30	PINV	Same as the m bit of PINVn/JPINV0,1 register.
29	Reserved	When read, the value after reset is read. When writing, write the value after reset.
28	PODC	Same as the m bit of PODCn/JPODC0,1 register.
27	PODCE	Same as the m bit of PODCEn/JPODCE0,1 register.
26	Reserved	When read, the value after reset is read. When writing, write the value after reset.
25	PUCC	Same as the m bit of PUCCn/JPUCC0,1 register.
24	PDSC	Same as the m bit of PDSCn/JPDSC0,1 register.
23	Reserved	When read, the value after reset is read. When writing, write the value after reset.
22	PISA	Same as the m bit of PISAn/JPISA0,1 register.
21	Reserved	When read, the value after reset is read. When writing, write the value after reset.
20	PIS	Same as the m bit of PISn/JPIS0,1 register.
19	PU	Same as the m bit of PUn/JPU0,1 register.
18	PD	Same as the m bit of PDn/JPD0,1 register.
17	PBDC	Same as the m bit of PBDCn/JPBDC0,1 register.
16	PIBC	Same as the m bit of PIBCn/JPIBC0,1 register.
15 to 13	Reserved	When read, the value after reset is read. When writing, write the value after reset.
12	P	Same as the m bit of Pn/JP0,1 register.
11 to 9	Reserved	When read, the value after reset is read. When writing, write the value after reset.
8	PPR	Same as the m bit of PPRn/JPPR0,1 register.

Table 2.41 PCRn_m Register Contents (2/2)

Bit Position	Bit Name	Function
7	Reserved	When read, the value after reset is read. When writing, write the value after reset.
6	PMC	Same as the m bit of PMCn/JPMC0,1 register.
5	PIPC	Same as the m bit of PIPCn/JPIPC0 register.
4	PM	Same as the m bit of PMn/JPM0,1 register.
3	PFCEAE	Same as the m bit of PFCEAEn register.
2	PFCAE	Same as the m bit of PFCAEn register.
1	PFCE	Same as the m bit of PFCEn register.
0	PFC	Same as the m bit of PFCn register.

Setting all attributes per each pin

Setting up to 16 pins at the same time per each attribute

	PCR register									
Other registers	PCRn_15	PCRn_14	PCRn_13	PCRn_12	…	PCRn_3	PCRn_2	PCRn_1	PCRn_0	Bit
—	—	—	—	—	…	—	—	—	—	31
PINVn	PINVn_15	PINVn_14	PINVn_13	PINVn_12	…	PINVn_3	PINVn_2	PINVn_1	PINVn_0	30
—	—	—	—	—	…	—	—	—	—	29
PODCn	PODCn_15	PODCn_14	PODCn_13	PODCn_12	…	PODCn_3	PODCn_2	PODCn_1	PODCn_0	28
PODCEn	PODCEn_15	PODCEn_14	PODCEn_13	PODCEn_12	…	PODCEn_3	PODCEn_2	PODCEn_1	PODCEn_0	27
—	—	—	—	—	…	—	—	—	—	26
PUCCn	PUCCn_15	PUCCn_14	PUCCn_13	PUCCn_12	…	PUCCn_3	PUCCn_2	PUCCn_1	PUCCn_0	25
PDSCn	PDSCn_15	PDSCn_14	PDSCn_13	PDSCn_12	…	PDSCn_3	PDSCn_2	PDSCn_1	PDSCn_0	24
—	—	—	—	—	…	—	—	—	—	23
—	—	—	—	—	…	—	—	—	—	22
—	—	—	—	—	…	—	—	—	—	21
PISn	PISn_15	PISn_14	PISn_13	PISn_12	…	PISn_3	PISn_2	PISn_1	PISn_0	20
PUn	PUn_15	PUn_14	PUn_13	PUn_12	…	PUn_3	PUn_2	PUn_1	PUn_0	19
PDn	PDn_15	PDn_14	PDn_13	PDn_12	…	PDn_3	PDn_2	PDn_1	PDn_0	18
PBDCn	PBDCn_15	PBDCn_14	PBDCn_13	PBDCn_12	…	PBDCn_3	PBDCn_2	PBDCn_1	PBDCn_0	17
PIBCn	PIBCn_15	PIBCn_14	PIBCn_13	PIBCn_12	…	PIBCn_3	PIBCn_2	PIBCn_1	PIBCn_0	16
—	—	—	—	—	…	—	—	—	—	15
—	—	—	—	—	…	—	—	—	—	14
—	—	—	—	—	…	—	—	—	—	13
Pn	Pn_15	Pn_14	Pn_13	Pn_12	…	Pn_3	Pn_2	Pn_1	Pn_0	12
—	—	—	—	—	…	—	—	—	—	11
—	—	—	—	—	…	—	—	—	—	10
—	—	—	—	—	…	—	—	—	—	9
PPRn	PPRn_15	PPRn_14	PPRn_13	PPRn_12	…	PPRn_3	PPRn_2	PPRn_1	PPRn_0	8
—	—	—	—	—	…	—	—	—	—	7
PMCn	PMCn_15	PMCn_14	PMCn_13	PMCn_12	…	PMCn_3	PMCn_2	PMCn_1	PMCn_0	6
PIPCn	PIPCn_15	PIPCn_14	PIPCn_13	PIPCn_12	…	PIPCn_3	PIPCn_2	PIPCn_1	PIPCn_0	5
PMn	PMn_15	PMn_14	PMn_13	PMn_12	…	PMn_3	PMn_2	PMn_1	PMn_0	4
—	—	—	—	—	…	—	—	—	—	3
—	—	—	—	—	…	—	—	—	—	2
PFCEn	PFCEn_15	PFCEn_14	PFCEn_13	PFCEn_12	…	PFCEn_3	PFCEn_2	PFCEn_1	PFCEn_0	1
PFCn	PFCn_15	PFCn_14	PFCn_13	PFCn_12	…	PFCn_3	PFCn_2	PFCn_1	PFCn_0	0

Figure 2.11 Relationship between Other Registers and PCR(Port Control Register)